# environmental-engineering: the Environmental Engineering program, user guide

**environmental-engineering-ai999x 1.6.0**, Rust build: command line, desktop app and browser. **Creator / engineer:** Ibnul Jaif Farabi · **Company:** AI999X · support@ai999x.com

Environmental Engineering covers Gaussian plume dispersion and plume rise, BOD and Streeter-Phelps oxygen sag, water and wastewater treatment design, activated sludge, disinfection CT, Henry's law and air stripping, noise and solid waste. This guide is about `environmental-engineering`, the model as a self-contained program for Windows and Linux: 132 capabilities in 14 topics, answered on your own computer with no network, no Python and nothing else to install. The same compiled model runs in the AI999X desktop app and in the browser on the model's page (https://www.ai999x.com/models/environmental-engineering-ai999x/); the Python package (`pip install environmental-engineering-ai999x`) has its own guide. Every command output printed here is what the program printed.

## Contents


- 1. Get it: desktop app, command line, browser, Python
- 2. Quick start
- 3. Writing inputs
- 4. Outcomes and exit codes
- 5. Scripts and agents: batch and JSON
- 6. Figures
- 7. In a web page or Node.js: WebAssembly
- 8. Every capability


## 1. Get it

Version 1.6.0 of the AI999X desktop app and of the command-line bundle. Checksums for every file: https://downloads.ai999x.com/desktop/1.6.0/SHA256SUMS; the newest version is always under https://downloads.ai999x.com/desktop/latest/. Full instructions: https://www.ai999x.com/desktop-cli/

| what | system | download |
|---|---|---|
| Desktop app: Environmental Engineering | Windows x86-64 | https://downloads.ai999x.com/desktop/1.6.0/ai999x-environmental-engineering-1.6.0-windows-x86_64.zip |
| Desktop app: Environmental Engineering | Linux x86-64 | https://downloads.ai999x.com/desktop/1.6.0/ai999x-environmental-engineering-1.6.0-linux-x86_64.tar.gz |
| Desktop app: all models | Windows x86-64 | https://downloads.ai999x.com/desktop/1.6.0/ai999x-1.6.0-windows-x86_64.zip |
| Desktop app: all models | Linux x86-64 | https://downloads.ai999x.com/desktop/1.6.0/ai999x-1.6.0-linux-x86_64.tar.gz |
| Command line | Windows x86-64 | https://downloads.ai999x.com/desktop/1.6.0/ai999x-cli-1.6.0-windows-x86_64.zip |
| Command line | Linux x86-64 | https://downloads.ai999x.com/desktop/1.6.0/ai999x-cli-1.6.0-linux-x86_64.tar.gz |

### Desktop app

Environmental Engineering in its own app: search, questions in words, input forms, live figures and explanations. Windows: Download ai999x-environmental-engineering-1.6.0-windows-x86_64.zip, right-click it → Extract All, then open the folder and start ai999x-environmental-engineering.exe. If Windows SmartScreen asks, choose More info → Run anyway. Linux:

```
curl -LO https://downloads.ai999x.com/desktop/1.6.0/ai999x-environmental-engineering-1.6.0-linux-x86_64.tar.gz
curl -LO https://downloads.ai999x.com/desktop/1.6.0/SHA256SUMS
sha256sum -c --ignore-missing SHA256SUMS
tar xzf ai999x-environmental-engineering-1.6.0-linux-x86_64.tar.gz
./ai999x-environmental-engineering-1.6.0-linux-x86_64/ai999x-environmental-engineering
```

It opens on this model: type a question or pick a capability, fill in the form and press Compute. The AI999X desktop app with every model works the same way (choose the model first). Windows: Download ai999x-1.6.0-windows-x86_64.zip, right-click it → Extract All, then open the folder and start ai999x.exe. If Windows SmartScreen asks, choose More info → Run anyway. Linux:

```
curl -LO https://downloads.ai999x.com/desktop/1.6.0/ai999x-1.6.0-linux-x86_64.tar.gz
curl -LO https://downloads.ai999x.com/desktop/1.6.0/SHA256SUMS
sha256sum -c --ignore-missing SHA256SUMS
tar xzf ai999x-1.6.0-linux-x86_64.tar.gz
./ai999x-1.6.0-linux-x86_64/ai999x
```

### Command line, Linux

The bundle holds one program per model; `environmental-engineering` is this one. The checksum line should print OK.

```
curl -LO https://downloads.ai999x.com/desktop/1.6.0/ai999x-cli-1.6.0-linux-x86_64.tar.gz
curl -LO https://downloads.ai999x.com/desktop/1.6.0/SHA256SUMS
sha256sum -c --ignore-missing SHA256SUMS
mkdir -p ~/.local/share && tar xzf ai999x-cli-1.6.0-linux-x86_64.tar.gz -C ~/.local/share
echo 'export PATH="$HOME/.local/share/ai999x-cli-1.6.0-linux-x86_64:$PATH"' >> ~/.bashrc &&
source ~/.bashrc
environmental-engineering --version
```

### Command line, Windows (PowerShell)

The fourth line should print True.

```
$F = "ai999x-cli-1.6.0-windows-x86_64.zip"
Invoke-WebRequest "https://downloads.ai999x.com/desktop/1.6.0/$F" -OutFile $F
Invoke-WebRequest "https://downloads.ai999x.com/desktop/1.6.0/SHA256SUMS" -OutFile SHA256SUMS
(Get-FileHash $F -Algorithm SHA256).Hash.ToLower() -eq ((Select-String -SimpleMatch $F
SHA256SUMS).Line -split ' ')[0]
Expand-Archive $F -DestinationPath "$env:LOCALAPPDATA\AI999X" -Force
$D = "$env:LOCALAPPDATA\AI999X\ai999x-cli-1.6.0-windows-x86_64"; $env:Path += ";$D"
[Environment]::SetEnvironmentVariable("Path", [Environment]::GetEnvironmentVariable("Path",
"User") + ";$D", "User")
environmental-engineering --version
```

### In your browser

Open https://www.ai999x.com/models/environmental-engineering-ai999x/, choose a capability and press Compute in your browser: the same program runs on your device as WebAssembly; nothing is sent anywhere.

### Python

```
pip install environmental-engineering-ai999x
```

## 2. Quick start

### 2.1 Check the install

Prints the model, its version and that this is the Rust build.

```
$ environmental-engineering --version
environmental-engineering-ai999x 1.6.0 (Rust build, ax-engine 1.6.0)
```

### 2.2 Ask in words

Write the question as you would say it: the program reads the quantities and their units, picks the capability and answers.

```
$ environmental-engineering ask "Streeter-Phelps oxygen deficit: kd = 0.23 1/d, ka = 0.46 1/d,
L0 = 10 mg/L, D0 = 1 mg/L, t = 2 d, DOsat = 9.1 mg/L"
Read as: Streeter-Phelps oxygen deficit and dissolved oxygen at a travel time
D = 0.002726165085 kg/m^3
DO = 0.006373834915 kg/m^3
D_mgL = 2.726165085
DO_mgL = 6.373834915
  by: Streeter-Phelps oxygen deficit and dissolved oxygen at a travel time  (D = kd L0/(ka -
kd) (e^(-kd t) - e^(-ka t)) + D0 e^(-ka t); for ka = kd, D = (kd L0 t + D0) e^(-kd t))
  D0 = 0.001 kg/m^3
  DOsat = 0.0091 kg/m^3
  L0 = 0.01 kg/m^3
  ka = 5.324074074e-06 1/s
  kd = 2.662037037e-06 1/s
  t = 172800 s
  assumes: first-order deoxygenation and reaeration; plug flow; deficit stays below saturation
```

### 2.3 Find a capability

Search the capabilities by what you want to know; the first column is the id to pass to info and solve.

```
$ environmental-engineering search "streeter-phelps oxygen deficit and dissolved oxygen at a
travel time"
enveng_streeter_phelps_deficit       Streeter-Phelps oxygen deficit and dissolved oxygen at a
travel time  (water quality / oxygen sag)
enveng_oxygen_sag_distance           Dissolved oxygen at a distance downstream (Streeter-
Phelps)  (water quality / oxygen sag)
enveng_streeter_phelps_critical_point Critical (minimum) dissolved oxygen of the oxygen sag
(water quality / oxygen sag)
enveng_aeration_do_curve             Dissolved oxygen during aeration and the transfer rate
(wastewater / aeration)
enveng_do_saturation                 Dissolved oxygen saturation concentration (Benson-Krause)
(water quality / dissolved oxygen)
enveng_kla_two_point                 Overall oxygen transfer coefficient KLa from two DO
readings  (wastewater / aeration)
enveng_theoretical_oxygen_demand     Theoretical oxygen demand of an organic compound  (water
quality / oxygen demand)
enveng_thod_solution                 Theoretical oxygen demand of a solution (mg/L)  (water
quality / oxygen demand)
… 12 more
```

### 2.4 List the capabilities

Every capability the program answers: its id (what you pass to solve) and its name.

```
$ environmental-engineering list
enveng_activated_sludge_cstr_design      Activated sludge (complete mix) effluent substrate and
biomass (Lawrence-McCarty)
enveng_aeration_do_curve                 Dissolved oxygen during aeration and the transfer rate
enveng_aeration_oxygen_requirement       Oxygen requirement of an activated-sludge process
enveng_air_stripping_tower               Packed-tower air stripping: stripping factor, transfer
units and height
enveng_anaerobic_methane_yield           Methane production from COD removed (anaerobic
digestion)
enveng_background_noise_correction       Source level after removing background noise
enveng_backwash_bed_expansion            Filter bed expansion during backwash
enveng_bod_dilution_test                 BOD from a dilution test (with optional seed
correction)
… 124 more
```

### 2.5 What a capability needs

The equation or method, every input and output with its SI unit. Read this before solve: it tells you the input names.

```
$ environmental-engineering info enveng_streeter_phelps_deficit
Streeter-Phelps oxygen deficit and dissolved oxygen at a travel time
  D = kd L0/(ka - kd) (e^(-kd t) - e^(-ka t)) + D0 e^(-ka t); for ka = kd, D = (kd L0 t + D0)
e^(-kd t)
  in   kd       deoxygenation rate constant [1/s]
  in   ka       reaeration rate constant [1/s]
  in   L0       ultimate BOD of the mixed river water at the outfall [kg/m^3]
  in   D0       oxygen deficit at the outfall (DOsat - DO) [kg/m^3]
  in   t        travel time from the outfall [s]
  in   DOsat    saturation dissolved oxygen [kg/m^3]
  out  D        oxygen deficit at t [kg/m^3]
  out  DO       dissolved oxygen at t [kg/m^3]
  out  D_mgL    deficit in mg/L []
  out  DO_mgL   dissolved oxygen in mg/L []
```

### 2.6 Solve

Inputs as name=value, each with any unit of the right kind. The answer lists every output, the method used, each input converted to SI, and the conditions the answer assumes.

```
$ environmental-engineering solve enveng_streeter_phelps_deficit "kd=0.23 1/day" "ka=0.46
1/day" "L0=10 mg/L" "D0=1 mg/L" "t=2 day" "DOsat=9.1 mg/L"
D = 0.002726165085 kg/m^3
DO = 0.006373834915 kg/m^3
D_mgL = 2.726165085
DO_mgL = 6.373834915
  by: Streeter-Phelps oxygen deficit and dissolved oxygen at a travel time  (D = kd L0/(ka -
kd) (e^(-kd t) - e^(-ka t)) + D0 e^(-ka t); for ka = kd, D = (kd L0 t + D0) e^(-kd t))
  kd = 2.662037037e-06 1/s
  ka = 5.324074074e-06 1/s
  L0 = 0.01 kg/m^3
  D0 = 0.001 kg/m^3
  t = 172800 s
  DOsat = 0.0091 kg/m^3
  assumes: first-order deoxygenation and reaeration; plug flow; deficit stays below saturation
```

### 2.7 Machine-readable answer

--json prints the same answer as one line of JSON (values in SI): for scripts and other programs.

```
$ environmental-engineering solve enveng_streeter_phelps_deficit "kd=0.23 1/day" "ka=0.46
1/day" "L0=10 mg/L" "D0=1 mg/L" "t=2 day" "DOsat=9.1 mg/L" --json
{"outcome":"SOLVED","message":"","value":0.00637383491469,"unit":"kg/m^3","solved_for":"DO","si":
[["kd",2.662037037037037e-6],["ka",5.324074074074074e-6],["L0",0.01],["D0",0.001],
["t",172800.0],["DOsat",0.0091]],"values":[["D",0.00272616508531,"kg/m^3"],
["DO",0.00637383491469,"kg/m^3"],["D_mgL",2.72616508531,""],
["DO_mgL",6.37383491469,""]],"target":"DO","phrase":"oxygen deficit at t","assumptions":
["first-order deoxygenation and reaeration","plug flow","deficit stays below
saturation"],"name":"Streeter-Phelps oxygen deficit and dissolved oxygen at a travel
time","equation":"D = kd L0/(ka - kd) (e^(-kd t) - e^(-ka t)) + D0 e^(-ka t); for ka = kd, D =
(kd L0 t + D0) e^(-kd t)"}
```

### 2.8 When an input is missing

No number is guessed: the answer names what is missing, shows a complete call, and the program exits with status 1.

```
$ environmental-engineering solve enveng_streeter_phelps_deficit "kd=0.23 1/day" "ka=0.46
1/day" "L0=10 mg/L" "D0=1 mg/L" "t=2 day"
NEEDS_INPUT: enveng_streeter_phelps_deficit needs DOsat (saturation dissolved oxygen). Example:
solve('enveng_streeter_phelps_deficit', kd='2.66204e-06 1/s', ka='5.32407e-06 1/s', L0='0.01
kg/m^3', D0='0.001 kg/m^3', t='172800 s', DOsat='0.0091 kg/m^3').
```

### 2.9 Many questions at once

One JSON request per line in, one JSON answer per line out, in order: for scripts, notebooks and agents (the program starts once). In PowerShell: Get-Content cases.jsonl | environmental-engineering batch

`cases.jsonl`:

```
{"key": "enveng_streeter_phelps_deficit", "inputs": {"kd": "0.23 1/day", "ka": "0.46 1/day",
"L0": "10 mg/L", "D0": "1 mg/L", "t": "2 day", "DOsat": "9.1 mg/L"}}
{"key": "enveng_briggs_plume_rise", "inputs": {"vs": "15 m/s", "d": "3 m", "Ts": "420 K", "Ta":
"288 K", "u": "5 m/s", "hs": "100 m"}}
```

```
$ environmental-engineering batch < cases.jsonl
{"outcome":"SOLVED","message":"","value":0.00637383491469,"unit":"kg/m^3","solved_for":"DO","si":
[["kd",2.662037037037037e-6],["ka",5.324074074074074e-6],["L0",0.01],["D0",0.001],
["t",172800.0],["DOsat",0.0091]],"values":[["D",0.00272616508531,"kg/m^3"],
["DO",0.00637383491469,"kg/m^3"],["D_mgL",2.72616508531,""],
["DO_mgL",6.37383491469,""]],"target":"DO","phrase":"oxygen deficit at t","assumptions":
["first-order deoxygenation and reaeration","plug flow","deficit stays below
saturation"],"name":"Streeter-Phelps oxygen deficit and dissolved oxygen at a travel
time","equation":"D = kd L0/(ka - kd) (e^(-kd t) - e^(-ka t)) + D0 e^(-ka t); for ka = kd, D =
(kd L0 t + D0) e^(-kd t)"}
{"outcome":"SOLVED","message":"","value":125.637535501,"unit":"m","solved_for":"dh","si":
[["vs",15.0],["d",3.0],["Ts",420.0],["Ta",288.0],["u",5.0],["hs",100.0]],"values":
[["F",104.0205375,""],["x_f",762.771747653,"m"],["dh",125.637535501,"m"],
["H",225.637535501,"m"]],"target":"dh","phrase":"buoyancy flux, m^4/s^3","assumptions":["Ts >
Ta","neutral or unstable air (classes A-D)","buoyancy-dominated plume"],"name":"Briggs plume
rise and effective stack height (neutral or unstable air)","equation":"F = g vs d^2 (Ts -
Ta)/(4 Ts); x_f = 49 F^(5/8) (F < 55) or 119 F^(2/5); dh = 1.6 F^(1/3) x_f^(2/3)/u"}
```

## 3. Writing inputs

Inputs are written `name=value`, one per argument; `environmental-engineering info <capability>` lists the names. Quote an argument that holds a space or brackets (bash, zsh and PowerShell all accept double quotes). The answer always states each input as it was read, converted to SI, so a misread unit is visible at once.

| kind | example | how it is read |
|---|---|---|
| a number with a unit | `"R=0.128 nm"` | any unit of the same kind; converted to SI |
| a number in SI | `R=1.28e-10` | no unit means the SI unit shown by info |
| a choice | `structure=FCC` | one of the values info lists |
| a list or vector | `"v=[1, 2, 3]" or "v=1, 2, 3"` | brackets keep it a list |

| kind | example | how it is read |
|---|---|---|
| a matrix | `"A=[[1, 2], [3, 4]]"` | one list per row |
| true / false | `inclusive=true` | for yes/no options |

An equation can be solved for any of its quantities: give the others and add `--target <symbol>`. Without `--target`, the program solves for the quantity the capability is named after.

## 4. Outcomes and exit codes

The program never guesses a number: when it cannot answer, it says why and exits with status 1, so a script can tell an answer from a refusal without reading the text.

| outcome | meaning | exit status |
|---|---|---|
| SOLVED | the answer: every output with its unit, the method, the inputs in SI, the conditions assumed | 0 |
| NEEDS_INPUT | an input is missing; the message names it and shows a complete call | 1 |
| INVALID_INPUT | a value or unit of the wrong kind, an unknown input name, or a value outside the conditions | 1 |
| AMBIGUOUS | the inputs allow two different answers; say which one is meant | 1 |
| OUT_OF_SCOPE | the model has no capability for this, or the result would not be meaningful | 1 |
| (usage) | a command the program does not know: it prints its usage | 2 |

## 5. Scripts and agents: batch and JSON

`environmental-engineering solve ... --json` prints one answer as one line of JSON. `environmental-engineering batch` reads one request per line on standard input and writes one JSON answer per line, in order, starting the program once for any number of requests. A request is `{"key": "<capability>", "inputs": {"name": "value", ...}, "target": null}`. In PowerShell, pipe the file in: `Get-Content cases.jsonl | environmental-engineering batch`. `environmental-engineering catalog` prints every capability with its inputs as JSON, to build forms or tool definitions.

| answer field | meaning |
|---|---|
| outcome | one of the outcomes above |
| message | why there is no answer (empty when SOLVED) |
| value, unit | the main result in SI units |
| values | every output: [symbol, value, unit] (procedures) |
| target | the quantity solved for |
| si | each numeric input converted to SI: [symbol, value] |
| assumptions | the conditions the answer relies on |
| name, equation | the capability and its equation or method |

```
import json, subprocess

reqs = [{"key": "<capability>", "inputs": {"name": "value"}}]
out = subprocess.run(["environmental-engineering", "batch"], input="\n".join(map(json.dumps,
reqs)),
                     capture_output=True, text=True, check=True).stdout
answers = [json.loads(line) for line in out.splitlines()]
```

## 6. Figures

`environmental-engineering figures <capability> name=value ... --out DIR` draws the live figures for those inputs and writes each as an SVG file into DIR (an existing folder); without `--out` the SVG text comes back as JSON. The desktop app and the model page draw the same figures as soon as you press Compute.

## 7. In a web page or Node.js: WebAssembly

The same model is a WebAssembly module, `environmental_engineering_ai999x.wasm` (https://www.ai999x.com/compute/rust/environmental_engineering_ai999x.wasm), with a small loader, `ax-wasm.js` (https://www.ai999x.com/compute/rust/ax-wasm.js). It runs in a browser, a Web Worker or Node.js, with no server.

```
<script src="ax-wasm.js"></script>
<script>
  AXWasm.load("environmental_engineering_ai999x.wasm").then(function (m) {
    var a = m.solve({key: "<capability>", inputs: {name: "value"}});
    console.log(a.outcome, a.values || a.value);
    console.log(m.catalog().capabilities.length + " capabilities");
  });
</script>
```

## 8. Every capability

By topic. For each: what it does, what it needs and gives, when it applies, and a command with what the program printed for it. `environmental-engineering info <id>` shows the same inputs and outputs in the terminal.

### Air pollution (25)

#### Atmospheric stability from the environmental lapse rate, `enveng_lapse_rate_stability`

The environmental lapse rate from the air temperature at two heights and the stability it gives against the dry adiabatic lapse rate (9.8 degC/km): superadiabatic (unstable), neutral, subadiabatic (stable) or an inversion.

$$\Gamma_{env} = -\frac{T_2 - T_1}{z_2 - z_1} \text{ vs } \Gamma_d = 9.8\ \text{°C/km}$$

| symbol | | meaning | SI unit |
|---|---|---|---|
| `T1` | in | air temperature at the lower height | K |
| `z1` | in | lower height | m |
| `T2` | in | air temperature at the upper height | K |
| `z2` | in | upper height | m |
| `ELR` | out | environmental lapse rate -dT/dz | K/m |
| `ELR_C_per_km` | out | lapse rate, degC/km | , |
| `stability` | out | stability: superadiabatic (unstable), neutral, subadiabatic (stable) or inversion (very stable) | , |

**Applies when:** dry air (dry adiabatic lapse rate 9.8 degC/km); z2 > z1; linear temperature profile between the two heights.

```
$ environmental-engineering solve enveng_lapse_rate_stability "T1=20 degC" "z1=0 m" "T2=14
degC" "z2=500 m"
ELR = 0.012 K/m
ELR_C_per_km = 12
stability = superadiabatic (unstable)
  by: Atmospheric stability from the environmental lapse rate  (ELR = -(T2 - T1)/(z2 - z1); ELR
> 9.8 degC/km superadiabatic, = 9.8 neutral, 0 to 9.8 subadiabatic, < 0 inversion)
  T1 = 293.15 K
  z1 = 0 m
  T2 = 287.15 K
  z2 = 500 m
  assumes: dry air (dry adiabatic lapse rate 9.8 degC/km); z2 > z1; linear temperature profile
between the two heights
```

In the worked example, T1 = 20 degC, z1 = 0 m, T2 = 14 degC, z2 = 500 m go through the equation above and give ELR = 0.012 K/m, ELR_C_per_km = 12, stability = superadiabatic (unstable). It applies only when dry air (dry adiabatic lapse rate 9.8 degC/km); z2 > z1; linear temperature profile between the two heights: outside that it says what is wrong instead of returning a number.

**Box model of an air basin (well-mixed, time-dependent), `enveng_box_model`**

The concentration over a city treated as a well-mixed box: emission flux per area, wind through a box of length L under a mixing height, background air; the concentration after time t and at steady state.

$$C(t) = C_{ss} + (C_0 - C_{ss})e^{-ut/L}, \quad C_{ss} = C_b + \frac{qL}{uH_m}$$

| symbol | | meaning | SI unit |
|---|---|---|---|
| `q` | in | emission rate per unit ground area | kg/(m^2*s) |
| `L` | in | box length along the wind | m |
| `u` | in | wind speed | m/s |
| `Hm` | in | mixing height | m |
| `Cb` | in | background (upwind) concentration | kg/m^3 |
| `C0` | in | concentration at t = 0 | kg/m^3 |
| `t` | in | elapsed time | s |
| `C` | out | concentration at time t | kg/m^3 |
| `C_ss` | out | steady-state concentration | kg/m^3 |
| `C_ugm3` | out | concentration at t in micrograms per cubic metre | , |

**Applies when:** well-mixed box; constant wind, mixing height and emissions.

```
$ environmental-engineering solve enveng_box_model "q=4 ug/(m^2*s)" "L=15000 m" "u=3 m/s"
"Hm=600 m" "Cb=20 ug/m^3" "C0=20 ug/m^3" "t=2 h"
C = 4.543574138e-08 kg/m^3
C_ss = 5.333333333e-08 kg/m^3
C_ugm3 = 45.43574138
  by: Box model of an air basin (well-mixed, time-dependent)  (dC/dt = q/Hm - u (C - Cb)/L, so
C = Css + (C0 - Css) e^(-u t/L) with Css = Cb + q L/(u Hm))
  q = 4e-09 kg/(m^2*s)
  L = 15000 m
  u = 3 m/s
  Hm = 600 m
  Cb = 2e-08 kg/m^3
  C0 = 2e-08 kg/m^3
  t = 7200 s
  assumes: well-mixed box; constant wind, mixing height and emissions
```

**What the worked example means:** This calculation determines the pollutant concentration in a well-mixed air basin after a specific time. It uses the emission rate per area, box length, wind speed, mixing height, background concentration, initial concentration, and elapsed time. The result shows how the concentration approaches its steady state, with the final value being 4.54357e-08 kg/m^3.

**Briggs dispersion coefficients sigma_y and sigma_z, `enveng_briggs_dispersion_coefficients`**

The horizontal and vertical plume spreads (Pasquill-Gifford curves in Briggs' formulas) at a downwind distance for a stability class, open-country or urban.

$$\sigma_y = \frac{a_y x}{\sqrt{1 + b_y x}}, \quad \sigma_z = a_z x \, (1 + b_z x)^{p_z}$$

| symbol | | meaning | SI unit |
|---|---|---|---|
| `x` | in | downwind distance from the source | m |
| `cls` | in | Pasquill stability class A-F (urban: A and B share A-B, E and F share E-F) | , |
| `terrain` | in | rural (open country) or urban coefficients | , |
| `sigma_y` | out | horizontal (crosswind) dispersion coefficient | m |
| `sigma_z` | out | vertical dispersion coefficient | m |

**Applies when:** 100 m <= x <= 10 km.

```
$ environmental-engineering solve enveng_briggs_dispersion_coefficients "x=1000 m" cls=d
terrain=rural
sigma_y = 76.27700714 m
sigma_z = 37.94733192 m
  by: Briggs dispersion coefficients sigma_y and sigma_z  (Briggs (1973) interpolation formulas
sigma = a x (1 + b x)^p for each class and terrain)
  x = 1000 m
  cls = d
  terrain = rural
  assumes: 100 m <= x <= 10 km
```

**What the worked example means:** This calculation determines the dispersion coefficients for a plume traveling a specific distance. Using Briggs interpolation formulas, the model computes the horizontal crosswind spread as 76.277 m and the vertical spread as 37.9473 m. These values represent the standard deviations of the pollutant

concentration distribution in the crosswind and vertical directions respectively, based on the atmospheric stability class and terrain type provided.

**Briggs plume rise and effective stack height (neutral or unstable air),** `enveng_briggs_plume_rise`

The buoyancy flux, the final plume rise by Briggs' 2/3 law (neutral or unstable air) and the effective stack height.

$$F = \frac{g v_s d^2 (T_s - T_a)}{4 T_s}, \quad \Delta h = \frac{1.6 F^{1/3} x_f^{2/3}}{u}$$

| symbol | | meaning | SI unit |
|---|---|---|---|
| `vs` | in | stack gas exit velocity | m/s |
| `d` | in | stack inside diameter at the exit | m |
| `Ts` | in | stack gas temperature | K |
| `Ta` | in | ambient air temperature | K |
| `u` | in | wind speed at the stack top | m/s |
| `hs` | in | physical stack height | m |
| `F` | out | buoyancy flux, m^4/s^3 | , |
| `x_f` | out | distance to final rise | m |
| `dh` | out | plume rise | m |
| `H` | out | effective stack height hs + dh | m |

**Applies when:** Ts > Ta; neutral or unstable air (classes A-D); buoyancy-dominated plume.

```
$ environmental-engineering solve enveng_briggs_plume_rise "vs=15 m/s" "d=3 m" "Ts=420 K"
"Ta=288 K" "u=5 m/s" "hs=100 m"
F = 104.0205375
x_f = 762.7717477 m
dh = 125.6375355 m
H = 225.6375355 m
  by: Briggs plume rise and effective stack height (neutral or unstable air)  (F = g vs d^2 (Ts
- Ta)/(4 Ts); x_f = 49 F^(5/8) (F < 55) or 119 F^(2/5); dh = 1.6 F^(1/3) x_f^(2/3)/u)
  vs = 15 m/s
  d = 3 m
  Ts = 420 K
  Ta = 288 K
  u = 5 m/s
  hs = 100 m
  assumes: Ts > Ta; neutral or unstable air (classes A-D); buoyancy-dominated plume
```

**What the worked example means:** This calculation determines the effective stack height by combining the physical structure with the buoyancy-driven plume rise. The buoyancy flux, calculated as 104.021, drives the vertical ascent of the smoke column in neutral or unstable atmospheric conditions. Using Briggs' method for final rise distance and subsequent elevation gain, the total plume rise is found to be 125.638 m. Adding this to the physical stack height yields an effective release height of 225.638 m, which represents the altitude at which pollutants are effectively dispersed into the atmosphere.

**Briggs plume rise in stable air,** `enveng_briggs_stable_plume_rise`

The final rise of a buoyant plume in stable air (classes E, F) from the buoyancy flux, wind speed and potential-temperature gradient, and the effective stack height.

$$\Delta h = 2.6\left(\frac{F}{us}\right)^{1/3}, \quad s = \frac{g}{T_a}\frac{\partial \theta}{\partial z}$$

| symbol | | meaning | SI unit |
|---|---|---|---|
| `vs` | in | stack gas exit velocity | m/s |
| `d` | in | stack inside diameter at the exit | m |
| `Ts` | in | stack gas temperature | K |
| `Ta` | in | ambient air temperature | K |
| `u` | in | wind speed at the stack top | m/s |
| `dtheta` | in | potential temperature gradient d(theta)/dz | K/m |
| `hs` | in | physical stack height | m |
| `F` | out | buoyancy flux, m^4/s^3 | , |
| `s` | out | stability parameter g/Ta dtheta/dz, 1/s^2 | , |
| `dh` | out | plume rise | m |
| `H` | out | effective stack height | m |

**Applies when:** Ts > Ta; stable air: dtheta/dz > 0.

```
$ environmental-engineering solve enveng_briggs_stable_plume_rise "vs=15 m/s" "d=3 m" "Ts=420
K" "Ta=288 K" "u=4 m/s" "dtheta=0.02 K/m" "hs=100 m"
F = 104.0205375
s = 0.0006810173611
dh = 87.55352869 m
H = 187.5535287 m
  by: Briggs plume rise in stable air  (s = (g/Ta) dtheta/dz; dh = 2.6 (F/(u s))^(1/3))
  vs = 15 m/s
  d = 3 m
  Ts = 420 K
  Ta = 288 K
  u = 4 m/s
  dtheta = 0.02 K/m
  hs = 100 m
  assumes: Ts > Ta; stable air: dtheta/dz > 0
```

**What the worked example means:** This calculation determines the final vertical rise of a buoyant smoke plume under stable atmospheric conditions. It uses the buoyancy flux derived from the exit velocity 15 m/s, stack diameter 3 m, and temperature difference between the gas 420 K and ambient air 288 K. The stability parameter 0.000681017 reflects how strongly the atmosphere resists vertical motion based on the potential temperature gradient 0.02 K/m. The wind speed 4 m/s at the stack top dilutes the plume's momentum. The computed plume rise 87.5535 m is added to the physical stack height 100 m to find the total effective release height 187.554 m, which dictates how high pollutants are lifted before dispersing.

### Cyclone cut diameter and fractional efficiency (Lapple), `enveng_cyclone_lapple`

The cut diameter (50 % efficiency) of a cyclone from the gas viscosity, inlet width, effective number of turns, inlet velocity and densities, and the collection efficiency of a particle size by Lapple's curve.

$$d_{50} = \sqrt{\frac{9\mu W}{2\pi N_e V_i(\rho_p - \rho_g)}}, \quad \eta = \frac{1}{1 + (d_{50}/d_p)^2}$$

| symbol | | meaning | SI unit |
|---|---|---|---|
| `mu` | in | gas viscosity | Pa*s |
| `W` | in | inlet width | m |
| `Ne` | in | effective number of turns | , |
| `Vi` | in | inlet gas velocity | m/s |
| `rho_p` | in | particle density | kg/m^3 |
| `rho_g` | in | gas density | kg/m^3 |
| `dp` | in | particle diameter | m |
| `d50` | out | cut diameter | m |
| `eta` | out | collection efficiency of particles of diameter dp | , |

**Applies when:** rho_p > rho_g.

```
$ environmental-engineering solve enveng_cyclone_lapple "mu=1.8e-05 Pa*s" "W=0.25 m" Ne=6
"Vi=15 m/s" "rho_p=1500 kg/m^3" "rho_g=1.2 kg/m^3" "dp=10 um"
d50 = 6.912648602e-06 m
eta = 0.6766599842
  by: Cyclone cut diameter and fractional efficiency (Lapple)  (d50 = sqrt(9 mu W/(2 pi Ne Vi
(rho_p - rho_g))); eta = 1/(1 + (d50/dp)^2))
  mu = 1.8e-05 Pa*s
  W = 0.25 m
  Ne = 6
  Vi = 15 m/s
  rho_p = 1500 kg/m^3
  rho_g = 1.2 kg/m^3
  dp = 1e-05 m
  assumes: rho_p > rho_g
```

**What the worked example means:** This calculation determines the cyclone's cut diameter, which is the particle size collected with fifty percent efficiency. Using Lapple's method, the result depends on gas viscosity, inlet width, number of turns, inlet velocity, and density differences. For the specified particle size of 10 um, the predicted fractional efficiency is 0.67666. A smaller cut diameter generally indicates higher separation capability for fine particles.

### Electrostatic precipitator efficiency (Deutsch-Anderson), `enveng_esp_efficiency`

The collection efficiency and penetration of an electrostatic precipitator from the drift velocity, collecting area and gas flow.

$$\eta = 1 - e^{-wA/Q}$$

| symbol | | meaning | SI unit |
|---|---|---|---|
| `w` | in | effective drift (migration) velocity | m/s |
| `A` | in | collecting plate area | m^2 |
| `Q` | in | gas flow rate | m^3/s |

| symbol | | meaning | SI unit |
|---|---|---|---|
| `eta` | out | collection efficiency (fraction) | , |
| `penetration` | out | fraction passing (1 - eta) | , |

**Applies when:** w, A, Q > 0.

```
$ environmental-engineering solve enveng_esp_efficiency "w=0.1 m/s" "A=5000 m^2" "Q=200 m^3/s"
eta = 0.9179150014
penetration = 0.08208499862
  by: Electrostatic precipitator efficiency (Deutsch-Anderson)  (eta = 1 - exp(-w A/Q))
  w = 0.1 m/s
  A = 5000 m^2
  Q = 200 m^3/s
  assumes: w, A, Q > 0
```

**What the worked example means:** This calculation determines the electrostatic precipitator performance using the Deutsch-Anderson equation. It takes the effective drift velocity of particles, the total collecting plate area, and the gas flow rate as inputs. The result shows that with a drift velocity of 0.1 m/s, an area of 5000 m^2, and a flow of 200 m^3/s, the device achieves a collection efficiency of 0.917915, meaning only 0.082085 of the particulate matter passes through uncollected.

**Electrostatic precipitator plate area for a target efficiency, `enveng_esp_area`**

The collecting plate area an electrostatic precipitator needs to reach a stated efficiency (Deutsch-Anderson).

$$A = -\frac{Q}{w}\ln(1-\eta)$$

| symbol | | meaning | SI unit |
|---|---|---|---|
| `w` | in | effective drift velocity | m/s |
| `Q` | in | gas flow rate | m^3/s |
| `eta` | in | required collection efficiency (fraction or %) | , |
| `A` | out | required collecting area | m^2 |

**Applies when:** 0 < eta < 1.

```
$ environmental-engineering solve enveng_esp_area "w=0.1 m/s" "Q=200 m^3/s" eta=0.99
A = 9210.340372 m^2
  by: Electrostatic precipitator plate area for a target efficiency  (A = -Q ln(1 - eta)/w)
  w = 0.1 m/s
  Q = 200 m^3/s
  eta = 0.99
  assumes: 0 < eta < 1
```

**What the worked example means:** This calculation determines the collecting plate area needed for an electrostatic precipitator to achieve a specific efficiency. Using the gas flow rate of 200 m^3/s, the effective drift velocity of 0.1 m/s, and the target collection efficiency of 0.99, the model applies the Deutsch-Anderson equation. The resulting required area is 9210.34 m^2, which represents the physical surface size necessary to capture the desired fraction of particulate matter from the gas stream.

**Gas concentration: mass concentration to ppm (by volume), `enveng_mass_concentration_to_ppm`**

The volume mixing ratio (ppm and ppb) of a gaseous pollutant from its mass concentration, molar mass, temperature and pressure (ideal gas).

$$\mathrm{ppm} = \frac{CRT}{PM} \times 10^{6}$$

| symbol | | meaning | SI unit |
|---|---|---|---|
| `C` | in | mass concentration | kg/m^3 |
| `M` | in | molar mass | kg/mol |
| `T` | in | air temperature | K |
| `P` | in | air pressure | Pa |
| `ppm` | out | volume mixing ratio in ppm | , |
| `ppb` | out | volume mixing ratio in ppb | , |

**Applies when:** ideal gas.

```
$ environmental-engineering solve enveng_mass_concentration_to_ppm "C=100 ug/m^3" "M=46.006
g/mol" "T=298.15 K" "P=101.325 kPa"
ppm = 0.05317872386
ppb = 53.17872386
  by: Gas concentration: mass concentration to ppm (by volume)  (ppm = C R T/(P M) 1e6)
  C = 1e-07 kg/m^3
  M = 0.046006 kg/mol
  T = 298.15 K
  P = 101325 Pa
  assumes: ideal gas
```

**What the worked example means:** This calculation determines the volume mixing ratio of a gaseous pollutant from its mass concentration using the ideal gas law. Given a mass concentration of 100 ug/m^3, a molar mass of 46.006 g/mol, an air temperature of 298.15 K, and an air pressure of 101.325 kPa, the resulting concentration is expressed as 0.0531787 parts per million or 53.1787 parts per billion. This conversion allows engineers to compare emission limits defined in mass units with atmospheric measurements typically reported in volumetric ratios.

**Gas concentration: ppm (by volume) to mass concentration, `enveng_ppm_to_mass_concentration`**

The mass concentration of a gaseous pollutant from its volume mixing ratio in ppm, its molar mass and the air temperature and pressure (ideal gas).

$$C = \mathrm{ppm} \times 10^{-6}\,\frac{PM}{RT}$$

| symbol | | meaning | SI unit |
|---|---|---|---|
| `ppm` | in | volume mixing ratio in parts per million | , |
| `M` | in | molar mass of the pollutant | kg/mol |
| `T` | in | air temperature | K |
| `P` | in | air pressure | Pa |
| `C` | out | mass concentration | kg/m^3 |
| `C_ugm3` | out | in micrograms per cubic metre | , |

| symbol | | meaning | SI unit |
|---|---|---|---|
| `C_mgm3` | out | in milligrams per cubic metre | , |

**Applies when:** ideal gas.

```
$ environmental-engineering solve enveng_ppm_to_mass_concentration ppm=0.14 "M=64.066 g/mol"
"T=298.15 K" "P=101.325 kPa"
C = 3.666091151e-07 kg/m^3
C_ugm3 = 366.6091151
C_mgm3 = 0.3666091151
  by: Gas concentration: ppm (by volume) to mass concentration  (C = ppm 1e-6 P M/(R T))
  ppm = 0.14
  M = 0.064066 kg/mol
  T = 298.15 K
  P = 101325 Pa
  assumes: ideal gas
```

**What the worked example means:** This calculation converts a volume mixing ratio of 0.14 into a mass concentration using the ideal gas law. Given the pollutant's molar mass of 64.066 g/mol, an air temperature of 298.15 K, and a pressure of 101.325 kPa, the resulting mass concentration is 3.66609e-07 kg/m^3. This value represents the actual weight of the pollutant present in each cubic metre of air, expressed as 366.609 or 0.366609.

**Gaussian plume concentration at a receptor (ground reflection), `enveng_gaussian_plume_concentration`**

The steady concentration downwind of a continuous point source at crosswind distance y and height z, given the dispersion coefficients at the receptor's distance and the effective stack height, with full reflection at the ground.

$$C = \frac{Q}{2\pi u \sigma_y \sigma_z} e^{-\frac{y^2}{2\sigma_y^2}} \left[ e^{-\frac{(z-H)^2}{2\sigma_z^2}} + e^{-\frac{(z+H)^2}{2\sigma_z^2}} \right]$$

| symbol | | meaning | SI unit |
|---|---|---|---|
| `Q` | in | pollutant emission rate | kg/s |
| `u` | in | mean wind speed at the stack height | m/s |
| `sigma_y` | in | horizontal dispersion coefficient at the receptor distance | m |
| `sigma_z` | in | vertical dispersion coefficient at the receptor distance | m |
| `H` | in | effective stack height (stack height plus plume rise) | m |
| `y` | in | crosswind distance from the plume centreline | m |
| `z` | in | receptor height above the ground | m |
| `C` | out | concentration at the receptor | kg/m^3 |
| `C_ugm3` | out | concentration in micrograms per cubic metre | , |

**Applies when:** Q, u, sigma_y, sigma_z > 0; H, z >= 0; steady state, flat terrain.

```
$ environmental-engineering solve enveng_gaussian_plume_concentration "Q=100 g/s" "u=4 m/s"
"sigma_y=60 m" "sigma_z=30 m" "H=50 m" "y=20 m" "z=0 m"
C = 1.042805659e-06 kg/m^3
C_ugm3 = 1042.805659
  by: Gaussian plume concentration at a receptor (ground reflection)  (C = Q/(2 pi u sy sz)
exp(-y^2/2sy^2) [exp(-(z-H)^2/2sz^2) + exp(-(z+H)^2/2sz^2)])
  Q = 0.1 kg/s
  u = 4 m/s
  sigma_y = 60 m
  sigma_z = 30 m
  H = 50 m
  y = 20 m
  z = 0 m
  assumes: Q, u, sigma_y, sigma_z > 0; H, z >= 0; steady state, flat terrain
```

**What the worked example means:** This calculation determines the steady-state pollutant concentration downwind of a continuous point source. It uses the emission rate, wind speed, dispersion coefficients, effective stack height, and receptor position to account for plume spread and ground reflection. The resulting concentration is 1.04281e-06 kg/m^3 kilograms per cubic metre, equivalent to 1042.81 micrograms per cubic metre.

**Gaussian plume concentration from the stability class, `enveng_plume_concentration_stability`**

The concentration at a receptor (x, y, z) downwind of an elevated point source, with the dispersion coefficients taken from Briggs' formulas for the stability class and terrain.

$$C = \frac{Q}{2\pi u \sigma_y(x)\sigma_z(x)} e^{-\frac{y^2}{2\sigma_y^2}} \left[ e^{-\frac{(z-H)^2}{2\sigma_z^2}} + e^{-\frac{(z+H)^2}{2\sigma_z^2}} \right]$$

| symbol | | meaning | SI unit |
|---|---|---|---|
| `Q` | in | pollutant emission rate | kg/s |
| `u` | in | mean wind speed at the stack height | m/s |
| `H` | in | effective stack height | m |
| `x` | in | downwind distance | m |
| `y` | in | crosswind distance from the centreline | m |
| `z` | in | receptor height | m |
| `cls` | in | Pasquill stability class A-F | , |
| `terrain` | in | rural or urban dispersion coefficients | , |
| `C` | out | concentration at the receptor | kg/m^3 |
| `C_ugm3` | out | concentration in micrograms per cubic metre | , |
| `sigma_y` | out | horizontal dispersion coefficient at x | m |
| `sigma_z` | out | vertical dispersion coefficient at x | m |

**Applies when:** 100 m <= x <= 10 km; Q, u > 0; H, z >= 0.

```
$ environmental-engineering solve enveng_plume_concentration_stability "Q=200 g/s" "u=5 m/s"
"H=80 m" "x=2000 m" "y=0 m" "z=0 m" cls=c terrain=rural
C = 3.935687085e-07 kg/m^3
C_ugm3 = 393.5687085
sigma_y = 200.8316044 m
sigma_z = 135.2246808 m
  by: Gaussian plume concentration from the stability class  (Briggs sigma_y(x), sigma_z(x) for
the class, then the reflected Gaussian plume)
  Q = 0.2 kg/s
  u = 5 m/s
  H = 80 m
  x = 2000 m
  y = 0 m
  z = 0 m
  cls = c
  terrain = rural
  assumes: 100 m <= x <= 10 km; Q, u > 0; H, z >= 0
```

**What the worked example means:** This calculation determines the pollutant concentration at a specific receptor location downwind of an elevated point source. It uses Briggs' formulas to derive the horizontal and vertical dispersion coefficients based on the stability class and terrain, resulting in values of 200.832 m and 135.225 m. These coefficients are then applied to the Gaussian plume equation along with the emission rate, wind speed, and geometric parameters to yield a final concentration of 3.93569e-07 kg/m^3. This value represents the mass of pollutant per unit volume of air at the specified coordinates.

**Gaussian plume trapped below an inversion lid (multiple reflections), `enveng_plume_mixing_lid`**

The concentration of a plume reflected both at the ground and at the base of an elevated inversion (mixing height), by the image-source series, with the fully mixed limit for comparison.

$$C = \frac{Q e^{-y^2/2\sigma_y^2}}{2\pi u \sigma_y \sigma_z} \sum_{n=-\infty}^{\infty} \left[ e^{-\frac{(z-H+2nL)^2}{2\sigma_z^2}} + e^{-\frac{(z+H+2nL)^2}{2\sigma_z^2}} \right]$$

| symbol | | meaning | SI unit |
|---|---|---|---|
| `Q` | in | emission rate | kg/s |
| `u` | in | mean wind speed | m/s |
| `sigma_y` | in | horizontal dispersion coefficient | m |
| `sigma_z` | in | vertical dispersion coefficient | m |
| `H` | in | effective stack height | m |
| `Lm` | in | mixing height (inversion base) | m |
| `y` | in | crosswind distance | m |
| `z` | in | receptor height | m |
| `C` | out | concentration with ground and lid reflections | kg/m^3 |
| `C_ugm3` | out | in micrograms per cubic metre | , |
| `C_mixed` | out | uniformly mixed limit Q/(sqrt(2 pi) u sy Lm) e^(-y^2/2sy^2) | kg/m^3 |

**Applies when:** H, z <= Lm; sigma_z <= 10 Lm.

```
$ environmental-engineering solve enveng_plume_mixing_lid "Q=100 g/s" "u=4 m/s" "sigma_y=900 m"
"sigma_z=600 m" "H=80 m" "Lm=500 m" "y=0 m" "z=0 m"
C = 2.219531498e-08 kg/m^3
C_ugm3 = 22.19531498
C_mixed = 2.216346002e-08 kg/m^3
  by: Gaussian plume trapped below an inversion lid (multiple reflections)  (image sources at z
= +-H + 2 n Lm summed until the terms vanish)
  Q = 0.1 kg/s
  u = 4 m/s
  sigma_y = 900 m
  sigma_z = 600 m
  H = 80 m
  Lm = 500 m
  y = 0 m
  z = 0 m
  assumes: H, z <= Lm; sigma_z <= 10 Lm
```

In the worked example, Q = 100 g/s, u = 4 m/s, sigma_y = 900 m, sigma_z = 600 m, H = 80 m, Lm = 500 m, y = 0 m, z = 0 m go through the equation above and give C = 2.219531498e-08 kg/m^3, C_ugm3 = 22.19531498, C_mixed = 2.216346002e-08 kg/m^3. It applies only when H, z <= Lm; sigma_z <= 10 Lm: outside that it says what is wrong instead of returning a number.

**Gravity settling chamber collection efficiency, `enveng_settling_chamber_efficiency`**

The Stokes settling velocity of a particle in a gas and the collection efficiency of a horizontal-flow settling chamber with N trays (laminar plug-flow model).

$$\eta = \frac{v_t L W N}{Q}, \quad v_t = \frac{g(\rho_p - \rho_g) d_p^2}{18\mu}$$

| symbol | | meaning | SI unit |
|---|---|---|---|
| `dp` | in | particle diameter | m |
| `rho_p` | in | particle density | kg/m^3 |
| `rho_g` | in | gas density | kg/m^3 |
| `mu` | in | gas viscosity | Pa*s |
| `L` | in | chamber length | m |
| `W` | in | chamber width | m |
| `Q` | in | gas flow rate | m^3/s |
| `N` | in | number of trays (1 for a single chamber) | , |
| `vt` | out | Stokes terminal settling velocity | m/s |
| `eta` | out | collection efficiency (fraction, at most 1) | , |
| `Re` | out | particle Reynolds number | , |

**Applies when:** Stokes regime Re < 1; laminar plug flow.

```
$ environmental-engineering solve enveng_settling_chamber_efficiency "dp=50 um" "rho_p=2000
kg/m^3" "rho_g=1.2 kg/m^3" "mu=1.8e-05 Pa*s" "L=5 m" "W=2 m" "Q=10 m^3/s" N=1
vt = 0.151246389 m/s
eta = 0.151246389
Re = 0.5041546301
  by: Gravity settling chamber collection efficiency  (vt = g (rho_p - rho_g) dp^2/(18 mu); eta
= min(1, vt L W N/Q))
  dp = 5e-05 m
  rho_p = 2000 kg/m^3
  rho_g = 1.2 kg/m^3
  mu = 1.8e-05 Pa*s
  L = 5 m
  W = 2 m
  Q = 10 m^3/s
  assumes: Stokes regime Re < 1; laminar plug flow
```

**What the worked example means:** This calculation determines the particle's terminal settling velocity using Stokes law, which yields 0.151246 m/s, and then computes the fraction of particles captured by a horizontal-flow settling chamber with multiple trays. The resulting collection efficiency is 0.151246, representing the proportion of incoming particulate matter that successfully settles out before exiting the system.

### Ground-level centreline concentration of a plume, `enveng_ground_centerline_concentration`

The ground-level concentration directly below the plume axis (y = 0, z = 0) from the emission rate, wind speed, dispersion coefficients and effective stack height (H = 0 for a ground-level source).

$$C(x,0,0) = \frac{Q}{\pi u \sigma_y \sigma_z} \exp\left(-\frac{H^2}{2\sigma_z^2}\right)$$

| symbol | | meaning | SI unit |
|---|---|---|---|
| `Q` | in | pollutant emission rate | kg/s |
| `u` | in | mean wind speed | m/s |
| `sigma_y` | in | horizontal dispersion coefficient | m |
| `sigma_z` | in | vertical dispersion coefficient | m |
| `H` | in | effective stack height (0 for a ground-level source) | m |
| `C` | out | ground-level centreline concentration | kg/m^3 |
| `C_ugm3` | out | concentration in micrograms per cubic metre | , |

**Applies when:** Q, u, sigma_y, sigma_z > 0; H >= 0.

```
$ environmental-engineering solve enveng_ground_centerline_concentration "Q=150 g/s" "u=5 m/s"
"sigma_y=80 m" "sigma_z=40 m" "H=60 m"
C = 9.688133431e-07 kg/m^3
C_ugm3 = 968.8133431
  by: Ground-level centreline concentration of a plume  (C = Q/(pi u sy sz) exp(-H^2/(2 sz^2)))
  Q = 0.15 kg/s
  u = 5 m/s
  sigma_y = 80 m
  sigma_z = 40 m
  H = 60 m
  assumes: Q, u, sigma_y, sigma_z > 0; H >= 0
```

**What the worked example means:** This calculation determines the pollutant concentration at ground level directly below the plume axis using the emission rate, wind speed, horizontal and vertical dispersion coefficients, and effective stack height. The resulting ground-level centerline concentration is 9.68813e-07 kg/m^3, which converts to 968.813 in micrograms per cubic meter. This value represents the exposure at the specific downwind location defined by the input parameters.

### Holland plume rise, `enveng_holland_plume_rise`

Plume rise by Holland's formula (momentum term plus a buoyancy term with the pressure in millibars) and the effective stack height.

$$\Delta h = \frac{v_s d}{u}\left(1.5 + 2.68 \times 10^{-3}\, p\, d\, \frac{T_s - T_a}{T_s}\right)$$

| symbol | | meaning | SI unit |
|---|---|---|---|
| `vs` | in | stack gas exit velocity | m/s |
| `d` | in | stack inside diameter | m |
| `u` | in | wind speed at the stack top | m/s |
| `p` | in | atmospheric pressure | Pa |
| `Ts` | in | stack gas temperature | K |
| `Ta` | in | ambient air temperature | K |
| `hs` | in | physical stack height | m |
| `dh` | out | plume rise | m |
| `H` | out | effective stack height | m |

**Applies when:** Ts > Ta.

```
$ environmental-engineering solve enveng_holland_plume_rise "vs=10 m/s" "d=2 m" "u=4 m/s"
"p=101.325 kPa" "Ts=400 K" "Ta=290 K" "hs=50 m"
dh = 14.9676525 m
H = 64.9676525 m
  by: Holland plume rise  (dh = vs d/u (1.5 + 2.68e-3 p[mbar] d (Ts - Ta)/Ts))
  vs = 10 m/s
  d = 2 m
  u = 4 m/s
  p = 101325 Pa
  Ts = 400 K
  Ta = 290 K
  hs = 50 m
  assumes: Ts > Ta
```

**What the worked example means:** This calculation determines how far the smoke plume rises above the physical stack using Holland's formula, which combines momentum from the exit velocity and buoyancy driven by temperature differences. The computed plume rise of 14.9677 m represents the vertical distance the hot gases travel upward due to their initial speed and heat compared to the surrounding air. Adding this rise to the physical stack height yields an effective stack height of 64.9677 m, which is the virtual elevation used for predicting downwind pollutant dispersion.

### Maximum ground-level concentration (constant sigma_y/sigma_z), `enveng_max_ground_concentration_ratio`

The largest ground-level centreline concentration of an elevated source when sigma_y/sigma_z is taken constant along the plume, and the vertical spread at which it occurs (sigma_z = H/sqrt 2).

$$C_{max} = \frac{2Q}{\pi e\, u H^2}\frac{\sigma_z}{\sigma_y}, \quad \sigma_z = \frac{H}{\sqrt{2}}$$

| symbol | | meaning | SI unit |
|---|---|---|---|
| `Q` | in | emission rate | kg/s |
| `u` | in | mean wind speed | m/s |
| `H` | in | effective stack height | m |
| `k` | in | ratio sigma_y/sigma_z (constant along the plume) | , |
| `C_max` | out | maximum ground-level concentration | kg/m^3 |
| `C_max_ugm3` | out | maximum concentration in micrograms per cubic metre | , |
| `sigma_z_max` | out | vertical dispersion coefficient at the maximum | m |

**Applies when:** Q, u, H, k > 0; sigma_y/sigma_z constant along the plume.

```
$ environmental-engineering solve enveng_max_ground_concentration_ratio "Q=100 g/s" "u=5 m/s"
"H=100 m" k=2
C_max = 2.341993261e-07 kg/m^3
C_max_ugm3 = 234.1993261
sigma_z_max = 70.71067812 m
  by: Maximum ground-level concentration (constant sigma_y/sigma_z)  (maximise C = Q/(pi u k
sz^2) exp(-H^2/2sz^2) over sz: sz = H/sqrt2, Cmax = 2Q/(pi e u H^2 k))
  Q = 0.1 kg/s
  u = 5 m/s
  H = 100 m
  k = 2
  assumes: Q, u, H, k > 0; sigma_y/sigma_z constant along the plume
```

**What the worked example means:** This calculation determines the highest concentration at ground level for an elevated source, assuming the ratio of horizontal to vertical dispersion remains constant. By optimizing the Gaussian plume equation with respect to the vertical dispersion coefficient, the peak occurs when the vertical spread equals the effective stack height divided by the square root of two. For the given emission rate, wind speed, and stack height, the resulting maximum concentration is 2.34199e-07 kg/m^3, which corresponds to 234.199 in micrograms per cubic metre. This value represents the worst-case exposure scenario along the centerline of the plume under these specific atmospheric conditions.

### Maximum ground-level concentration and its distance (Briggs coefficients), `enveng_max_ground_concentration_briggs`

The distance of the largest ground-level centreline concentration of an elevated source and that concentration, found exactly on Briggs' dispersion curves for the stability class (no constant-ratio shortcut).

$$\frac{d\ln C}{dx} = -\frac{\sigma_y'}{\sigma_y} + \frac{\sigma_z'}{\sigma_z}\left(\frac{H^2}{\sigma_z^2} - 1\right) = 0$$

| symbol | | meaning | SI unit |
|---|---|---|---|
| `Q` | in | emission rate | kg/s |

| symbol | | meaning | SI unit |
|---|---|---|---|
| `u` | in | mean wind speed | m/s |
| `H` | in | effective stack height | m |
| `cls` | in | Pasquill stability class A-F | , |
| `terrain` | in | rural or urban dispersion coefficients | , |
| `x_max` | out | downwind distance of the maximum | m |
| `C_max` | out | maximum ground-level concentration | kg/m^3 |
| `C_max_ugm3` | out | maximum concentration in micrograms per cubic metre | , |
| `sigma_y` | out | sigma_y at the maximum | m |
| `sigma_z` | out | sigma_z at the maximum | m |

**Applies when:** the maximum lies between 100 m and 10 km.

```
$ environmental-engineering solve enveng_max_ground_concentration_briggs "Q=100 g/s" "u=5 m/s"
"H=50 m" cls=c terrain=rural
x_max = 459.3930757 m
C_max = 1.333568186e-06 kg/m^3
C_max_ugm3 = 1333.568186
sigma_y = 49.41102781 m
sigma_z = 35.17120532 m
  by: Maximum ground-level concentration and its distance (Briggs coefficients)  (root of d ln
C/dx = -sy'/sy + (sz'/sz)(H^2/sz^2 - 1) by a log-spaced scan and bisection)
  Q = 0.1 kg/s
  u = 5 m/s
  H = 50 m
  cls = c
  terrain = rural
  assumes: the maximum lies between 100 m and 10 km
```

**What the worked example means:** This calculation determines the downwind location and magnitude of the highest pollutant concentration at ground level for an elevated source. By analyzing Briggs' dispersion coefficients, it finds where the rate of change in concentration equals zero. The result shows that with an effective stack height of 50 m and stability class c, the peak concentration reaches 1333.57 micrograms per cubic metre at 459.393 m metres downwind. This precise method avoids simplified assumptions, ensuring accuracy for regulatory or safety assessments.

### Overall efficiency of collection devices in series, `enveng_collection_efficiency_series`

The overall collection efficiency of air pollution control devices in series (a cyclone before a baghouse or an ESP), 1 - prod(1 - eta_i), the fraction that penetrates, and the outlet concentration for a given inlet concentration.

$$\eta_{overall} = 1 - \prod_{i} (1 - \eta_i)$$

| symbol | | meaning | SI unit |
|---|---|---|---|
| `eta` | in | efficiency of each device as a fraction (0.9 for 90%) | , |
| `C_in` | in | inlet concentration (optional) (optional) | kg/m^3 |
| `eta_total` | out | overall efficiency 1 - prod(1 - eta_i) | , |
| `eta_total_percent` | out | overall efficiency, % | , |

| symbol | | meaning | SI unit |
|---|---|---|---|
| `penetration` | out | fraction passing all devices prod(1 - eta_i) | , |
| `C_out` | out | outlet concentration | kg/m^3 |
| `C_out_mgm3` | out | outlet concentration, mg/m^3 | , |

**Applies when:** devices in series, each efficiency on what reaches it; 0 <= eta_i <= 1 as fractions.

```
$ environmental-engineering solve enveng_collection_efficiency_series "eta=[0.8, 0.99]"
eta_total = 0.998
eta_total_percent = 99.8
penetration = 0.002
  by: Overall efficiency of collection devices in series  (P = prod(1 - eta_i); eta = 1 - P;
C_out = C_in P)
  assumes: devices in series, each efficiency on what reaches it; 0 <= eta_i <= 1 as fractions
```

In the worked example, eta = [0.8, 0.99] go through the equation above and give eta_total = 0.998, eta_total_percent = 99.8, penetration = 0.002. It applies only when devices in series, each efficiency on what reaches it; 0 <= eta_i <= 1 as fractions: outside that it says what is wrong instead of returning a number.

**Pasquill-Turner atmospheric stability class, `enveng_stability_class`**

The Pasquill stability class (A very unstable to F stable) from the surface wind speed and the daytime insolation or night-time cloud cover.

$$\text{class} = T(u,\ \text{insolation or cloud cover})$$

| symbol | | meaning | SI unit |
|---|---|---|---|
| `u` | in | surface wind speed at 10 m | m/s |
| `condition` | in | daytime insolation (strong, moderate, slight), overcast (day or night), night_cloudy (at least 4/8 low cloud) or night_clear (at most 3/8 cloud) | , |
| `stability` | out | Pasquill stability class (A-F; A-B etc. for intermediate cases) | , |

**Applies when:** u > 0; no class is defined at night with u below 2 m/s.

```
$ environmental-engineering solve enveng_stability_class "u=2.5 m/s" condition=strong
stability = A-B
  by: Pasquill-Turner atmospheric stability class  (Turner's table: wind-speed rows < 2, 2-3,
3-5, 5-6, > 6 m/s against insolation or cloud-cover columns; heavy overcast is D)
  u = 2.5 m/s
  condition = strong
  assumes: u > 0; no class is defined at night with u below 2 m/s
```

**What the worked example means:** This calculation determines the atmospheric stability class, represented as A-B, by applying Turner's classification method. The result is derived from the surface wind speed of 2.5 m/s and the specified sky condition or insolation level strong. This class indicates how rapidly pollutants disperse vertically in the atmosphere, with values ranging from very unstable to stable.

**Stack plume: effective height, dispersion and ground-level concentration, `enveng_stack_plume_ground_concentration`**

The whole chain for a buoyant stack in neutral or unstable air: Briggs plume rise to the effective height, Briggs dispersion coefficients at the receptor distance, then the ground-level centreline concentration.

$$C = \frac{Q}{\pi u \sigma_y \sigma_z} \exp\left(-\frac{(h_s + \Delta h)^2}{2\sigma_z^2}\right)$$

| symbol | | meaning | SI unit |
|---|---|---|---|
| `Q` | in | emission rate | kg/s |
| `u` | in | wind speed at the stack top | m/s |
| `vs` | in | stack gas exit velocity | m/s |
| `d` | in | stack inside diameter at the exit | m |
| `Ts` | in | stack gas temperature | K |
| `Ta` | in | ambient air temperature | K |
| `hs` | in | physical stack height | m |
| `x` | in | downwind distance of the receptor | m |
| `cls` | in | Pasquill stability class A-D | , |
| `terrain` | in | rural or urban dispersion coefficients | , |
| `H` | out | effective stack height | m |
| `sigma_y` | out | sigma_y at x | m |
| `sigma_z` | out | sigma_z at x | m |
| `C` | out | ground-level centreline concentration at x | kg/m^3 |
| `C_ugm3` | out | in micrograms per cubic metre | , |

**Applies when:** Ts > Ta; class A-D; 100 m <= x <= 10 km.

```
$ environmental-engineering solve enveng_stack_plume_ground_concentration "Q=100 g/s" "u=6 m/s"
"vs=10 m/s" "d=1.5 m" "Ts=400 K" "Ta=290 K" "hs=50 m" "x=3000 m" cls=c terrain=rural
H = 77.44719963 m
sigma_y = 289.4291464 m
sigma_z = 189.7366596 m
C = 8.888445322e-08 kg/m^3
C_ugm3 = 88.88445322
  by: Stack plume: effective height, dispersion and ground-level concentration  (Briggs final
rise (2/3 law) -> H; Briggs sigma_y, sigma_z at x; C = Q/(pi u sy sz) exp(-H^2/2sz^2))
  Q = 0.1 kg/s
  u = 6 m/s
  vs = 10 m/s
  d = 1.5 m
  Ts = 400 K
  Ta = 290 K
  hs = 50 m
  x = 3000 m
  cls = c
  terrain = rural
  assumes: Ts > Ta; class A-D; 100 m <= x <= 10 km
```

**What the worked example means:** This calculation determines the pollutant concentration at a receptor downwind of a stack. It first computes the effective stack height, which results in 77.4472 m, by adding the plume rise to the physical stack height. Then, it evaluates the horizontal and vertical dispersion coefficients at the

specified distance, yielding 289.429 m for crosswind spread and 189.737 m for vertical spread. Finally, these values are used to find the ground-level centerline concentration, calculated as 8.88845e-08 kg/m^3 or 88.8845. The result represents the mass of pollutant per unit volume of air at the specific location.

**Steady indoor air concentration (well-mixed room box model),** `enveng_indoor_air_steady_state`

The steady concentration of a pollutant in a well-mixed room from its indoor emission rate, the ventilation rate, the outdoor concentration and a first-order decay rate, C = (Qv Camb + S)/(Qv + k V), with the air changes per hour.

$$C_{ss} = \frac{Q_v C_{amb} + S}{Q_v + kV}$$

| symbol | | meaning | SI unit |
|---|---|---|---|
| `S` | in | indoor emission rate | kg/s |
| `Qv` | in | ventilation (outdoor air) rate | m^3/s |
| `V` | in | room volume | m^3 |
| `Camb` | in | outdoor (ambient) concentration (default 0) (optional) | kg/m^3 |
| `k` | in | first-order decay (reaction, deposition) rate (default 0) (optional) | 1/s |
| `C` | out | steady indoor concentration | kg/m^3 |
| `C_ugm3` | out | in ug/m^3 | , |
| `C_mgm3` | out | in mg/m^3 | , |
| `ach` | out | air changes per hour Qv/V | , |
| `tau_h` | out | time constant V/(Qv + k V), h | , |

**Applies when:** well-mixed room at steady state; Camb = 0 and k = 0 unless given; something enters the room.

```
$ environmental-engineering solve enveng_indoor_air_steady_state "S=100 mg/h" "Qv=100 m^3/h"
"V=200 m^3" "Camb=10 ug/m^3" "k=0.5 1/h"
C = 5.05e-07 kg/m^3
C_ugm3 = 505
C_mgm3 = 0.505
ach = 0.5
tau_h = 1
  by: Steady indoor air concentration (well-mixed room box model)  (C = (Qv Camb + S)/(Qv + k
V); ach = Qv/V)
  S = 2.777777778e-08 kg/s
  Qv = 0.02777777778 m^3/s
  V = 200 m^3
  Camb = 1e-08 kg/m^3
  k = 0.0001388888889 1/s
  assumes: well-mixed room at steady state; Camb = 0 and k = 0 unless given; something enters
the room
```

In the worked example, S = 100 mg/h, Qv = 100 m^3/h, V = 200 m^3, Camb = 10 ug/m^3, k = 0.5 1/h go through the equation above and give C = 5.05e-07 kg/m^3, C_ugm3 = 505, C_mgm3 = 0.505, ach = 0.5. It applies only when well-mixed room at steady state; Camb = 0 and k = 0 unless given; something enters the room: outside that it says what is wrong instead of returning a number.

**Stoichiometric oxygen for complete combustion of a fuel,** `enveng_combustion_oxygen`

The oxygen needed to burn a fuel CxHyOz (or pure carbon) completely to CO2 and water, per mole and per kilogram of fuel, the CO2 and water formed per kilogram, and the total oxygen for a given mass of fuel.

Double subscripts: use braces to clarify

| symbol | | meaning | SI unit |
|---|---|---|---|
| `formula` | in | fuel formula with C, H, O only (e.g. CH4, C2H5OH, C) | , |
| `m` | in | mass of fuel burned (for the total oxygen) (optional) | kg |
| `O2_per_kg` | out | kg O2 per kg fuel | , |
| `o2_per_mol` | out | mol O2 per mol fuel x + y/4 - z/2 | , |
| `molar_mass` | out | molar mass of the fuel, g/mol | , |
| `CO2_per_kg` | out | kg CO2 formed per kg fuel | , |
| `H2O_per_kg` | out | kg water formed per kg fuel | , |
| `O2_mass` | out | oxygen for the given mass of fuel | kg |

**Applies when:** complete combustion to CO2 and H2O; elements C, H, O only.

```
$ environmental-engineering solve enveng_combustion_oxygen formula=C "m=1000 kg"
o2_per_mol = 1
molar_mass = 12.0107
O2_per_kg = 2.664191096
CO2_per_kg = 3.664191096
H2O_per_kg = 0
O2_mass = 2664.191096 kg
  by: Stoichiometric oxygen for complete combustion of a fuel  (CxHyOz + (x + y/4 - z/2) O2 ->
x CO2 + y/2 H2O; O2/fuel = (x + y/4 - z/2) M_O2 / M_fuel)
  m = 1000 kg
  formula = C
  assumes: complete combustion to CO2 and H2O; elements C, H, O only
```

In the worked example, formula = C, m = 1000 kg go through the equation above and give o2_per_mol = 1, molar_mass = 12.0107, O2_per_kg = 2.664191096, CO2_per_kg = 3.664191096. It applies only when complete combustion to CO2 and H2O; elements C, H, O only: outside that it says what is wrong instead of returning a number.

### Sulfur dioxide emission from fuel sulfur, `enveng_so2_emission`

The SO2 emission rate from a fuel burning rate and its sulfur mass fraction (all sulfur to SO2), before and after a control device.

$$E = \dot{m}_{fuel}\, S\, \frac{M_{SO_2}}{M_S}(1-\eta)$$

| symbol | | meaning | SI unit |
|---|---|---|---|
| `m_fuel` | in | fuel burning rate | kg/s |
| `S` | in | sulfur mass fraction of the fuel (fraction or %) | , |
| `eta` | in | SO2 control (removal) efficiency (fraction or %; 0 if uncontrolled) | , |
| `E` | out | controlled SO2 emission rate | kg/s |

| symbol | | meaning | SI unit |
|---|---|---|---|
| `E_uncontrolled` | out | uncontrolled SO2 emission rate | kg/s |
| `E_kg_per_day` | out | controlled emission in kg per day | , |

**Applies when:** all fuel sulfur oxidised to SO2.

```
$ environmental-engineering solve enveng_so2_emission "m_fuel=36 t/h" S=0.02 eta=0.9
E = 0.03995870887 kg/s
E_uncontrolled = 0.3995870887 kg/s
E_kg_per_day = 3452.432447
  by: Sulfur dioxide emission from fuel sulfur  (E = m_fuel S (M_SO2/M_S)(1 - eta))
  m_fuel = 10 kg/s
  S = 0.02
  eta = 0.9
  assumes: all fuel sulfur oxidised to SO2
```

**What the worked example means:** This calculation determines the mass flow rate of sulfur dioxide emitted into the atmosphere. It starts with the uncontrolled emission potential, which depends directly on how much fuel is burned and how much sulfur that fuel contains. The result assumes all sulfur converts to sulfur dioxide. A control efficiency factor then reduces this amount to produce the final controlled emission rate of 0.0399587 kg/s. If no control device is present, the efficiency is zero, and the output equals the uncontrolled rate.

### Wind speed at height (power law), `enveng_wind_power_law`

The wind speed at another height from a measured speed with the power-law profile u2 = u1 (z2/z1)^p (p from the stability class and terrain, e.g. 0.15 rural neutral).

$$u_2 = u_1 \left( \frac{z_2}{z_1} \right)^{p}$$

| symbol | | meaning | SI unit |
|---|---|---|---|
| `u1` | in | measured wind speed | m/s |
| `z1` | in | measurement height | m |
| `z2` | in | height of interest (e.g. the stack top) | m |
| `p` | in | power-law exponent (0 < p < 1) | , |
| `u2` | out | wind speed at z2 | m/s |

**Applies when:** 0 < p < 1.

```
$ environmental-engineering solve enveng_wind_power_law "u1=4 m/s" "z1=10 m" "z2=100 m" p=0.15
u2 = 5.650150178 m/s
  by: Wind speed at height (power law)  (u2 = u1 (z2/z1)^p)
  u1 = 4 m/s
  z1 = 10 m
  z2 = 100 m
  p = 0.15
  assumes: 0 < p < 1
```

**What the worked example means:** This calculation determines the wind speed at a specific height using the power law profile. It takes a known wind speed measurement, 4 m/s, taken at height 10 m, and adjusts it to the height of interest, 100 m, using an exponent 0.15 that reflects terrain and atmospheric stability. The resulting wind

speed at the new height is 5.65015 m/s. This method assumes a neutral or stable boundary layer where wind speed changes predictably with altitude according to the chosen exponent.

## Biosolids (1)

### Volatile solids reduction in digestion (Van Kleeck), `enveng_volatile_solids_reduction`

The fraction of the volatile solids destroyed by sludge digestion from the volatile fractions of the feed and the digested sludge by the Van Kleeck equation, VSR = (VS_in - VS_out)/(VS_in - VS_in VS_out), which assumes the fixed solids pass unchanged.

$$VSR = \frac{VS_{in} - VS_{out}}{VS_{in} - VS_{in}\,VS_{out}}$$

| symbol | | meaning | SI unit |
|---|---|---|---|
| `VS_in` | in | volatile fraction of the feed solids (fraction or %) | , |
| `VS_out` | in | volatile fraction of the digested solids (fraction or %) | , |
| `VSR` | out | volatile solids reduction (fraction) | , |
| `VSR_percent` | out | volatile solids reduction, % | , |
| `TS_remaining` | out | total solids left per unit of feed solids | , |

**Applies when:** fixed (inorganic) solids conserved; 0 < VS_out <= VS_in, VS_out < 1.

```
$ environmental-engineering solve enveng_volatile_solids_reduction VS_in=0.75 VS_out=0.55
VSR = 0.5925925926
VSR_percent = 59.25925926
TS_remaining = 0.5555555556
  by: Volatile solids reduction in digestion (Van Kleeck)  (VSR = (VS_in - VS_out)/(VS_in -
VS_in VS_out))
  VS_in = 0.75
  VS_out = 0.55
  assumes: fixed (inorganic) solids conserved; 0 < VS_out <= VS_in, VS_out < 1
```

In the worked example, VS_in = 0.75, VS_out = 0.55 go through the equation above and give VSR = 0.5925925926, VSR_percent = 59.25925926, TS_remaining = 0.5555555556. It applies only when fixed (inorganic) solids conserved; 0 < VS_out <= VS_in, VS_out < 1: outside that it says what is wrong instead of returning a number.

## Fundamentals (3)

### Population projection by the arithmetic and geometric methods, `enveng_population_projection`

The population projected to a year from two censuses by the arithmetic method (constant increase per year) and by the geometric method (constant percentage growth: the continuous rate and the equivalent compound annual rate).

$$P_t = P_2 + \frac{P_2 - P_1}{t_2 - t_1}(t - t_2), \qquad P_t = P_2\, e^{k_g (t - t_2)},\ k_g = \frac{\ln(P_2/P_1)}{t_2 - t_1}$$

| symbol | | meaning | SI unit |
|---|---|---|---|
| `t1` | in | year of the first census | , |
| `P1` | in | population at the first census | , |
| `t2` | in | year of the second census | , |
| `P2` | in | population at the second census | , |
| `t` | in | year of the projection | , |
| `k_a` | out | arithmetic increase, persons per year | , |
| `P_arith` | out | arithmetic projection | , |
| `k_g` | out | geometric (continuous) growth rate, per year | , |
| `r_g` | out | equivalent compound growth rate per year | , |
| `P_geom` | out | geometric projection | , |

**Applies when:** t2 > t1; positive census populations; interval and projection span at most 200 years; the arithmetic projection must stay positive.

```
$ environmental-engineering solve enveng_population_projection t1=2000 P1=50000 t2=2020
P2=70000 t=2040
k_a = 1000
P_arith = 90000
k_g = 0.01682361183
r_g = 0.01696592575
P_geom = 98000
  by: Population projection by the arithmetic and geometric methods  (k_a = (P2 - P1)/(t2 -
t1), P = P2 + k_a (t - t2); k_g = ln(P2/P1)/(t2 - t1), P = P2 exp(k_g (t - t2)))
  t1 = 2000
  P1 = 50000
  t2 = 2020
  P2 = 70000
  t = 2040
  assumes: t2 > t1; positive census populations; interval and projection span at most 200
years; the arithmetic projection must stay positive
```

In the worked example, t1 = 2000, P1 = 50000, t2 = 2020, P2 = 70000, t = 2040 go through the equation above and give k_a = 1000, P_arith = 90000, k_g = 0.01682361183, r_g = 0.01696592575. It applies only when t2 > t1; positive census populations; interval and projection span at most 200 years; the arithmetic projection must stay positive: outside that it says what is wrong instead of returning a number.

**Population projection by the logistic method (three censuses), `enveng_population_logistic`**

The saturation population and the logistic curve P = Ps/(1 + m exp(-k (t - t0))) through three equally spaced censuses P0, P1, P2 (interval n years), and the population it projects for a year.

$$P_s = \frac{2P_0P_1P_2 - P_1^2(P_0 + P_2)}{P_0P_2 - P_1^2}, \quad P_t = \frac{P_s}{1 + m\,e^{-k(t - t_0)}}$$

| symbol | | meaning | SI unit |
|---|---|---|---|
| `t0` | in | year of the first census | , |
| `n` | in | interval between the censuses in years | , |
| `P0` | in | population at the first census | , |

| symbol | | meaning | SI unit |
|---|---|---|---|
| `P1` | in | population one interval later | , |
| `P2` | in | population two intervals later | , |
| `t` | in | year of the projection | , |
| `P_sat` | out | saturation population | , |
| `m` | out | logistic constant (Ps - P0)/P0 | , |
| `k` | out | logistic rate constant, per year | , |
| `P_t` | out | projected population | , |

**Applies when:** three equally spaced censuses with P0 < P1 < P2; growth slowing: P1^2 > P0 P2; saturation population at most 100 times the last census.

```
$ environmental-engineering solve enveng_population_logistic t0=2000 n=10 P0=40000 P1=60000
P2=75000 t=2030
P_sat = 90000
m = 1.25
k = 0.09162907319
P_t = 83333.33333
  by: Population projection by the logistic method (three censuses)  (Ps = (2 P0 P1 P2 - P1^2
(P0 + P2))/(P0 P2 - P1^2); m = (Ps - P0)/P0; k = ln(P1 (Ps - P0)/(P0 (Ps - P1)))/n)
  t0 = 2000
  n = 10
  P0 = 40000
  P1 = 60000
  P2 = 75000
  t = 2030
  assumes: three equally spaced censuses with P0 < P1 < P2; growth slowing: P1^2 > P0 P2;
saturation population at most 100 times the last census
```

In the worked example, t0 = 2000, n = 10, P0 = 40000, P1 = 60000, P2 = 75000, t = 2030 go through the equation above and give P_sat = 90000, m = 1.25, k = 0.09162907319, P_t = 83333.33333. It applies only when three equally spaced censuses with P0 < P1 < P2; growth slowing: P1^2 > P0 P2; saturation population at most 100 times the last census: outside that it says what is wrong instead of returning a number.

### Water and wastewater demand with peaking factors, `enveng_water_demand`

The average-day, maximum-day and peak-hour water demand of a population from the per-capita demand and the peaking factors (both relative to the average day) and, with the fraction of water returned as wastewater, the average wastewater flow and its peak by Harmon's factor 1 + 14/(4 + sqrt(P/1000)).

$$Q_{avg} = P\,q, \quad Q_{md} = f_{md} Q_{avg}, \quad Q_{ph} = f_{ph} Q_{avg}, \quad PF = 1 + \frac{14}{4 + \sqrt{P/1000}}$$

| symbol | | meaning | SI unit |
|---|---|---|---|
| `pop` | in | population served | , |
| `q` | in | average per-capita demand in L per person per day | , |
| `f_md` | in | maximum-day factor (maximum day / average day) | , |
| `f_ph` | in | peak-hour factor (peak hour / average day) | , |
| `r_ww` | in | fraction of the water returned as wastewater (optional) (optional) | , |

| symbol | | meaning | SI unit |
|---|---|---|---|
| `Q_avg` | out | average-day demand | m^3/s |
| `Q_avg_m3_per_day` | out | average-day demand, m^3/d | , |
| `Q_maxday` | out | maximum-day demand | m^3/s |
| `Q_maxday_m3_per_day` | out | maximum-day demand, m^3/d | , |
| `Q_peakhour` | out | peak-hour demand | m^3/s |
| `Q_peakhour_m3_per_day` | out | peak-hour demand as a daily rate, m^3/d | , |
| `Q_peakhour_L_per_s` | out | peak-hour demand, L/s | , |
| `Q_ww_avg_m3_per_day` | out | average wastewater flow, m^3/d | , |
| `harmon_peak_factor` | out | Harmon peaking factor | , |
| `Q_ww_peak_m3_per_day` | out | peak wastewater flow, m^3/d | , |

**Applies when:** pop a whole number >= 1; 20 <= q <= 2000 L per person per day; 1 <= f_md <= 4; f_md <= f_ph <= 8 (both relative to the average day); 0 < r_ww <= 1.

```
$ environmental-engineering solve enveng_water_demand pop=50000 q=300 f_md=1.8 f_ph=3
Q_avg = 0.1736111111 m^3/s
Q_avg_m3_per_day = 15000
Q_maxday = 0.3125 m^3/s
Q_maxday_m3_per_day = 27000
Q_peakhour = 0.5208333333 m^3/s
Q_peakhour_m3_per_day = 45000
Q_peakhour_L_per_s = 520.8333333
  by: Water and wastewater demand with peaking factors  (Q_avg = P q; Q_md = f_md Q_avg; Q_ph =
f_ph Q_avg; Q_ww = r Q_avg; PF = 1 + 14/(4 + sqrt(P/1000)))
  q = 300
  f_md = 1.8
  f_ph = 3
  assumes: pop a whole number >= 1; 20 <= q <= 2000 L per person per day; 1 <= f_md <= 4; f_md
<= f_ph <= 8 (both relative to the average day); 0 < r_ww <= 1
```

In the worked example, pop = 50000, q = 300, f_md = 1.8, f_ph = 3 go through the equation above and give Q_avg = 0.1736111111 m^3/s, Q_avg_m3_per_day = 15000, Q_maxday = 0.3125 m^3/s, Q_maxday_m3_per_day = 27000. It applies only when pop a whole number >= 1; 20 <= q <= 2000 L per person per day; 1 <= f_md <= 4; f_md <= f_ph <= 8 (both relative to the average day); 0 < r_ww <= 1: outside that it says what is wrong instead of returning a number.

## Gas transfer (6)

### Dissolved concentration in equilibrium with a gas (Henry's law), `enveng_henry_law_equilibrium`

The equilibrium molar (and, with the molar mass, mass) concentration of a gas dissolved in water from its partial pressure and the Henry's volatility constant.

$$C = \frac{p}{H}$$

| symbol | | meaning | SI unit |
|---|---|---|---|
| `p` | in | partial pressure of the gas | Pa |
| `H` | in | Henry's volatility constant | Pa*m^3/mol |

| symbol | | meaning | SI unit |
|---|---|---|---|
| `M` | in | molar mass (for the mass concentration) (optional) | kg/mol |
| `C` | out | dissolved molar concentration | mol/m^3 |
| `C_mass` | out | dissolved mass concentration | kg/m^3 |
| `C_mgL` | out | in mg/L | , |

**Applies when:** dilute solution, equilibrium.

```
$ environmental-engineering solve enveng_henry_law_equilibrium "p=0.21 atm" "H=0.79
atm*m^3/mol" "M=31.9988 g/mol"
C = 0.2658227848 mol/m^3
C_mass = 0.008506010127 kg/m^3
C_mgL = 8.506010127
  by: Dissolved concentration in equilibrium with a gas (Henry's law)  (C = p/H)
  p = 21278.25 Pa
  H = 80046.75 Pa*m^3/mol
  M = 0.0319988 kg/mol
  assumes: dilute solution, equilibrium
```

**What the worked example means:** This calculation determines the equilibrium concentration of a gas in water using Henry's law. By dividing the partial pressure of the gas, which is 0.21 atm, by the Henry volatility constant, which is 0.79 atm*m^3/mol, this gives the molar concentration 0.265823 mol/m^3. To express this as a mass concentration, the molar value is multiplied by the molar mass 31.9988 g/mol, resulting in 0.00850601 kg/m^3. This represents how much of the gas dissolves into the liquid phase under these specific conditions.

**Dissolved gas concentration from Henry's solubility constant (C = KH P), `enveng_henry_law_solubility`**

The equilibrium concentration of a gas dissolved in water from its partial pressure and Henry's solubility constant KH in mol/(L atm), as mol/L and, with the gas molar mass, as mg/L.

$$C = K_H\,P, \qquad C_m = K_H\,P\,M$$

| symbol | | meaning | SI unit |
|---|---|---|---|
| `KH` | in | Henry's solubility constant | mol/(m^3*Pa) |
| `P` | in | partial pressure of the gas above the water | Pa |
| `M` | in | molar mass of the gas (for mg/L) (optional) | kg/mol |
| `C` | out | dissolved concentration KH P | mol/m^3 |
| `C_molL` | out | in mol/L | , |
| `C_mass` | out | dissolved mass concentration | kg/m^3 |
| `C_mgL` | out | in mg/L | , |

**Applies when:** KH, P > 0; dilute solution at equilibrium, KH at the water temperature.

```
$ environmental-engineering solve enveng_henry_law_solubility "KH=0.0013 mol/(L*atm)" "P=0.21
atm" "M=32 g/mol"
C = 0.273 mol/m^3
C_molL = 0.000273
C_mass = 0.008736 kg/m^3
C_mgL = 8.736
  by: Dissolved gas concentration from Henry's solubility constant (C = KH P)  (C = KH P; mass
concentration C M)
  KH = 1.283000247e-05 mol/(m^3*Pa)
  P = 21278.25 Pa
  M = 0.032 kg/mol
  assumes: KH, P > 0; dilute solution at equilibrium, KH at the water temperature
```

In the worked example, KH = 0.0013 mol/(L*atm), P = 0.21 atm, M = 32 g/mol go through the equation above and give C = 0.273 mol/m^3, C_molL = 0.000273, C_mass = 0.008736 kg/m^3, C_mgL = 8.736. It applies only when KH, P > 0; dilute solution at equilibrium, KH at the water temperature: outside that it says what is wrong instead of returning a number.

**Henry's law constant conversion between forms, `enveng_henry_constant_conversion`**

A Henry's law constant converted between its common forms: volatility (Pa m^3/mol, atm m^3/mol), dimensionless gas/liquid ratio at temperature T, mole-fraction form (atm) and solubility forms (mol/(m^3 Pa), mol/(L atm)).

$$H_{cc} = \frac{H_{pc}}{RT}, \quad H_{px} = H_{pc}\, c_w, \quad H_{cp} = \frac{1}{H_{pc}}$$

| symbol | | meaning | SI unit |
|---|---|---|---|
| `H` | in | the Henry's constant value in the stated form | , |
| `form` | in | form of H: pa_m3_per_mol, atm_m3_per_mol, dimensionless (Cg/Cl), atm (p = H x), mol_per_m3_pa or m_per_atm (mol/(L atm)) | , |
| `T` | in | temperature (for the dimensionless form) | K |
| `H_pa_m3_per_mol` | out | volatility constant, Pa m^3/mol | , |
| `H_atm_m3_per_mol` | out | volatility constant, atm m^3/mol | , |
| `H_dimensionless` | out | dimensionless gas/liquid ratio at T | , |
| `H_atm` | out | mole-fraction form, atm | , |
| `H_mol_per_m3_pa` | out | solubility constant, mol/(m^3 Pa) | , |
| `H_m_per_atm` | out | solubility constant, mol/(L atm) | , |

**Applies when:** dilute aqueous solution (water 55344.59 mol/m^3).

```
$ environmental-engineering solve enveng_henry_constant_conversion H=0.0055 form=atm_m3_per_mol
"T=298.15 K"
H_pa_m3_per_mol = 557.2875
H_atm_m3_per_mol = 0.0055
H_dimensionless = 0.2248072449
H_atm = 304.395245
H_mol_per_m3_pa = 0.001794405939
H_m_per_atm = 0.1818181818
  by: Henry's law constant conversion between forms  (all forms through the volatility constant
Hpc: Hcc = Hpc/(R T), Hpx = Hpc rho_w, Hcp = 1/Hpc)
  H = 0.0055
  T = 298.15 K
  form = atm_m3_per_mol
  assumes: dilute aqueous solution (water 55344.59 mol/m^3)
```

**What the worked example means:** This calculation converts the input Henry's law constant 0.0055 into all standard forms. The primary volatility constant is determined as 557.288, which allows derivation of the dimensionless gas-to-liquid concentration ratio 0.224807 at temperature 298.15 K. Additionally, the mole-fraction form 304.395 and solubility constants 0.00179441 and 0.181818 are computed. These values describe how the chemical partitions between air and water under dilute aqueous conditions.

**Overall liquid-phase transfer coefficient and flux (two-film theory), `enveng_two_film_transfer`**

The overall liquid-side mass-transfer coefficient from the liquid- and gas-film coefficients and the dimensionless Henry's constant, the share of resistance in the liquid film, and the flux from water to air (negative when the gas dissolves).

$$\frac{1}{K_L} = \frac{1}{k_l} + \frac{1}{H k_g}, \quad N = K_L \left( C - \frac{C_g}{H} \right)$$

| symbol | | meaning | SI unit |
|---|---|---|---|
| `kl` | in | liquid-film coefficient | m/s |
| `kg` | in | gas-film coefficient | m/s |
| `H` | in | dimensionless Henry's constant (Cg/Cl) | , |
| `C` | in | bulk liquid concentration | kg/m^3 |
| `Cg` | in | bulk gas concentration | kg/m^3 |
| `KL` | out | overall liquid-phase coefficient | m/s |
| `flux` | out | flux from liquid to gas | kg/(m^2*s) |
| `liquid_resistance_fraction` | out | fraction of the resistance in the liquid film | , |

**Applies when:** linear equilibrium.

```
$ environmental-engineering solve enveng_two_film_transfer "kl=2e-05 m/s" "kg=0.005 m/s" H=0.2
"C=1 mg/L" "Cg=0 ug/m^3"
KL = 1.960784314e-05 m/s
flux = 1.960784314e-08 kg/(m^2*s)
liquid_resistance_fraction = 0.9803921569
  by: Overall liquid-phase transfer coefficient and flux (two-film theory)  (1/KL = 1/kl + 1/(H
kg); flux = KL (C - Cg/H))
  kl = 2e-05 m/s
  kg = 0.005 m/s
  H = 0.2
  C = 0.001 kg/m^3
  Cg = 0 kg/m^3
  assumes: linear equilibrium
```

**What the worked example means:** This calculation determines the overall liquid-phase mass-transfer coefficient, denoted as KL, by combining the individual liquid-film coefficient kl and gas-film coefficient kg with the dimensionless Henry's constant H. The resulting coefficient KL equals 1.96078e-05 m/s. Using this value, the net flux from the liquid to the gas phase is computed as 1.96078e-08 kg/(m^2*s), which represents the rate of mass transfer per unit area. Additionally, the analysis reveals that the liquid film accounts for 0.980392 of the total resistance to mass transfer, indicating how much of the barrier to diffusion occurs on the water side versus the air side.

**Packed-tower air stripping: stripping factor, transfer units and height, `enveng_air_stripping_tower`**

The stripping factor, number of transfer units, packed height and minimum air-to-water ratio of a countercurrent packed tower stripping a volatile compound with clean air.

$$NTU = \frac{R}{R-1}\ln\left[\frac{(C_{in}/C_{out})(R-1)+1}{R}\right], \quad Z = HTU \cdot NTU$$

| symbol | | meaning | SI unit |
|---|---|---|---|
| `Qw` | in | water flow | m^3/s |
| `Qa` | in | air flow | m^3/s |
| `H` | in | dimensionless Henry's constant (Cg/Cl) | , |
| `Cin` | in | influent concentration | kg/m^3 |
| `Cout` | in | required effluent concentration | kg/m^3 |
| `HTU` | in | height of a transfer unit | m |
| `R` | out | stripping factor H Qa/Qw | , |
| `NTU` | out | number of transfer units | , |
| `Z` | out | packed height HTU NTU | m |
| `air_to_water_min` | out | minimum air-to-water ratio (Cin - Cout)/(H Cin) | , |

**Applies when:** Qa/Qw above the minimum; clean inlet air; dilute, linear equilibrium.

```
$ environmental-engineering solve enveng_air_stripping_tower "Qw=0.05 m^3/s" "Qa=1.5 m^3/s"
H=0.2 "Cin=200 ug/L" "Cout=5 ug/L" "HTU=1 m"
R = 6
NTU = 4.213854527
Z = 4.213854527 m
air_to_water_min = 4.875
  by: Packed-tower air stripping: stripping factor, transfer units and height  (R = H Qa/Qw;
NTU = R/(R - 1) ln[((Cin/Cout)(R - 1) + 1)/R] (Cin/Cout - 1 when R = 1); Z = HTU NTU)
  Qw = 0.05 m^3/s
  Qa = 1.5 m^3/s
  H = 0.2
  Cin = 0.0002 kg/m^3
  Cout = 5e-06 kg/m^3
  HTU = 1 m
  assumes: Qa/Qw above the minimum; clean inlet air; dilute, linear equilibrium
```

**What the worked example means:** This calculation determines the required packed tower height, 4.21385 m, based on the stripping factor, 6, and the number of transfer units, 4.21385. The stripping factor compares the air-to-water ratio against equilibrium constraints, while the number of transfer units reflects the difficulty of removing the contaminant from influent concentration 200 ug/L to effluent concentration 5 ug/L. Multiplying the height of a transfer unit, 1 m, by the number of transfer units yields the total physical height needed for effective treatment.

**Temperature dependence of Henry's constant (van 't Hoff), `enveng_henry_temperature_dependence`**

The Henry's solubility constant at a temperature T from its value at a reference temperature and the van 't Hoff coefficient B = -d ln Hcp / d(1/T) (the form tabulated in solubility compilations).

$$H^{cp}(T) = H^{cp}(T_{ref}) \exp\left[B\left(\frac{1}{T} - \frac{1}{T_{ref}}\right)\right]$$

| symbol | | meaning | SI unit |
|---|---|---|---|
| `Hcp` | in | Henry's solubility constant at Tref | mol/(m^3*Pa) |
| `B` | in | temperature coefficient d ln Hcp/d(1/T), in K | , |
| `T` | in | temperature of interest | K |
| `Tref` | in | reference temperature (usually 298.15 K) | K |
| `Hcp_T` | out | solubility constant at T | mol/(m^3*Pa) |
| `Hpc_T` | out | volatility constant 1/Hcp at T | Pa*m^3/mol |

**Applies when:** constant enthalpy of solution over the range.

```
$ environmental-engineering solve enveng_henry_temperature_dependence "Hcp=1.3e-05
mol/(m^3*Pa)" B=1500 "T=283.15 K" "Tref=298.15 K"
Hcp_T = 1.697039196e-05 mol/(m^3*Pa)
Hpc_T = 58926.15813 Pa*m^3/mol
  by: Temperature dependence of Henry's constant (van 't Hoff)  (Hcp(T) = Hcp(Tref) exp(B (1/T
- 1/Tref)))
  Hcp = 1.3e-05 mol/(m^3*Pa)
  B = 1500
  T = 283.15 K
  Tref = 298.15 K
  assumes: constant enthalpy of solution over the range
```

**What the worked example means:** This calculation determines the temperature-dependent Henry's law constant using the van 't Hoff equation. Starting from the reference solubility constant 1.3e-05 mol/(m^3*Pa) at 298.15 K, the model applies the exponential factor derived from the coefficient 1500 and the difference in inverse temperatures between 283.15 K and 298.15 K. The resulting solubility constant 1.69704e-05 mol/(m^3*Pa) indicates how readily the substance dissolves in water at 283.15 K, while its reciprocal, the volatility constant 58926.2 Pa*m^3/mol, reflects the tendency of the substance to partition into the air phase.

## Groundwater (2)

### Darcy's law: groundwater flow and seepage velocity, `enveng_darcy_seepage_velocity`

Groundwater flow through an aquifer by Darcy's law: the Darcy flux q = K i, the flow Q = K i A through an area and the seepage (average linear) velocity v_s = K i / n_e; the gradient may be given as a head difference over a length.

$$Q = K\,i\,A, \qquad v_s = \frac{K\,i}{n_e}, \qquad i = \frac{\Delta h}{L}$$

| symbol | | meaning | SI unit |
|---|---|---|---|
| `K` | in | hydraulic conductivity | m/s |
| `i` | in | hydraulic gradient dh/L (dimensionless) (optional) | , |
| `dh` | in | head difference along the flow path (optional) | m |
| `L` | in | length of the flow path (optional) | m |
| `A` | in | cross-sectional area of flow (optional) | m^2 |
| `n_e` | in | effective porosity (fraction or %) (optional) | , |
| `i` | out | hydraulic gradient | , |
| `q` | out | Darcy flux (specific discharge) K i | m/s |
| `q_m_per_day` | out | Darcy flux, m/d | , |
| `Q` | out | flow K i A | m^3/s |
| `Q_m3_per_day` | out | flow, m^3/d | , |
| `v_s` | out | seepage (average linear) velocity K i / n_e | m/s |
| `v_s_m_per_day` | out | seepage velocity, m/d | , |

**Applies when:** laminar flow (Darcy's law); K, i > 0; 0 < n_e < 1.

```
$ environmental-engineering solve enveng_darcy_seepage_velocity "K=10 m/day" i=0.002 "A=500
m^2" n_e=0.25
i = 0.002
q = 2.314814815e-07 m/s
q_m_per_day = 0.02
Q = 0.0001157407407 m^3/s
Q_m3_per_day = 10
v_s = 9.259259259e-07 m/s
v_s_m_per_day = 0.08
  by: Darcy's law: groundwater flow and seepage velocity  (i = dh/L; q = K i; Q = q A; v_s =
q/n_e)
  K = 0.0001157407407 m/s
  i = 0.002
  A = 500 m^2
  n_e = 0.25
  assumes: laminar flow (Darcy's law); K, i > 0; 0 < n_e < 1
```

In the worked example, K = 10 m/day, i = 0.002, A = 500 m^2, n_e = 0.25 go through the equation above and give i = 0.002, q = 2.314814815e-07 m/s, q_m_per_day = 0.02, Q = 0.0001157407407 m^3/s. It applies only when laminar flow (Darcy's law); K, i > 0; 0 < n_e < 1: outside that it says what is wrong instead of returning a number.

**Retardation factor and contaminant velocity in groundwater, `enveng_retardation_factor`**

The retardation factor of a sorbing contaminant in an aquifer, R = 1 + rho_b Kd / n (Kd given or Koc foc), and the velocity at which the contaminant migrates, v_c = v / R (R may also be given directly).

$$R = 1 + \frac{\rho_b K_d}{n}, \quad K_d = K_{oc} f_{oc}, \quad v_c = \frac{v}{R}$$

| symbol | | meaning | SI unit |
|---|---|---|---|
| `rho_b` | in | dry bulk density of the aquifer material (optional) | kg/m^3 |
| `n` | in | porosity (fraction or %) (optional) | , |
| `Kd` | in | distribution coefficient (optional) | m^3/kg |
| `Koc` | in | organic-carbon partition coefficient (optional) | m^3/kg |
| `foc` | in | fraction of organic carbon (fraction or %) (optional) | , |
| `R` | in | retardation factor, if already known (optional) | , |
| `v` | in | groundwater (seepage) velocity (optional) | m/s |
| `R` | out | retardation factor | , |
| `vc` | out | contaminant migration velocity v/R | m/s |
| `vc_m_per_day` | out | contaminant velocity, m/d | , |
| `v_m_per_day` | out | groundwater velocity, m/d | , |
| `Kd` | out | distribution coefficient Koc foc | m^3/kg |
| `Kd_L_per_kg` | out | Kd in L/kg | , |

**Applies when:** linear, reversible, instantaneous sorption; 0 < n < 1; R >= 1.

```
$ environmental-engineering solve enveng_retardation_factor "rho_b=1.6 g/cm^3" n=0.3 "Koc=100
L/kg" foc=0.01 "v=0.5 m/day"
Kd = 0.001 m^3/kg
Kd_L_per_kg = 1
R = 6.333333333
vc = 9.137426901e-07 m/s
vc_m_per_day = 0.07894736842
v_m_per_day = 0.5
  by: Retardation factor and contaminant velocity in groundwater  (Kd = Koc foc; R = 1 + rho_b
Kd / n; v_c = v / R)
  rho_b = 1600 kg/m^3
  n = 0.3
  Koc = 0.1 m^3/kg
  foc = 0.01
  v = 5.787037037e-06 m/s
  assumes: linear, reversible, instantaneous sorption; 0 < n < 1; R >= 1
```

In the worked example, rho_b = 1.6 g/cm^3, n = 0.3, Koc = 100 L/kg, foc = 0.01, v = 0.5 m/day go through the equation above and give Kd = 0.001 m^3/kg, Kd_L_per_kg = 1, R = 6.333333333, vc = 9.137426901e-07 m/s. It applies only when linear, reversible, instantaneous sorption; 0 < n < 1; R >= 1: outside that it says what is wrong instead of returning a number.

## Hydraulics (3)

### Hazen-Williams equation for a full pipe (SI): velocity, flow and head loss, `enveng_hazen_williams`

Water flow in a full circular pipe by the Hazen-Williams formula in SI units, v = 0.849 C R^0.63 S^0.54 (R = D/4): the velocity and flow from the slope of the energy line, or the slope from the flow, and the head loss over a pipe length.

$$v = 0.849\,C\,R^{0.63}S^{0.54}, \qquad Q = vA, \qquad h_f = S\,L$$

| symbol | | meaning | SI unit |
|---|---|---|---|
| `C` | in | Hazen-Williams roughness coefficient | , |
| `D` | in | pipe diameter | m |
| `S` | in | slope of the energy line hf/L (fraction or %) (optional) | , |
| `Q` | in | flow rate (optional) | m^3/s |
| `L` | in | pipe length (for the head loss) (optional) | m |
| `v` | out | mean velocity | m/s |
| `Q` | out | flow | m^3/s |
| `S` | out | slope of the energy line hf/L | , |
| `hf` | out | head loss over L | m |
| `R` | out | hydraulic radius D/4 | m |
| `A` | out | pipe area | m^2 |

**Applies when:** water, turbulent flow, full circular pipe; SI form (0.849); exactly one of S, Q; S < 1.

```
$ environmental-engineering solve enveng_hazen_williams C=100 "D=0.3 m" S=0.002
v = 0.5790985584 m/s
Q = 0.04093406498 m^3/s
S = 0.002
R = 0.075 m
A = 0.07068583471 m^2
  by: Hazen-Williams equation for a full pipe (SI): velocity, flow and head loss  (R = D/4; v =
0.849 C R^0.63 S^0.54; Q = v A; from Q: S = (v/(0.849 C R^0.63))^(1/0.54); hf = S L)
  C = 100
  D = 0.3 m
  S = 0.002
  assumes: water, turbulent flow, full circular pipe; SI form (0.849); exactly one of S, Q; S <
1
```

In the worked example, C = 100, D = 0.3 m, S = 0.002 go through the equation above and give v = 0.5790985584 m/s, Q = 0.04093406498 m^3/s, S = 0.002, R = 0.075 m. It applies only when water, turbulent flow, full circular pipe; SI form (0.849); exactly one of S, Q; S < 1: outside that it says what is wrong instead of returning a number.

**Manning's equation: velocity and flow in a channel or a full pipe (SI), `enveng_manning_flow`**

The mean velocity and flow by Manning's equation in SI units, v = (1/n) R^(2/3) S^(1/2), for a rectangular channel (width and flow depth) or a circular pipe flowing full (diameter), with the hydraulic radius, area and wetted perimeter.

$$v = \frac{1}{n} R^{2/3} S^{1/2}, \qquad Q = vA, \qquad R = \frac{A}{P}$$

| symbol | | meaning | SI unit |
|---|---|---|---|
| `n` | in | Manning roughness coefficient | , |
| `S` | in | slope of the channel or energy line (fraction or %) | , |
| `W` | in | channel width (rectangular channel) (optional) | m |
| `y` | in | flow depth (rectangular channel) (optional) | m |
| `D` | in | pipe diameter (circular pipe flowing full) (optional) | m |
| `v` | out | mean velocity (1/n) R^(2/3) S^(1/2) | m/s |
| `Q` | out | flow v A | m^3/s |
| `Q_L_per_s` | out | flow, L/s | , |
| `R` | out | hydraulic radius A/P | m |
| `A` | out | flow area | m^2 |
| `P` | out | wetted perimeter | m |

**Applies when:** uniform flow, SI units (factor 1); 0 < S < 1; rectangular channel (W, y) or a full circular pipe (D).

```
$ environmental-engineering solve enveng_manning_flow n=0.013 S=0.001 "D=0.6 m"
v = 0.6867270488 m/s
Q = 0.1941674986 m^3/s
Q_L_per_s = 194.1674986
R = 0.15 m
A = 0.2827433388 m^2
P = 1.884955592 m
  by: Manning's equation: velocity and flow in a channel or a full pipe (SI)  (R = A/P
(rectangle: A = W y, P = W + 2y; full pipe: R = D/4); v = R^(2/3) S^(1/2)/n; Q = v A)
  n = 0.013
  S = 0.001
  D = 0.6 m
  assumes: uniform flow, SI units (factor 1); 0 < S < 1; rectangular channel (W, y) or a full
circular pipe (D)
```

In the worked example, n = 0.013, S = 0.001, D = 0.6 m go through the equation above and give v = 0.6867270488 m/s, Q = 0.1941674986 m^3/s, Q_L_per_s = 194.1674986, R = 0.15 m. It applies only when uniform flow, SI units (factor 1); 0 < S < 1; rectangular channel (W, y) or a full circular pipe (D): outside that it says what is wrong instead of returning a number.

**Pump power from flow and head, with energy and electricity cost, `enveng_pump_power`**

The power a pump draws to lift a flow Q through a total head H with efficiency eta, P = rho g Q H / eta (water 1000 kg/m^3 unless a density is given), in kW, and the energy and electricity cost over a number of operating hours at a price per kWh.

$$P = \frac{\rho g Q H}{\eta}, \qquad E = P t, \qquad \text{cost} = E \cdot \text{price}$$

| symbol | | meaning | SI unit |
|---|---|---|---|
| `Q` | in | flow pumped | m^3/s |
| `H` | in | total dynamic head | m |
| `eta` | in | pump (wire-to-water) efficiency (fraction or %) | , |
| `rho` | in | density of the liquid (default water, 1000 kg/m^3) (optional) | kg/m^3 |
| `hours` | in | operating time (optional) | s |
| `price` | in | electricity price per kWh (optional) | , |
| `P` | out | power drawn to lift the water: the input (brake, shaft or motor) power the pump requires, rho g Q H / eta | W |
| `P_kW` | out | power drawn, kW | , |
| `P_water` | out | water (hydraulic) power, the useful output power imparted to the water, rho g Q H | W |
| `P_water_kW` | out | water power, kW | , |
| `energy_kWh` | out | energy over the operating time, kWh | , |
| `cost` | out | electricity cost, in the price's currency | , |

**Applies when:** g = 9.80665 m/s^2; 0 < eta <= 1; rho = 1000 kg/m^3 unless given; a price needs the operating hours.

```
$ environmental-engineering solve enveng_pump_power "Q=0.1 m^3/s" "H=30 m" eta=0.75
P = 39226.6 W
P_kW = 39.2266
P_water = 29419.95 W
P_water_kW = 29.41995
  by: Pump power from flow and head, with energy and electricity cost  (P = rho g Q H / eta; E
= P t; cost = E x price)
  Q = 0.1 m^3/s
  H = 30 m
  eta = 0.75
  assumes: g = 9.80665 m/s^2; 0 < eta <= 1; rho = 1000 kg/m^3 unless given; a price needs the
operating hours
```

In the worked example, Q = 0.1 m^3/s, H = 30 m, eta = 0.75 go through the equation above and give P = 39226.6 W, P_kW = 39.2266, P_water = 29419.95 W, P_water_kW = 29.41995. It applies only when g = 9.80665 m/s^2; 0 < eta <= 1; rho = 1000 kg/m^3 unless given; a price needs the operating hours: outside that it says what is wrong instead of returning a number.

## Hydrology (2)

### Rational method: peak runoff Q = C i A, `enveng_rational_method`

The peak storm runoff from a small catchment by the rational method, Q = C i A, from the runoff coefficient, the rainfall intensity for the time of concentration (mm/h) and the drainage area (m^2, km^2 or ha), in m^3/s.

$$Q = C\,i\,A$$

| symbol | | meaning | SI unit |
|---|---|---|---|
| `C` | in | runoff coefficient (fraction or %) | , |
| `i` | in | rainfall intensity | m/s |
| `A` | in | drainage area | m^2 |
| `Q` | out | peak runoff C i A | m^3/s |
| `Q_L_per_s` | out | peak runoff, L/s | , |
| `Q_m3_per_h` | out | peak runoff, m^3/h | , |

**Applies when:** 0 < C <= 1; i for a duration equal to the time of concentration; small catchment.

```
$ environmental-engineering solve enveng_rational_method C=0.6 "i=50 mm/h" "A=0.25 km^2"
Q = 2.083333333 m^3/s
Q_L_per_s = 2083.333333
Q_m3_per_h = 7500
  by: Rational method: peak runoff Q = C i A  (Q = C i A (SI); with i in mm/h and A in ha, Q =
C i A / 360 m^3/s)
  C = 0.6
  i = 1.388888889e-05 m/s
  A = 250000 m^2
  assumes: 0 < C <= 1; i for a duration equal to the time of concentration; small catchment
```

In the worked example, C = 0.6, i = 50 mm/h, A = 0.25 km^2 go through the equation above and give Q = 2.083333333 m^3/s, Q_L_per_s = 2083.333333, Q_m3_per_h = 7500. It applies only when 0 < C <= 1; i for a duration equal to the time of concentration; small catchment: outside that it says what is wrong instead of returning a number.

#### Return period, exceedance probability and risk over a design life, `enveng_return_period_risk`

The annual exceedance probability of an event with return period T, p = 1/T, and the probability that it is equalled or exceeded at least once in n years, 1 - (1 - 1/T)^n; or the return period needed for an acceptable risk over a design life.

$$p = \frac{1}{T}, \qquad P_n = 1 - \left(1 - \frac{1}{T}\right)^n$$

| symbol | | meaning | SI unit |
|---|---|---|---|
| `T` | in | return period, years (optional) | , |
| `n` | in | number of years (design life) (optional) | , |
| `risk` | in | acceptable probability of at least one exceedance in n years (fraction or %) (optional) | , |
| `p` | out | annual exceedance probability 1/T | , |
| `p_percent` | out | annual exceedance probability, % | , |
| `T` | out | return period, years | , |
| `P_n` | out | probability of at least one exceedance in n years | , |
| `P_n_percent` | out | that probability, % | , |
| `P_none` | out | probability of no exceedance in n years | , |

**Applies when:** independent years; T >= 1 yr; n a whole number of years; T, or a risk with n.

```
$ environmental-engineering solve enveng_return_period_risk T=100 n=30
p = 0.01
p_percent = 1
T = 100
P_n = 0.2602996266
P_n_percent = 26.02996266
P_none = 0.7397003734
  by: Return period, exceedance probability and risk over a design life  (p = 1/T; P(at least
one in n years) = 1 - (1 - p)^n; for a risk R: p = 1 - (1 - R)^(1/n), T = 1/p)
  T = 100
  n = 30
  assumes: independent years; T >= 1 yr; n a whole number of years; T, or a risk with n
```

In the worked example, T = 100, n = 30 go through the equation above and give p = 0.01, p_percent = 1, T = 100, P_n = 0.2602996266. It applies only when independent years; T >= 1 yr; n a whole number of years; T, or a risk with n: outside that it says what is wrong instead of returning a number.

### Noise (7)

#### Adding sound levels in decibels, `enveng_decibel_addition`

The combined sound pressure level of several incoherent sources, added on an energy basis.

$$L = 10 \log_{10} \sum_i 10^{L_i/10}$$

| symbol | | meaning | SI unit |
|---|---|---|---|
| `L` | in | sound levels in dB | , |
| `L_total` | out | combined level, dB | , |

**Applies when:** incoherent sources.

```
$ environmental-engineering solve enveng_decibel_addition "L=[85, 85]"
L_total = 88.01029996
  by: Adding sound levels in decibels  (L = 10 log10(sum 10^(Li/10)))
  assumes: incoherent sources
```

**What the worked example means:** This calculation determines the total sound pressure level from multiple incoherent sources by converting each individual decibel value into its linear energy equivalent, summing those energies, and then converting the total back to a logarithmic decibel scale. The result, 88.0103 dB, represents the perceived loudness when all sources operate simultaneously, which is higher than any single source but follows logarithmic addition rules rather than simple arithmetic summation.

### Day-night (Ldn) and day-evening-night (Lden) sound levels, `enveng_day_night_level`

The day-night average level (10 dB night penalty, 22:00-07:00) or the day-evening-night level (5 dB evening 19:00-23:00, 10 dB night 23:00-07:00) from 24 hourly equivalent levels.

$$L_{dn} = 10 \log_{10} \frac{1}{24} \left( \sum_{day} 10^{L_h/10} + \sum_{night} 10^{(L_h+10)/10} \right)$$

| symbol | | meaning | SI unit |
|---|---|---|---|
| `L` | in | 24 hourly Leq values in dB, the first for 00:00-01:00 | , |
| `metric` | in | ldn or lden | , |
| `L` | out | Ldn or Lden, dB | , |

**Applies when:** exactly 24 hourly levels.

```
$ environmental-engineering solve enveng_day_night_level "L=[50, 50, 50, 50, 50, 50, 50, 65,
65, 65, 65, 65, 65, 65, 65, 65, 65, 65, 65, 60, 60, 60, 55, 55]" metric=ldn
L = 63.54363682
  by: Day-night (Ldn) and day-evening-night (Lden) sound levels  (L = 10 log10((1/24) sum
10^((L_h + penalty_h)/10)))
  metric = ldn
  assumes: exactly 24 hourly levels
```

**What the worked example means:** This calculation determines the day-night or day-evening-night sound level using the provided twenty-four hourly equivalent levels. The method applies specific penalties to evening and night hours before averaging the energy values logarithmically. The resulting 63.5436 indicates the overall noise exposure adjusted for these time-of-day sensitivities, offering a single metric that reflects human annoyance potential better than a simple average.

### Equivalent continuous sound level Leq, `enveng_equivalent_sound_level`

The equivalent continuous sound level of a period made of intervals at different levels (energy average weighted by duration).

$$L_{eq} = 10 \log_{10} \frac{\sum_i t_i 10^{L_i/10}}{\sum_i t_i}$$

| symbol | | meaning | SI unit |
|---|---|---|---|
| `L` | in | levels of the intervals in dB | , |
| `t` | in | durations of the intervals (any one unit) | , |
| `Leq` | out | equivalent continuous level, dB | , |

**Applies when:** one duration per level.

```
$ environmental-engineering solve enveng_equivalent_sound_level "L=[70, 85, 60]" "t=[4, 1, 3]"
Leq = 76.52279911
  by: Equivalent continuous sound level Leq  (Leq = 10 log10(sum t_i 10^(L_i/10)/sum t_i))
  assumes: one duration per level
```

**What the worked example means:** This calculation determines the equivalent continuous sound level, which represents a single steady noise level containing the same acoustic energy as the varying noise environment over the measurement period. The result, 76.5228 dB, is derived by weighting each interval's sound pressure level, given as [70, 85, 60] dB, by its corresponding duration, provided as [4, 1, 3]. This method effectively averages the logarithmic decibel values on an energy basis rather than a simple arithmetic mean, reflecting how human hearing perceives cumulative noise exposure.

**Occupational noise dose and 8-hour TWA (OSHA, 5 dB exchange), `enveng_occupational_noise_dose`**

The daily noise dose (per cent of the permissible exposure) and the 8-hour time-weighted average from levels and exposure times, with the 90 dBA criterion, 5 dB exchange rate and 80 dBA threshold.

$$D = 100 \sum_i \frac{C_i}{T_i},\ T_i = \frac{8}{2^{(L_i - 90)/5}},\ TWA = 16.61 \log_{10} \frac{D}{100} + 90$$

| symbol | | meaning | SI unit |
|---|---|---|---|
| `L` | in | A-weighted levels in dBA | , |
| `hours` | in | exposure time at each level in hours | , |
| `dose_percent` | out | noise dose, per cent | , |
| `TWA` | out | 8-hour time-weighted average, dBA | , |

**Applies when:** levels below 80 dBA excluded.

```
$ environmental-engineering solve enveng_occupational_noise_dose "L=[90, 95, 85]" "hours=[4, 2,
2]"
dose_percent = 112.5
TWA = 90.8496434
  by: Occupational noise dose and 8-hour TWA (OSHA, 5 dB exchange)  (T = 8/2^((L - 90)/5); D =
100 sum C/T; TWA = 16.61 log10(D/100) + 90)
  assumes: levels below 80 dBA excluded
```

In the worked example, L = [90, 95, 85], hours = [4, 2, 2] go through the equation above and give dose_percent = 112.5, TWA = 90.8496434. It applies only when levels below 80 dBA excluded: outside that it says what is wrong instead of returning a number.

### Sound level at another distance (point or line source), `enveng_noise_distance_attenuation`

The sound level at a second distance from a point source (6 dB per doubling) or a line source such as a busy road (3 dB per doubling), free field.

$$L_2 = L_1 - 20 \log_{10} \frac{r_2}{r_1} \ \text{(point)}$$

| symbol | | meaning | SI unit |
|---|---|---|---|
| `L1` | in | level at distance r1, dB | , |
| `r1` | in | first distance | m |
| `r2` | in | second distance | m |
| `source` | in | point or line source | , |
| `L2` | out | level at distance r2, dB | , |

**Applies when:** free field, no absorption.

```
$ environmental-engineering solve enveng_noise_distance_attenuation L1=90 "r1=10 m" "r2=80 m"
source=point
L2 = 71.93820026
  by: Sound level at another distance (point or line source)  (L2 = L1 - 20 log10(r2/r1)
(point), L1 - 10 log10(r2/r1) (line))
  L1 = 90
  r1 = 10 m
  r2 = 80 m
  source = point
  assumes: free field, no absorption
```

**What the worked example means:** This calculation determines the sound pressure level at a specific distance from a noise source, assuming free-field conditions with no atmospheric absorption. Using the provided initial level of 90 measured at 10 m, and the target distance of 80 m, the model applies geometric divergence rules for either a point or line source as indicated by point. The resulting level at the new location is 71.9382. This output represents the expected acoustic intensity after accounting for the change in distance, illustrating how sound energy spreads out as it travels away from the origin.

### Sound pressure level from sound power level, `enveng_sound_power_to_pressure`

The sound pressure level at a distance from a source of given sound power level and directivity factor (1 free field, 2 on a hard floor, 4 at a wall-floor edge, 8 in a corner).

$$L_p = L_W + 10 \log_{10} \frac{Q}{4 \pi r^2}$$

| symbol | | meaning | SI unit |
|---|---|---|---|
| `Lw` | in | sound power level, dB re 1 pW | , |
| `r` | in | distance from the source | m |
| `Qd` | in | directivity factor (1, 2, 4 or 8) | , |
| `Lp` | out | sound pressure level, dB | , |

**Applies when:** far field, no absorption; rho c taken as 400 rayl (Lp = LI).

```
$ environmental-engineering solve enveng_sound_power_to_pressure Lw=110 "r=20 m" Qd=2
Lp = 75.9976014
  by: Sound pressure level from sound power level  (Lp = Lw + 10 log10(Q/(4 pi r^2)))
  Lw = 110
  r = 20 m
  Qd = 2
  assumes: far field, no absorption; rho c taken as 400 rayl (Lp = LI)
```

**What the worked example means:** This calculation determines the sound pressure level, denoted as Lp, based on the source's sound power level Lw, the directivity factor Qd, and the distance r from the source. The result, 75.9976, represents the acoustic intensity perceived at that specific location. This value accounts for how the sound energy spreads out in space and whether the environment reflects sound, such as near a hard floor or wall, which affects the overall loudness experienced by a listener.

#### Source level after removing background noise, `enveng_background_noise_correction`

The sound level of a source alone from the total measured level and the background level (energy subtraction).

$$L_s = 10 \log_{10}\left(10^{L_t/10} - 10^{L_b/10}\right)$$

| symbol | | meaning | SI unit |
|---|---|---|---|
| `Lt` | in | total measured level, dB | , |
| `Lb` | in | background level, dB | , |
| `L_source` | out | level of the source alone, dB | , |

**Applies when:** Lt > Lb.

```
$ environmental-engineering solve enveng_background_noise_correction Lt=85 Lb=80
L_source = 83.34911461
  by: Source level after removing background noise  (Ls = 10 log10(10^(Lt/10) - 10^(Lb/10)))
  Lt = 85
  Lb = 80
  assumes: Lt > Lb
```

**What the worked example means:** This calculation determines the sound pressure level of a specific source in isolation. It uses the total measured level, 85, and the background level, 80. Because decibels are logarithmic units representing power ratios, simple subtraction of the numbers does not work. Instead, the method converts both levels back to linear energy values, subtracts the background energy from the total energy, and then converts the remaining energy back into a decibel scale. The resulting value, 83.3491, represents how loud the source would be if the background noise were completely absent.

## Reactors (2)

#### First-order decay in a batch reactor: C = C0 exp(-k t) and the half-life, `enveng_first_order_decay`

First-order decay in a batch (or plug-flow) reactor, C = C0 exp(-k t): the concentration after a time, the initial concentration, the rate constant or the time, whichever one of the four is not given, with the half-life t1/2 = ln 2 / k (the rate may be given as a half-life instead of k), the fraction remaining and the per cent removed.

$$C = C_0\, e^{-kt}, \qquad k = \frac{\ln(C_0/C)}{t}, \qquad t_{1/2} = \frac{\ln 2}{k}$$

| symbol | | meaning | SI unit |
|---|---|---|---|
| `C0` | in | initial concentration (optional) | kg/m^3 |
| `C` | in | concentration after the time t (optional) | kg/m^3 |
| `k` | in | first-order rate constant (base e) (optional) | 1/s |
| `t` | in | reaction time (optional) | s |
| `t_half` | in | half-life (in place of k) (optional) | s |
| `C` | out | concentration after the time t, C0 exp(-k t) | kg/m^3 |
| `C_mgL` | out | in mg/L | , |
| `C0` | out | initial concentration | kg/m^3 |
| `C0_mgL` | out | initial concentration, mg/L | , |
| `k` | out | first-order rate constant ln(C0/C)/t | 1/s |
| `k_per_day` | out | rate constant, 1/d | , |
| `k_per_h` | out | rate constant, 1/h | , |
| `t` | out | reaction time ln(C0/C)/k | s |
| `t_days` | out | reaction time, days | , |
| `t_h` | out | reaction time, hours | , |
| `t_half` | out | half-life ln 2/k | s |
| `t_half_days` | out | half-life, days | , |
| `t_half_h` | out | half-life, hours | , |
| `fraction_remaining` | out | fraction remaining C/C0 | , |
| `removal_percent` | out | per cent removed (1 - C/C0) x 100 | , |

**Applies when:** first-order kinetics, constant k (batch reactor or plug flow); exactly three of C0, C, k (or t_half), t; 0 < C < C0 when both are given; k t <= 700.

```
$ environmental-engineering solve enveng_first_order_decay "C0=100 mg/L" "k=0.2 1/day" "t=120
h"
C = 0.03678794412 kg/m^3
C_mgL = 36.78794412
C0 = 0.1 kg/m^3
C0_mgL = 100
k = 2.314814815e-06 1/s
k_per_day = 0.2
k_per_h = 0.008333333333
t = 432000 s
t_days = 5
t_h = 120
t_half = 299439.582 s
t_half_days = 3.465735903
t_half_h = 83.17766167
fraction_remaining = 0.3678794412
removal_percent = 63.21205588
  by: First-order decay in a batch reactor: C = C0 exp(-k t) and the half-life  (C = C0 e^(-k
t); C0 = C e^(k t); k = ln(C0/C)/t; t = ln(C0/C)/k; t1/2 = ln 2/k (k = ln 2/t1/2 when the half-
life is given))
  C0 = 0.1 kg/m^3
  k = 2.314814815e-06 1/s
  t = 432000 s
  assumes: first-order kinetics, constant k (batch reactor or plug flow); exactly three of C0,
C, k (or t_half), t; 0 < C < C0 when both are given; k t <= 700
```

In the worked example, C0 = 100 mg/L, k = 0.2 1/day, t = 120 h go through the equation above and give C = 0.03678794412 kg/m^3, C_mgL = 36.78794412, C0 = 0.1 kg/m^3, C0_mgL = 100. It applies only when first-order kinetics, constant k (batch reactor or plug flow); exactly three of C0, C, k (or t_half), t; 0 < C < C0 when both are given; k t <= 700: outside that it says what is wrong instead of returning a number.

**First-order reaction in N stirred tanks in series and in plug flow, `enveng_cmfr_series`**

The effluent concentration of a first-order reaction through N identical completely mixed flow reactors (CMFRs) in series sharing a total residence time, Cout = Cin/(1 + k theta/N)^N, with a single CMFR at steady state (N = 1, the default: Cout = Cin/(1 + k theta)) and the plug-flow limit Cin exp(-k theta) for comparison; theta given directly or as the volume over the flow, V/Q; with the fraction and per cent removed.

$$C_{out} = \frac{C_{in}}{(1 + k\theta/N)^N}, \qquad \theta = \frac{V}{Q}, \qquad C_{PFR} = C_{in}\, e^{-k\theta}$$

| symbol | | meaning | SI unit |
|---|---|---|---|
| `Cin` | in | influent concentration | kg/m^3 |
| `k` | in | first-order rate constant | 1/s |
| `theta` | in | total hydraulic residence time of all the tanks (or give V and Q) (optional) | s |
| `N` | in | number of identical tanks in series (default 1) (optional) | , |
| `V` | in | total reactor volume (with Q, in place of theta) (optional) | m^3 |
| `Q` | in | flow through the reactor (with V, in place of theta) (optional) | m^3/s |
| `Cout` | out | effluent of the N tanks Cin/(1 + k theta/N)^N | kg/m^3 |
| `Cout_mgL` | out | in mg/L | , |
| `removal` | out | fraction removed by the N tanks | , |

| symbol | | meaning | SI unit |
|---|---|---|---|
| `removal_percent` | out | per cent removed | , |
| `Cout_one` | out | effluent of one CMFR of the same total volume | kg/m^3 |
| `Cout_one_mgL` | out | in mg/L | , |
| `Cout_pfr` | out | effluent of a plug-flow reactor Cin exp(-k theta) | kg/m^3 |
| `Cout_pfr_mgL` | out | in mg/L | , |
| `removal_pfr` | out | fraction removed in plug flow | , |
| `theta_each` | out | residence time of each tank theta/N | s |
| `theta_each_h` | out | residence time of each tank, h | , |

**Applies when:** steady state, first-order decay; identical tanks sharing the total residence time theta; 1 <= N <= 1000; theta, or the volume V and the flow Q (theta = V/Q), not both.

```
$ environmental-engineering solve enveng_cmfr_series "Cin=100 mg/L" "k=4.8 1/day" "theta=10 h"
N=4
Cout = 0.01975308642 kg/m^3
Cout_mgL = 19.75308642
removal = 0.8024691358
removal_percent = 80.24691358
Cout_one = 0.03333333333 kg/m^3
Cout_one_mgL = 33.33333333
Cout_pfr = 0.01353352832 kg/m^3
Cout_pfr_mgL = 13.53352832
removal_pfr = 0.8646647168
theta_each = 9000 s
theta_each_h = 2.5
  by: First-order reaction in N stirred tanks in series and in plug flow  (theta = V/Q when V
and Q are given; Cout = Cin/(1 + k theta/N)^N; one CMFR: Cin/(1 + k theta); plug flow: Cin
exp(-k theta))
  Cin = 0.1 kg/m^3
  k = 5.555555556e-05 1/s
  theta = 36000 s
  assumes: steady state, first-order decay; identical tanks sharing the total residence time
theta; 1 <= N <= 1000; theta, or the volume V and the flow Q (theta = V/Q), not both
```

In the worked example, Cin = 100 mg/L, k = 4.8 1/day, theta = 10 h, N = 4 go through the equation above and give Cout = 0.01975308642 kg/m^3, Cout_mgL = 19.75308642, removal = 0.8024691358, removal_percent = 80.24691358. It applies only when steady state, first-order decay; identical tanks sharing the total residence time theta; 1 <= N <= 1000; theta, or the volume V and the flow Q (theta = V/Q), not both: outside that it says what is wrong instead of returning a number.

## Risk assessment (1)

### Chronic daily intake, hazard quotient and cancer risk, `enveng_chronic_daily_intake_risk`

The chronic daily intake of a contaminant in drinking water, CDI = C IR EF ED / (BW AT) (a continuous lifetime exposure, CDI = C IR / BW, when EF, ED and AT are not given), with the hazard quotient CDI/RfD and the lifetime cancer risk CDI SF when the reference dose or slope factor is given.

$$CDI = \frac{C \cdot IR \cdot EF \cdot ED}{BW \cdot AT}, \quad HQ = \frac{CDI}{RfD}, \quad \mathrm{risk} = CDI \cdot SF$$

| symbol | | meaning | SI unit |
|---|---|---|---|
| `C` | in | concentration in the water | kg/m^3 |
| `IR` | in | intake (ingestion) rate | m^3/s |
| `BW` | in | body weight | kg |
| `EF` | in | exposure frequency, days per year (optional) | , |
| `ED` | in | exposure duration, years (optional) | , |
| `AT` | in | averaging time, days (optional) | , |
| `RfD` | in | reference dose, mg/(kg d) (optional) | , |
| `SF` | in | cancer slope factor, (mg/(kg d))^-1 (optional) | , |
| `CDI` | out | chronic daily intake, mg/(kg d) | , |
| `HQ` | out | hazard quotient CDI/RfD | , |
| `risk` | out | lifetime excess cancer risk CDI SF | , |

**Applies when:** EF, ED and AT together or none (continuous lifetime); EF <= 365 d/yr, EF ED <= AT; ED in years, AT in days (70 yr x 365 = 25,550 d); linear low-dose risk up to 0.01.

```
$ environmental-engineering solve enveng_chronic_daily_intake_risk "C=0.05 mg/L" "IR=2 L/day"
"BW=70 kg" RfD=0.02
CDI = 0.001428571429
HQ = 0.07142857143
  by: Chronic daily intake, hazard quotient and cancer risk  (CDI = C IR EF ED / (BW AT) [mg/L,
L/d, d/yr, yr, kg, d]; HQ = CDI/RfD; risk = CDI SF)
  C = 5e-05 kg/m^3
  IR = 2.314814815e-08 m^3/s
  BW = 70 kg
  RfD = 0.02
  assumes: EF, ED and AT together or none (continuous lifetime); EF <= 365 d/yr, EF ED <= AT;
ED in years, AT in days (70 yr x 365 = 25,550 d); linear low-dose risk up to 0.01
```

In the worked example, C = 0.05 mg/L, IR = 2 L/day, BW = 70 kg, RfD = 0.02 go through the equation above and give CDI = 0.001428571429, HQ = 0.07142857143. It applies only when EF, ED and AT together or none (continuous lifetime); EF <= 365 d/yr, EF ED <= AT; ED in years, AT in days (70 yr x 365 = 25,550 d); linear low-dose risk up to 0.01: outside that it says what is wrong instead of returning a number.

## Solid waste (8)

### Compaction ratio of solid waste, `enveng_landfill_compaction_ratio`

The compaction ratio of solid waste (compacted over initial density, or initial over compacted volume) and the volume reduction it gives.

$$CR = \frac{\rho_c}{\rho_i} = \frac{V_i}{V_c}, \qquad \Delta V = 1 - \frac{1}{CR}$$

| symbol | | meaning | SI unit |
|---|---|---|---|
| `rho_i` | in | density as delivered (loose) (optional) | kg/m^3 |
| `rho_c` | in | density after compaction (optional) | kg/m^3 |
| `V_i` | in | volume before compaction (optional) | m^3 |

| symbol | | meaning | SI unit |
|---|---|---|---|
| `V_c` | in | volume after compaction (optional) | m^3 |
| `CR` | out | compaction ratio | , |
| `volume_reduction` | out | fraction of the volume removed 1 - 1/CR | , |
| `volume_reduction_percent` | out | volume reduction in per cent | , |

**Applies when:** two densities or two volumes; the same mass of waste before and after; CR >= 1.

```
$ environmental-engineering solve enveng_landfill_compaction_ratio "rho_i=120 kg/m^3"
"rho_c=600 kg/m^3"
CR = 5
volume_reduction = 0.8
volume_reduction_percent = 80
  by: Compaction ratio of solid waste  (CR = rho_c/rho_i = V_i/V_c (mass conserved); reduction
= 1 - V_c/V_i)
  rho_i = 120 kg/m^3
  rho_c = 600 kg/m^3
  assumes: two densities or two volumes; the same mass of waste before and after; CR >= 1
```

In the worked example, rho_i = 120 kg/m^3, rho_c = 600 kg/m^3 go through the equation above and give CR = 5, volume_reduction = 0.8, volume_reduction_percent = 80. It applies only when two densities or two volumes; the same mass of waste before and after; CR >= 1: outside that it says what is wrong instead of returning a number.

**Heating value of waste from its elemental composition (modified Dulong), `enveng_dulong_heating_value`**

The higher heating value of a waste or fuel from its dry mass fractions of carbon, hydrogen, oxygen, sulfur and nitrogen by the modified Dulong formula.

$$HHV = 337C + 1419\left(H - \frac{O}{8}\right) + 93S + 23N \ \ [\mathrm{kJ/kg},\ \%]$$

| symbol | | meaning | SI unit |
|---|---|---|---|
| `C` | in | carbon mass fraction (fraction or %) | , |
| `H` | in | hydrogen mass fraction | , |
| `O` | in | oxygen mass fraction | , |
| `S` | in | sulfur mass fraction | , |
| `N` | in | nitrogen mass fraction | , |
| `HHV` | out | higher heating value | J/kg |
| `HHV_kJ_per_kg` | out | in kJ/kg | , |
| `HHV_MJ_per_kg` | out | in MJ/kg | , |

**Applies when:** fractions on the same (dry) basis, sum at most 1.

```
$ environmental-engineering solve enveng_dulong_heating_value C=0.48 H=0.064 O=0.376 S=0.004
N=0.008
HHV = 18643900 J/kg
HHV_kJ_per_kg = 18643.9
HHV_MJ_per_kg = 18.6439
  by: Heating value of waste from its elemental composition (modified Dulong)  (HHV (kJ/kg) =
337 C + 1419 (H - O/8) + 93 S + 23 N, mass per cent)
  C = 0.48
  H = 0.064
  O = 0.376
  S = 0.004
  N = 0.008
  assumes: fractions on the same (dry) basis, sum at most 1
```

**What the worked example means:** This calculation determines the higher heating value of a waste sample using the modified Dulong formula. It uses the dry mass fractions of carbon, hydrogen, oxygen, sulfur, and nitrogen to estimate the energy content. The resulting higher heating value is 18643.9, which indicates the total heat released per unit mass when the fuel is completely combusted under standard conditions.

### Heating value on a dry basis from the as-received value and the moisture, `enveng_heating_value_dry_basis`

The higher heating value of a fuel or waste on a dry basis from its as-received value and moisture fraction, HHV_dry = HHV_ar/(1 - w) (or the reverse), and on a dry ash-free basis when the ash fraction is given.

$$HHV_{dry} = \frac{HHV_{ar}}{1 - w}, \qquad HHV_{daf} = \frac{HHV_{ar}}{1 - w - a}$$

| symbol | | meaning | SI unit |
|---|---|---|---|
| `w` | in | moisture fraction as received, wet basis (fraction or %) | , |
| `HHV_ar` | in | heating value as received (optional) | J/kg |
| `HHV_dry` | in | heating value on a dry basis (optional) | J/kg |
| `ash` | in | ash fraction as received (fraction or %) (optional) | , |
| `HHV_dry` | out | dry-basis heating value HHV_ar/(1 - w) | J/kg |
| `HHV_ar` | out | as-received heating value | J/kg |
| `HHV_dry_MJ_per_kg` | out | dry basis, MJ/kg | , |
| `HHV_ar_MJ_per_kg` | out | as received, MJ/kg | , |
| `HHV_daf` | out | dry ash-free heating value HHV_ar/(1 - w - ash) | J/kg |
| `HHV_daf_MJ_per_kg` | out | dry ash-free, MJ/kg | , |

**Applies when:** 0 <= w < 1 (wet-basis moisture); exactly one of HHV_ar, HHV_dry; w + ash < 1.

```
$ environmental-engineering solve enveng_heating_value_dry_basis w=0.25 "HHV_ar=12000 kJ/kg"
HHV_dry = 16000000 J/kg
HHV_ar = 12000000 J/kg
HHV_dry_MJ_per_kg = 16
HHV_ar_MJ_per_kg = 12
  by: Heating value on a dry basis from the as-received value and the moisture  (HHV_dry =
HHV_ar/(1 - w); HHV_daf = HHV_ar/(1 - w - ash))
  w = 0.25
  HHV_ar = 12000000 J/kg
  assumes: 0 <= w < 1 (wet-basis moisture); exactly one of HHV_ar, HHV_dry; w + ash < 1
```

In the worked example, w = 0.25, HHV_ar = 12000 kJ/kg go through the equation above and give HHV_dry = 16000000 J/kg, HHV_ar = 12000000 J/kg, HHV_dry_MJ_per_kg = 16, HHV_ar_MJ_per_kg = 12. It applies only when 0 <= w < 1 (wet-basis moisture); exactly one of HHV_ar, HHV_dry; w + ash < 1: outside that it says what is wrong instead of returning a number.

**Landfill life with daily cover and waste growth, `enveng_landfill_life`**

The landfill volume used in the first year (waste at its in-place density plus cover soil) and the life of a landfill of given capacity when the waste grows at a constant yearly rate.

$$n = \frac{\ln\left(1 + \frac{Cg}{V_0}\right)}{\ln(1+g)}, \quad V_0 = \frac{M(1+c)}{\rho}$$

| symbol | | meaning | SI unit |
|---|---|---|---|
| `capacity` | in | available landfill volume | m^3 |
| `M` | in | waste delivered in the first year | kg/s |
| `rho` | in | in-place (compacted) waste density | kg/m^3 |
| `cover` | in | cover soil volume per volume of waste (e.g. 0.25) | , |
| `g` | in | yearly growth of the waste quantity (fraction or %; 0 for none) | , |
| `V_first_year` | out | volume used in the first year, m^3 | , |
| `life_years` | out | landfill life, years | , |

**Applies when:** growth compounded yearly, life read continuously.

```
$ environmental-engineering solve enveng_landfill_life "capacity=2000000 m^3" "M=50000 t/yr"
"rho=600 kg/m^3" cover=0.25 g=0
V_first_year = 104166.6667
life_years = 19.2
  by: Landfill life with daily cover and waste growth  (V0 = M (1 + cover)/rho per year;
capacity = V0 ((1 + g)^n - 1)/g, so n = ln(1 + capacity g/V0)/ln(1 + g))
  capacity = 2000000 m^3
  M = 1.584404391 kg/s
  rho = 600 kg/m^3
  cover = 0.25
  g = 0
  assumes: growth compounded yearly, life read continuously
```

**What the worked example means:** This calculation determines the operational lifespan of a landfill by first establishing the volume consumed in the initial year. The first-year volume, denoted as 104167, is derived from the waste delivery rate 50000 t/yr, adjusted for cover soil 0.25, and divided by the compacted density 600 kg/m^3.

Using this base volume and the annual growth factor 0, the model computes how many years 19.2 it will take to fill the available capacity 2e+06 m^3. This result indicates when the site will reach its physical limit under current operating assumptions.

**Landfill methane generation (first-order decay, constant acceptance),** `enveng_landfill_gas_generation`

The methane generation rate and the cumulative methane of a landfill receiving waste at a constant rate for its operating life, each deposit decaying at first order (LandGEM-type model in continuous form), and the landfill gas flow at 50 % methane.

$$Q(t) = \int_0^{\min(t,t_c)} k L_0 M\, e^{-k(t-\tau)}\, d\tau = M L_0 \left( e^{-k(t-\min(t,t_c))} - e^{-kt} \right)$$

| symbol | | meaning | SI unit |
|---|---|---|---|
| `M` | in | waste acceptance rate | kg/s |
| `L0` | in | methane generation potential | m^3/kg |
| `k` | in | methane generation rate constant | 1/s |
| `tc` | in | operating (filling) period | s |
| `t` | in | time since opening | s |
| `Q_CH4` | out | methane generation rate at t | m^3/s |
| `Q_CH4_m3_per_year` | out | in m^3 per year | , |
| `V_CH4` | out | cumulative methane generated up to t | m^3 |
| `Q_LFG` | out | landfill gas rate at 50 % methane | m^3/s |

**Applies when:** constant acceptance from opening to tc; first-order decay of every deposit.

```
$ environmental-engineering solve enveng_landfill_gas_generation "M=100000 t/yr" "L0=100 m^3/t"
"k=0.05 1/yr" "tc=20 yr" "t=25 yr"
Q_CH4 = 0.1559991844 m^3/s
Q_CH4_m3_per_year = 4922959.862
V_CH4 = 101540802.8 m^3
Q_LFG = 0.3119983688 m^3/s
  by: Landfill methane generation (first-order decay, constant acceptance)  (Q(t) = M L0 (e^(-k
(t - min(t, tc))) - e^(-k t)); cumulative volume by integrating Q)
  M = 3.168808781 kg/s
  L0 = 0.1 m^3/kg
  k = 1.584404391e-09 1/s
  tc = 631152000 s
  t = 788940000 s
  assumes: constant acceptance from opening to tc; first-order decay of every deposit
```

**What the worked example means:** This calculation determines the current methane generation rate for a landfill that has received waste at a constant acceptance rate of 100000 t/yr since its opening. Using a first-order decay model with a rate constant of 0.05 1/yr and a methane generation potential of 100 m^3/t, the system yields an instantaneous methane flow of 0.155999 m^3/s. This rate translates to 4.92296e+06 annually. The total amount of methane produced from the start of operations until the current time of 25 yr is calculated as 1.01541e+08 m^3. Based on these figures, the total landfill gas flow, assuming fifty percent methane content, is estimated at 0.311998 m^3/s.

**Landfill volume per day and per year with cover soil, and the landfill life,** `enveng_landfill_daily_volume`

The landfill volume used each day and each year by waste placed at its compacted density with daily cover soil, V = mass / compacted density x (1 + cover ratio), and, for a given capacity, the life of the landfill (constant waste rate).

$$V = \frac{\dot{m}}{\rho_c}\,(1 + c), \qquad n = \frac{V_{cap}}{365\,V_{day}}$$

| symbol | | meaning | SI unit |
|---|---|---|---|
| `m` | in | waste placed per day | kg/s |
| `rho` | in | compacted (in-place) density of the waste | kg/m^3 |
| `cover` | in | cover soil volume per volume of compacted waste (fraction or %; default 0) (optional) | , |
| `capacity` | in | available landfill volume (for the life) (optional) | m^3 |
| `V` | out | landfill volume used, waste plus cover | m^3/s |
| `V_m3_per_day` | out | landfill volume used per day, m^3 | , |
| `V_m3_per_year` | out | landfill volume used per 365-day year, m^3 | , |
| `V_waste_m3_per_day` | out | compacted waste alone per day, m^3 | , |
| `V_cover_m3_per_day` | out | cover soil per day, m^3 | , |
| `life_years` | out | landfill life, years | , |
| `life_days` | out | landfill life, days | , |

**Applies when:** constant waste rate, every day of a 365-day year; cover as a volume ratio to the compacted waste, 0 to 1; no settlement or decomposition.

```
$ environmental-engineering solve enveng_landfill_daily_volume "m=200 t/day" "rho=600 kg/m^3"
cover=0.25
V = 0.004822530864 m^3/s
V_m3_per_day = 416.6666667
V_m3_per_year = 152083.3333
V_waste_m3_per_day = 333.3333333
V_cover_m3_per_day = 83.33333333
  by: Landfill volume per day and per year with cover soil, and the landfill life  (V = m/rho
(1 + cover); per year x 365; life = capacity / yearly volume)
  m = 2.314814815 kg/s
  rho = 600 kg/m^3
  cover = 0.25
  assumes: constant waste rate, every day of a 365-day year; cover as a volume ratio to the
compacted waste, 0 to 1; no settlement or decomposition
```

In the worked example, m = 200 t/day, rho = 600 kg/m^3, cover = 0.25 go through the equation above and give V = 0.004822530864 m^3/s, V_m3_per_day = 416.6666667, V_m3_per_year = 152083.3333, V_waste_m3_per_day = 333.3333333. It applies only when constant waste rate, every day of a 365-day year; cover as a volume ratio to the compacted waste, 0 to 1; no settlement or decomposition: outside that it says what is wrong instead of returning a number.

### Moisture content of mixed solid waste, `enveng_waste_moisture_content`

The moisture content (wet and dry basis) and the dry mass of a waste mixture from the masses of its components and their moisture fractions.

$$M_{wet} = \frac{\sum m_i w_i}{\sum m_i}$$

| symbol | | meaning | SI unit |
|---|---|---|---|
| `m` | in | wet masses of the components (any one unit) | , |
| `w` | in | moisture fractions of the components, wet basis | , |
| `moisture_wet` | out | moisture content, wet basis | , |
| `dry_mass` | out | dry mass, in the unit of m | , |
| `moisture_dry` | out | moisture content, dry basis | , |

**Applies when:** moisture fractions below 1.

```
$ environmental-engineering solve enveng_waste_moisture_content "m=[40, 25, 15, 20]" "w=[0.7,
0.06, 0.02, 0.2]"
moisture_wet = 0.338
dry_mass = 66.2
moisture_dry = 0.5105740181
  by: Moisture content of mixed solid waste  (water = sum m_i w_i; M_wet = water/sum m; M_dry =
water/(sum m - water))
  assumes: moisture fractions below 1
```

**What the worked example means:** This calculation determines the overall moisture content of a mixed solid waste stream on both wet and dry bases, as well as its total dry mass. It sums the water mass contributed by each component based on their individual wet masses [40, 25, 15, 20] and moisture fractions [0.7, 0.06, 0.02, 0.2], then divides this total water by the sum of all wet masses to find the wet basis result 0.338. The dry mass 66.2 is derived by subtracting the total water from the total wet mass, which allows for the calculation of the dry basis moisture content 0.510574. These values are essential for assessing the handling, storage, and processing requirements of the waste mixture.

**Municipal solid waste generation and collection trucks, `enveng_solid_waste_generation`**

The daily and yearly mass of municipal solid waste of a population, its compacted volume and the number of truck loads per day.

$$\dot{m} = P r, \quad n = \left\lceil \frac{\dot{m}}{\rho V_t} \right\rceil$$

| symbol | | meaning | SI unit |
|---|---|---|---|
| `pop` | in | population served | , |
| `rate` | in | generation rate in kg per person per day | , |
| `rho` | in | compacted density in the truck | kg/m^3 |
| `Vt` | in | truck body volume | m^3 |
| `m` | out | waste generation | kg/s |
| `m_kg_per_day` | out | in kg/d | , |
| `m_t_per_year` | out | in t per 365-day year | , |
| `V_m3_per_day` | out | compacted volume per day, m^3 | , |
| `trucks_per_day` | out | truck loads per day (rounded up) | , |

**Applies when:** full truck loads at the stated density.

```
$ environmental-engineering solve enveng_solid_waste_generation pop=50000 rate=1.5 "rho=400
kg/m^3" "Vt=15 m^3"
m = 0.8680555556 kg/s
m_kg_per_day = 75000
m_t_per_year = 27375
V_m3_per_day = 187.5
trucks_per_day = 13
  by: Municipal solid waste generation and collection trucks  (m = P r; V = m/rho; trucks =
ceil(V/Vt))
  rate = 1.5
  rho = 400 kg/m^3
  Vt = 15 m^3
  assumes: full truck loads at the stated density
```

**What the worked example means:** This calculation determines the municipal solid waste generation rate based on population and per capita production. The total daily mass of waste is computed as 75000. This mass is converted into a compacted volume of 187.5 cubic meters using the specified density. Finally, the number of collection trucks needed to transport this volume in full loads is determined to be 13.

## Wastewater (19)

### Activated sludge (complete mix) effluent substrate and biomass (Lawrence-McCarty), `enveng_activated_sludge_cstr_design`

The effluent soluble substrate, the mixed-liquor biomass and the observed yield of a complete-mix activated-sludge reactor with recycle, from the kinetic coefficients, the SRT and HRT and the influent substrate.

$$S = \frac{K_s(1 + k_d\theta_c)}{\theta_c(Yk - k_d) - 1}, \quad X = \frac{\theta_c Y(S_0 - S)}{\theta(1 + k_d\theta_c)}$$

| symbol | | meaning | SI unit |
|---|---|---|---|
| `Y` | in | true yield coefficient, kg VSS per kg BOD | , |
| `k` | in | maximum specific substrate utilisation rate | 1/s |
| `Ks` | in | half-saturation constant | kg/m^3 |
| `kd` | in | endogenous decay coefficient | 1/s |
| `SRT` | in | solids retention time | s |
| `HRT` | in | hydraulic retention time | s |
| `S0` | in | influent substrate (BOD) | kg/m^3 |
| `S` | out | effluent soluble substrate | kg/m^3 |
| `S_mgL` | out | in mg/L | , |
| `X` | out | biomass concentration in the reactor | kg/m^3 |
| `X_mgL` | out | in mg/L | , |
| `Y_obs` | out | observed yield Y/(1 + kd SRT) | , |

**Applies when:** steady state, complete mix; SRT above the washout limit; S < S0.

```
$ environmental-engineering solve enveng_activated_sludge_cstr_design Y=0.5 "k=5 1/day" "Ks=60
mg/L" "kd=0.06 1/day" "SRT=10 day" "HRT=6 h" "S0=250 mg/L"
S = 0.004102564103 kg/m^3
S_mgL = 4.102564103
X = 3.073717949 kg/m^3
X_mgL = 3073.717949
Y_obs = 0.3125
  by: Activated sludge (complete mix) effluent substrate and biomass (Lawrence-McCarty)  (S =
Ks (1 + kd SRT)/(SRT (Y k - kd) - 1); X = (SRT/HRT) Y (S0 - S)/(1 + kd SRT))
  Y = 0.5
  k = 5.787037037e-05 1/s
  Ks = 0.06 kg/m^3
  kd = 6.944444444e-07 1/s
  SRT = 864000 s
  HRT = 21600 s
  S0 = 0.25 kg/m^3
  assumes: steady state, complete mix; SRT above the washout limit; S < S0
```

**What the worked example means:** This calculation determines the steady-state performance of a complete-mix activated-sludge reactor. Using the kinetic coefficients, solids retention time, hydraulic retention time, and influent substrate, it computes the effluent soluble substrate at 0.00410256 kg/m^3 and the mixed-liquor biomass at 3.07372 kg/m^3. The result indicates how effectively the system removes organic matter and maintains microbial mass under the specified operational conditions.

**Dissolved oxygen during aeration and the transfer rate, `enveng_aeration_do_curve`**

The dissolved oxygen after an aeration time from the starting DO, the saturation DO and KLa, and the oxygen transfer rate KLa (Cs - C) at that moment.

$$C = C_s - (C_s - C_0)e^{-K_L a\, t}$$

| symbol | | meaning | SI unit |
|---|---|---|---|
| `Cs` | in | saturation DO | kg/m^3 |
| `C0` | in | starting DO | kg/m^3 |
| `KLa` | in | volumetric transfer coefficient | 1/s |
| `t` | in | aeration time | s |
| `C` | out | DO at t | kg/m^3 |
| `C_mgL` | out | in mg/L | , |
| `OTR` | out | oxygen transfer rate at t | kg/(m^3*s) |
| `OTR_mg_per_L_h` | out | transfer rate in mg/(L h) | , |

**Applies when:** C0 <= Cs; no oxygen uptake.

```
$ environmental-engineering solve enveng_aeration_do_curve "Cs=9.1 mg/L" "C0=1 mg/L" "KLa=10
1/h" "t=10 min"
C = 0.007570107617 kg/m^3
C_mgL = 7.570107617
OTR = 4.249701064e-06 kg/(m^3*s)
OTR_mg_per_L_h = 15.29892383
  by: Dissolved oxygen during aeration and the transfer rate  (dC/dt = KLa (Cs - C), C = Cs -
(Cs - C0) e^(-KLa t))
  Cs = 0.0091 kg/m^3
  C0 = 0.001 kg/m^3
  KLa = 0.002777777778 1/s
  t = 600 s
  assumes: C0 <= Cs; no oxygen uptake
```

**What the worked example means:** This calculation determines the dissolved oxygen concentration at time t using the first-order kinetic model. Starting from an initial concentration of 1 mg/L, the system approaches the saturation limit of 9.1 mg/L with a volumetric transfer coefficient of 10 1/h. The resulting concentration is 0.00757011 kg/m^3, which converts to 7.57011 for practical reporting. Additionally, the instantaneous oxygen transfer rate at this moment is calculated as 4.2497e-06 kg/(m^3*s), equivalent to 15.2989. This output indicates how quickly oxygen is being transferred into the water at that specific point in time.

**Field (actual) oxygen transfer rate of an aeration system, `enveng_field_oxygen_transfer`**

The actual oxygen transfer rate in the process from the standard (clean water, 20 degC, zero DO) rate, corrected by alpha, beta, the temperature coefficient, the site saturation and the operating DO.

$$AOTR = SOTR\,\alpha\,\theta^{T-20}\,\frac{\beta C_{s,T} F - C_L}{C_{s,20}}$$

| symbol | | meaning | SI unit |
|---|---|---|---|
| `SOTR` | in | standard oxygen transfer rate | kg/s |
| `alpha` | in | alpha factor (process/clean water KLa) | , |
| `beta` | in | beta factor (process/clean water saturation) | , |
| `theta` | in | temperature coefficient (1.024) | , |
| `CsT` | in | clean-water saturation DO at the process temperature | kg/m^3 |
| `Cs20` | in | clean-water saturation DO at 20 degC | kg/m^3 |
| `CL` | in | operating DO | kg/m^3 |
| `T` | in | process temperature | K |
| `Fp` | in | pressure (elevation) correction of saturation (optional) | , |
| `AOTR` | out | actual oxygen transfer rate | kg/s |
| `AOTR_kg_per_h` | out | in kg O2/h | , |
| `ratio` | out | AOTR/SOTR | , |

**Applies when:** beta CsT Fp > CL.

```
$ environmental-engineering solve enveng_field_oxygen_transfer "SOTR=100 kg/h" alpha=0.6
beta=0.95 theta=1.024 "CsT=8.26 mg/L" "Cs20=9.09 mg/L" "CL=2 mg/L" "T=25 degC"
AOTR = 0.01207029108 kg/s
AOTR_kg_per_h = 43.45304789
ratio = 0.4345304789
  by: Field (actual) oxygen transfer rate of an aeration system  (AOTR = SOTR alpha theta^(T -
20) (beta CsT Fp - CL)/Cs20)
  SOTR = 0.02777777778 kg/s
  alpha = 0.6
  beta = 0.95
  theta = 1.024
  CsT = 0.00826 kg/m^3
  Cs20 = 0.00909 kg/m^3
  CL = 0.002 kg/m^3
  T = 298.15 K
  assumes: beta CsT Fp > CL
```

**What the worked example means:** This calculation determines the field or actual oxygen transfer rate by adjusting the standard oxygen transfer rate for site-specific conditions. The result, 0.0120703 kg/s, represents the effective mass of oxygen transferred per second under current operating parameters. This value is derived from the standard rate 100 kg/h, modified by the process factors alpha and beta, the temperature coefficient theta raised to the power of the difference between the process temperature T and twenty degrees Celsius, and the ratio of the corrected saturation concentration minus the operating dissolved oxygen level to the standard saturation concentration at twenty degrees Celsius. The hourly equivalent is given as 43.453.

**Food-to-microorganism ratio (F/M),** `enveng_food_to_microorganism_ratio`

The food-to-microorganism ratio of an aeration tank from the flow, influent BOD, tank volume and mixed-liquor volatile suspended solids.

$$\frac{F}{M} = \frac{QS_0}{VX}$$

| symbol | | meaning | SI unit |
|---|---|---|---|
| `Q` | in | influent flow | m^3/s |
| `S0` | in | influent BOD (or COD) | kg/m^3 |
| `V` | in | aeration tank volume | m^3 |
| `X` | in | mixed liquor volatile suspended solids (MLVSS) | kg/m^3 |
| `FM` | out | food-to-microorganism (F/M) ratio | 1/s |
| `FM_per_day` | out | in kg BOD/(kg MLVSS d) | , |

**Applies when:** Q, S0, V, X > 0.

```
$ environmental-engineering solve enveng_food_to_microorganism_ratio "Q=10000 m^3/day" "S0=200
mg/L" "V=3000 m^3" "X=2500 mg/L"
FM = 3.086419753e-06 1/s
FM_per_day = 0.2666666667
  by: Food-to-microorganism ratio (F/M)  (F/M = Q S0/(V X))
  Q = 0.1157407407 m^3/s
  S0 = 0.2 kg/m^3
  V = 3000 m^3
  X = 2.5 kg/m^3
  assumes: Q, S0, V, X > 0
```

**What the worked example means:** This calculation determines the Food-to-Microorganism ratio for an aeration tank. It uses the influent flow 10000 m^3/day, influent biochemical oxygen demand 200 mg/L, tank volume 3000 m^3, and mixed liquor volatile suspended solids 2500 mg/L. The resulting ratio of 3.08642e-06 1/s indicates the balance between available organic matter and active biomass, which is critical for controlling treatment efficiency and sludge characteristics.

### Hydraulic retention time and volumetric organic loading, `enveng_hrt_organic_loading`

The hydraulic retention time of a reactor and its volumetric organic (BOD) loading rate from the flow, influent BOD and volume.

$$\theta = \frac{V}{Q}, \quad L_v = \frac{Q S_0}{V}$$

| symbol | | meaning | SI unit |
|---|---|---|---|
| `Q` | in | influent flow | m^3/s |
| `S0` | in | influent BOD | kg/m^3 |
| `V` | in | reactor volume | m^3 |
| `HRT` | out | hydraulic retention time | s |
| `HRT_h` | out | HRT in hours | , |
| `OLR` | out | volumetric organic loading rate | kg/(m^3*s) |
| `OLR_kg_per_m3_day` | out | loading in kg BOD/(m^3 d) | , |

**Applies when:** Q, S0, V > 0.

```
$ environmental-engineering solve enveng_hrt_organic_loading "Q=10000 m^3/day" "S0=200 mg/L"
"V=3000 m^3"
HRT = 25920 s
HRT_h = 7.2
OLR = 7.716049383e-06 kg/(m^3*s)
OLR_kg_per_m3_day = 0.6666666667
  by: Hydraulic retention time and volumetric organic loading  (HRT = V/Q; OLR = Q S0/V)
  Q = 0.1157407407 m^3/s
  S0 = 0.2 kg/m^3
  V = 3000 m^3
  assumes: Q, S0, V > 0
```

**What the worked example means:** This calculation determines the hydraulic retention time and volumetric organic loading rate for a reactor. Using the influent flow of 10000 m^3/day, influent biochemical oxygen demand of 200 mg/L, and reactor volume of 3000 m^3, the system yields a hydraulic retention time of 25920 s and a

loading rate of 7.71605e-06 kg/(m^3*s). The hydraulic retention time indicates how long water remains in the tank, while the loading rate shows the mass of organic matter applied per unit volume of the reactor.

**KLa fitted to a reaeration test, `enveng_kla_regression`**

The volumetric oxygen-transfer coefficient KLa from a series of dissolved-oxygen readings during a reaeration test, by linear regression of ln(Cs - C) against time.

$$\ln(C_s - C) = \ln(C_s - C_0) - K_L a\, t$$

| symbol | | meaning | SI unit |
|---|---|---|---|
| `t` | in | times in minutes | , |
| `C` | in | DO readings in mg/L (same order) | , |
| `Cs` | in | saturation DO | kg/m^3 |
| `KLa_per_min` | out | KLa in 1/min | , |
| `KLa` | out | KLa | 1/s |
| `KLa_per_h` | out | KLa in 1/h | , |
| `r2` | out | coefficient of determination | , |

**Applies when:** every reading below Cs; at least three distinct times.

```
$ environmental-engineering solve enveng_kla_regression "t=[2, 4, 6, 8, 10, 12]" "C=[2.5, 4.1,
5.4, 6.3, 7, 7.6]" "Cs=9.1 mg/L"
KLa_per_min = 0.1469891116
KLa = 0.002449818526 1/s
KLa_per_h = 8.819346694
r2 = 0.9991806961
  by: KLa fitted to a reaeration test  (least squares of ln(Cs - C) = ln(Cs - C0) - KLa t)
  Cs = 0.0091 kg/m^3
  assumes: every reading below Cs; at least three distinct times
```

**What the worked example means:** This calculation determines the volumetric oxygen-transfer coefficient KLa by performing a least-squares linear regression on the natural logarithm of the dissolved oxygen deficit against time. Using the saturation concentration 9.1 mg/L and the measured readings [2.5, 4.1, 5.4, 6.3, 7, 7.6] at times [2, 4, 6, 8, 10, 12], the method isolates the exponential decay rate. The resulting coefficient 0.00244982 1/s represents the speed at which oxygen transfers into the water, while the correlation statistic 0.999181 indicates how well the first-order model fits the observed reaeration behavior.

**Mean cell residence time (sludge age) from the solids masses, `enveng_mean_cell_residence_time`**

The mean cell residence time (solids retention time, sludge age) of an activated-sludge system as the mass of solids held in the basin over the mass leaving per day in the waste sludge and the effluent.

$$\theta_c = \frac{M_{basin}}{\dot{M}_w + \dot{M}_e}$$

| symbol | | meaning | SI unit |
|---|---|---|---|
| `M` | in | mass of solids (MLVSS or MLSS) in the aeration basin | kg |
| `Mw` | in | mass of solids wasted per day | kg/s |

| symbol | | meaning | SI unit |
|---|---|---|---|
| `Me` | in | mass of solids lost in the effluent per day | kg/s |
| `theta_c` | out | mean cell residence time | s |
| `theta_c_days` | out | in days | , |

**Applies when:** steady state; M > 0, Mw + Me > 0.

```
$ environmental-engineering solve enveng_mean_cell_residence_time "M=3000 kg" "Mw=300 kg/day"
"Me=20 kg/day"
theta_c = 810000 s
theta_c_days = 9.375
  by: Mean cell residence time (sludge age) from the solids masses  (theta_c = M / (Mw + Me))
  M = 3000 kg
  Mw = 0.003472222222 kg/s
  Me = 0.0002314814815 kg/s
  assumes: steady state; M > 0, Mw + Me > 0
```

In the worked example, M = 3000 kg, Mw = 300 kg/day, Me = 20 kg/day go through the equation above and give theta_c = 810000 s, theta_c_days = 9.375. It applies only when steady state; M > 0, Mw + Me > 0: outside that it says what is wrong instead of returning a number.

**Methane production from COD removed (anaerobic digestion), `enveng_anaerobic_methane_yield`**

The methane gas flow from an anaerobic process from the COD removed less the COD converted to biomass (1.42 kg COD per kg VSS), at a stated temperature and pressure (64 g COD per mole of methane).

$$Q_{CH_4} = \frac{\mathrm{COD} - 1.42 P_x}{64\ \mathrm{g/mol}} \frac{RT}{P}$$

| symbol | | meaning | SI unit |
|---|---|---|---|
| `COD` | in | COD removed | kg/s |
| `Px` | in | biomass (VSS) produced | kg/s |
| `T` | in | gas temperature | K |
| `P` | in | gas pressure | Pa |
| `Q_CH4` | out | methane flow at T and P | m^3/s |
| `Q_CH4_m3_per_day` | out | in m^3/d at T and P | , |
| `Q_CH4_stp_m3_per_day` | out | in m^3/d at 0 degC and 1 atm | , |

**Applies when:** ideal gas; COD balance.

```
$ environmental-engineering solve enveng_anaerobic_methane_yield "COD=5000 kg/day" "Px=300
kg/day" "T=308.15 K" "P=101.325 kPa"
Q_CH4 = 0.02091695562 m^3/s
Q_CH4_m3_per_day = 1807.224965
Q_CH4_stp_m3_per_day = 1601.958459
  by: Methane production from COD removed (anaerobic digestion)  (n = (COD - 1.42 Px)/(64
g/mol); Q = n R T/P)
  COD = 0.05787037037 kg/s
  Px = 0.003472222222 kg/s
  T = 308.15 K
  P = 101325 Pa
  assumes: ideal gas; COD balance
```

**What the worked example means:** This calculation determines the volumetric flow of methane gas produced from anaerobic digestion. It starts with the total chemical oxygen demand removed, which is 5000 kg/day, and subtracts the portion converted into new biomass cells, represented by 300 kg/day. The remaining mass is divided by sixty-four grams per mole to find the number of moles of methane generated. This molar quantity is then multiplied by the ideal gas constant, the absolute temperature of 308.15 K, and divided by the pressure of 101.325 kPa to yield the final gas volume flow of 0.020917 m^3/s.

**Minimum (washout) solids retention time, `enveng_minimum_srt`**

The washout SRT of a complete-mix biological reactor: below it the organisms are lost faster than they grow on the influent substrate.

$$\frac{1}{\theta_{c,min}} = \frac{Y k S_0}{K_s + S_0} - k_d$$

| symbol | | meaning | SI unit |
|---|---|---|---|
| `Y` | in | true yield coefficient, kg VSS per kg BOD | , |
| `k` | in | maximum specific substrate utilisation rate | 1/s |
| `Ks` | in | half-saturation constant | kg/m^3 |
| `kd` | in | endogenous decay coefficient | 1/s |
| `S0` | in | influent substrate | kg/m^3 |
| `SRT_min` | out | minimum SRT | s |
| `SRT_min_days` | out | minimum SRT in days | , |

**Applies when:** Y k S0/(Ks + S0) > kd.

```
$ environmental-engineering solve enveng_minimum_srt Y=0.5 "k=5 1/day" "Ks=60 mg/L" "kd=0.06
1/day" "S0=250 mg/L"
SRT_min = 44168.86544 s
SRT_min_days = 0.5112137203
  by: Minimum (washout) solids retention time  (1/SRT_min = Y k S0/(Ks + S0) - kd)
  Y = 0.5
  k = 5.787037037e-05 1/s
  Ks = 0.06 kg/m^3
  kd = 6.944444444e-07 1/s
  S0 = 0.25 kg/m^3
  assumes: Y k S0/(Ks + S0) > kd
```

**What the worked example means:** This calculation determines the minimum solids retention time, or washout SRT, for a complete-mix biological reactor. It uses the true yield coefficient (0.5), maximum specific substrate utilisation rate (5 1/day), half-saturation constant (60 mg/L), endogenous decay coefficient (0.06 1/day), and influent substrate concentration (250 mg/L). The result indicates that if the actual solids retention time drops below 44168.9 s, microorganisms will be washed out of the system faster than they can reproduce, leading to treatment failure.

### Monod specific growth rate, `enveng_monod_growth_rate`

The specific growth rate of microorganisms at a substrate concentration by the Monod equation, and the net rate when the endogenous decay coefficient is given.

$$\mu = \frac{\mu_{max} S}{K_s + S}$$

| symbol | | meaning | SI unit |
|---|---|---|---|
| `mu_max` | in | maximum specific growth rate | 1/s |
| `Ks` | in | half-saturation constant | kg/m^3 |
| `S` | in | substrate concentration | kg/m^3 |
| `kd` | in | endogenous decay coefficient (optional) | 1/s |
| `mu` | out | specific growth rate | 1/s |
| `mu_per_day` | out | in 1/d | , |
| `mu_net` | out | net specific growth rate mu - kd | 1/s |
| `mu_net_per_day` | out | net rate in 1/d | , |

**Applies when:** Monod kinetics.

```
$ environmental-engineering solve enveng_monod_growth_rate "mu_max=3 1/day" "Ks=60 mg/L" "S=10
mg/L"
mu = 4.96031746e-06 1/s
mu_per_day = 0.4285714286
  by: Monod specific growth rate  (mu = mu_max S/(Ks + S); mu_net = mu - kd)
  mu_max = 3.472222222e-05 1/s
  Ks = 0.06 kg/m^3
  S = 0.01 kg/m^3
  assumes: Monod kinetics
```

In the worked example, mu_max = 3 1/day, Ks = 60 mg/L, S = 10 mg/L go through the equation above and give mu = 4.96031746e-06 1/s, mu_per_day = 0.4285714286. It applies only when Monod kinetics: outside that it says what is wrong instead of returning a number.

### Overall oxygen transfer coefficient KLa from two DO readings, `enveng_kla_two_point`

The volumetric mass-transfer coefficient KLa from two dissolved-oxygen readings during reaeration toward a known saturation concentration.

$$K_L a = \frac{\ln \frac{C_s - C_1}{C_s - C_2}}{t_2 - t_1}$$

| symbol | | meaning | SI unit |
|---|---|---|---|
| `Cs` | in | saturation DO | kg/m^3 |
| `C1` | in | DO at the first time | kg/m^3 |
| `C2` | in | DO at the second time | kg/m^3 |
| `t1` | in | first time | s |
| `t2` | in | second time | s |
| `KLa` | out | KLa | 1/s |
| `KLa_per_h` | out | KLa in 1/h | , |

**Applies when:** C1 < C2 < Cs; t2 > t1.

```
$ environmental-engineering solve enveng_kla_two_point "Cs=9.1 mg/L" "C1=2 mg/L" "C2=6 mg/L"
"t1=2 min" "t2=10 min"
KLa = 0.001726443068 1/s
KLa_per_h = 6.215195044
  by: Overall oxygen transfer coefficient KLa from two DO readings  (KLa = ln((Cs - C1)/(Cs -
C2))/(t2 - t1))
  Cs = 0.0091 kg/m^3
  C1 = 0.002 kg/m^3
  C2 = 0.006 kg/m^3
  t1 = 120 s
  t2 = 600 s
  assumes: C1 < C2 < Cs; t2 > t1
```

**What the worked example means:** This calculation determines the volumetric mass-transfer coefficient, KLa, based on two dissolved oxygen readings taken during a reaeration process. Using the saturation concentration of 9.1 mg/L, the initial reading of 2 mg/L at time 2 min, and the subsequent reading of 6 mg/L at time 10 min, the model computes the rate of oxygen uptake. The resulting coefficient of 0.00172644 1/s indicates how quickly the water approaches equilibrium with the atmosphere under these specific conditions.

**Oxygen requirement of an activated-sludge process, `enveng_aeration_oxygen_requirement`**

The carbonaceous oxygen requirement: ultimate BOD removed (BOD5 removed divided by the BOD5/BODu ratio) less 1.42 times the biomass wasted.

$$\dot{m}_{O_2} = \frac{Q(S_0 - S)}{f} - 1.42 P_x$$

| symbol | | meaning | SI unit |
|---|---|---|---|
| `Q` | in | flow | m^3/s |
| `S0` | in | influent BOD5 | kg/m^3 |
| `S` | in | effluent soluble BOD5 | kg/m^3 |
| `f` | in | BOD5/BODu ratio (e.g. 0.68) | , |
| `Px` | in | biomass wasted (VSS) | kg/s |
| `O2` | out | oxygen required | kg/s |
| `O2_kg_per_day` | out | in kg O2/d | , |

**Applies when:** carbonaceous demand only (no nitrification).

```
$ environmental-engineering solve enveng_aeration_oxygen_requirement "Q=10000 m^3/day" "S0=250
mg/L" "S=5 mg/L" f=0.68 "Px=800 kg/day"
O2 = 0.02855255991 kg/s
O2_kg_per_day = 2466.941176
  by: Oxygen requirement of an activated-sludge process  (O2 = Q (S0 - S)/f - 1.42 Px)
  Q = 0.1157407407 m^3/s
  S0 = 0.25 kg/m^3
  S = 0.005 kg/m^3
  f = 0.68
  Px = 0.009259259259 kg/s
  assumes: carbonaceous demand only (no nitrification)
```

**What the worked example means:** This calculation determines the oxygen needed for the activated-sludge process by subtracting the oxygen equivalent of wasted biomass from the ultimate biochemical oxygen demand removed. Using a flow of 10000 m^3/day, influent BOD5 of 250 mg/L, and effluent soluble BOD5 of 5 mg/L, divided by the ratio 0.68, and subtracting 1.42 times the biomass wasted of 800 kg/day, the result is an oxygen requirement of 0.0285526 kg/s. This value represents the daily load as 2466.94, assuming only carbonaceous demand is considered.

**Return activated sludge ratio, `enveng_return_sludge_ratio`**

The return sludge ratio and flow needed to hold the mixed-liquor concentration, from a solids balance around the aeration tank (influent solids neglected).

$$R = \frac{Q_r}{Q} = \frac{X}{X_r - X}$$

| symbol | | meaning | SI unit |
|---|---|---|---|
| `X` | in | mixed liquor suspended solids | kg/m^3 |
| `Xr` | in | return sludge concentration | kg/m^3 |
| `Q` | in | influent flow | m^3/s |
| `R` | out | return ratio Qr/Q | , |
| `Qr` | out | return sludge flow | m^3/s |

**Applies when:** Xr > X; influent solids negligible.

```
$ environmental-engineering solve enveng_return_sludge_ratio "X=3000 mg/L" "Xr=10000 mg/L"
"Q=10000 m^3/day"
R = 0.4285714286
Qr = 0.0496031746 m^3/s
  by: Return activated sludge ratio  ((Q + Qr) X = Qr Xr, so R = X/(Xr - X))
  X = 3 kg/m^3
  Xr = 10 kg/m^3
  Q = 0.1157407407 m^3/s
  assumes: Xr > X; influent solids negligible
```

**What the worked example means:** This calculation determines the return activated sludge ratio and the corresponding return sludge flow rate needed to maintain the desired mixed liquor suspended solids concentration in the aeration tank. Using a solids balance around the tank where influent solids are neglected, the ratio is derived from the relationship between the mixed liquor concentration and the return sludge concentration. The resulting ratio of 0.428571 indicates how much return flow is needed relative to the influent flow, while the absolute

return flow of 0.0496032 m^3/s represents the volumetric rate at which concentrated sludge must be pumped back into the system.

**Sludge volume after thickening or dewatering, `enveng_sludge_thickening`**

The sludge volume after thickening or dewatering from one solids fraction to another (same specific gravity), and the fractional volume reduction.

$$V_2 = V_1 \frac{P_1}{P_2}$$

| symbol | | meaning | SI unit |
|---|---|---|---|
| `V1` | in | sludge volume rate before | m^3/s |
| `P1` | in | solids fraction before (fraction or %) | , |
| `P2` | in | solids fraction after (fraction or %) | , |
| `V2` | out | sludge volume rate after | m^3/s |
| `reduction` | out | fractional volume reduction | , |

**Applies when:** P2 > P1; same specific gravity.

```
$ environmental-engineering solve enveng_sludge_thickening "V1=100 m^3/day" P1=0.01 P2=0.05
V2 = 0.0002314814815 m^3/s
reduction = 0.8
  by: Sludge volume after thickening or dewatering  (V2 = V1 P1/P2)
  V1 = 0.001157407407 m^3/s
  P1 = 0.01
  P2 = 0.05
  assumes: P2 > P1; same specific gravity
```

**What the worked example means:** This calculation determines the new sludge volume rate after thickening or dewatering. By applying the solids balance principle, the initial volume rate of 100 m^3/day is adjusted for the change in solids fraction from 0.01 to 0.05. The resulting volume rate is 0.000231481 m^3/s, which indicates how much space the sludge occupies after processing. Additionally, the fractional volume reduction of 0.8 quantifies the efficiency of the process in shrinking the liquid waste stream.

**Sludge volume index (SVI), `enveng_sludge_volume_index`**

The sludge volume index from the 30-minute settled volume and the mixed-liquor suspended solids, with the customary estimate of the return sludge concentration (10^6/SVI mg/L).

$$SVI = \frac{V_{30}}{MLSS}$$

| symbol | | meaning | SI unit |
|---|---|---|---|
| `V30` | in | settled sludge volume after 30 minutes, mL/L | , |
| `MLSS` | in | mixed liquor suspended solids | kg/m^3 |
| `SVI` | out | sludge volume index, mL/g | , |
| `Xr_estimate` | out | estimated maximum return sludge concentration | kg/m^3 |
| `Xr_estimate_mgL` | out | estimate in mg/L | , |

**Applies when:** V30 <= 1000 mL/L.

```
$ environmental-engineering solve enveng_sludge_volume_index V30=300 "MLSS=2500 mg/L"
SVI = 120
Xr_estimate = 8.333333333 kg/m^3
Xr_estimate_mgL = 8333.333333
  by: Sludge volume index (SVI)  (SVI = V30 (mL/L)/MLSS (g/L))
  V30 = 300
  MLSS = 2.5 kg/m^3
  assumes: V30 <= 1000 mL/L
```

**What the worked example means:** This calculation determines the sludge volume index by dividing the thirty minute settled volume, given as 300, by the mixed liquor suspended solids concentration, provided as 2500 mg/L. The resulting value of 120 represents the volume in milliliters occupied by one gram of dried sludge after settling. This metric helps operators assess how well the biological floc settles and compacts within the clarifier, which is essential for maintaining proper return activated sludge concentrations.

**Solids retention time (mean cell residence time),** `enveng_solids_retention_time`

The solids retention time (sludge age, mean cell residence time) of an activated-sludge system from the biomass in the tank and the biomass leaving in the waste sludge and the effluent; the effluent solids are neglected (Xe = 0) when not given.

$$\theta_c = \frac{VX}{Q_w X_w + Q_e X_e}$$

| symbol | | meaning | SI unit |
|---|---|---|---|
| `V` | in | aeration tank volume | m^3 |
| `X` | in | MLVSS in the tank | kg/m^3 |
| `Qw` | in | waste sludge flow | m^3/s |
| `Xw` | in | VSS in the waste sludge | kg/m^3 |
| `Qe` | in | effluent flow (needed only with Xe) (optional) | m^3/s |
| `Xe` | in | VSS in the effluent (default 0: effluent solids neglected) (optional) | kg/m^3 |
| `SRT` | out | solids retention time | s |
| `SRT_days` | out | SRT in days | , |

**Applies when:** steady state; Xe = 0 (effluent solids neglected) unless given; Xe > 0 needs Qe.

```
$ environmental-engineering solve enveng_solids_retention_time "V=3000 m^3" "X=2500 mg/L"
"Qw=100 m^3/day" "Xw=10000 mg/L" "Qe=9900 m^3/day" "Xe=15 mg/L"
SRT = 564214.1924 s
SRT_days = 6.530256857
  by: Solids retention time (mean cell residence time)  (SRT = V X/(Qw Xw + Qe Xe))
  V = 3000 m^3
  X = 2.5 kg/m^3
  Qw = 0.001157407407 m^3/s
  Xw = 10 kg/m^3
  Qe = 0.1145833333 m^3/s
  Xe = 0.015 kg/m^3
  assumes: steady state; Xe = 0 (effluent solids neglected) unless given; Xe > 0 needs Qe
```

**What the worked example means:** This calculation determines the mean cell residence time, or sludge age, for an activated-sludge system. It uses the aeration tank volume of 3000 m^3 and the mixed liquor volatile suspended solids concentration of 2500 mg/L to find the total biomass in the reactor. This value is divided by the sum of biomass leaving via waste sludge flow 100 m^3/day at concentration 10000 mg/L and effluent flow 9900 m^3/day at concentration 15 mg/L. The resulting solids retention time of 564214 s indicates how long microorganisms remain in the system on average, which governs biological treatment efficiency and sludge production rates.

**Trickling filter BOD removal efficiency (NRC equation), `enveng_nrc_trickling_filter`**

The BOD removal efficiency of a single-stage stone-media trickling filter (with its settling tank) by the NRC equation, from the BOD load, the filter volume and the recirculation ratio.

$$E = \frac{100}{1 + 0.4432\sqrt{W/(VF)}}, \quad F = \frac{1 + R}{(1 + R/10)^2}$$

| symbol | | meaning | SI unit |
|---|---|---|---|
| `W` | in | BOD load to the filter (excluding recycle) | kg/s |
| `V` | in | filter media volume | m^3 |
| `R` | in | recirculation ratio Qr/Q | , |
| `F` | out | recirculation factor (1 + R)/(1 + R/10)^2 | , |
| `E` | out | BOD removal efficiency (fraction) | , |
| `E_percent` | out | efficiency in per cent | , |

**Applies when:** first-stage stone filter at 20 degC; W in kg/d and V in m^3 inside the formula constant.

```
$ environmental-engineering solve enveng_nrc_trickling_filter "W=800 kg/day" "V=1000 m^3" R=2
F = 2.083333333
E = 0.7845346289
E_percent = 78.45346289
  by: Trickling filter BOD removal efficiency (NRC equation)  (F = (1 + R)/(1 + R/10)^2; E =
1/(1 + 0.4432 sqrt(W/(V F))))
  W = 0.009259259259 kg/s
  V = 1000 m^3
  R = 2
  assumes: first-stage stone filter at 20 degC; W in kg/d and V in m^3 inside the formula
constant
```

**What the worked example means:** This calculation determines the BOD removal efficiency for a single-stage stone-media trickling filter using the National Research Council equation. It uses the BOD load of 800 kg/day, the filter media volume of 1000 m^3, and the recirculation ratio of 2 to compute a recirculation factor of 2.08333. The resulting efficiency is 0.784535 as a fraction, or 78.4535 when expressed as a percentage, indicating how effectively the system removes organic matter under these specific operating conditions.

**Volume of wet sludge from dry solids, `enveng_sludge_volume`**

The volume (flow) of wet sludge from the dry-solids mass rate, the solids fraction and the specific gravity of the sludge (water 1000 kg/m^3).

$$V = \frac{M_s}{\rho_w S_{sl} P_s}$$

| symbol | | meaning | SI unit |
|---|---|---|---|
| `Ms` | in | dry solids mass rate | kg/s |
| `Ps` | in | solids fraction of the wet sludge (fraction or %) | , |
| `Ssl` | in | specific gravity of the wet sludge | , |
| `V` | out | wet sludge volume rate | m^3/s |
| `V_m3_per_day` | out | in m^3/d | , |

**Applies when:** 0 < Ps <= 1.

```
$ environmental-engineering solve enveng_sludge_volume "Ms=1000 kg/day" Ps=0.04 Ssl=1.02
V = 0.0002836782861 m^3/s
V_m3_per_day = 24.50980392
  by: Volume of wet sludge from dry solids  (V = Ms/(rho_w Ssl Ps))
  Ms = 0.01157407407 kg/s
  Ps = 0.04
  Ssl = 1.02
  assumes: 0 < Ps <= 1
```

**What the worked example means:** This calculation determines the volumetric flow rate of wet sludge based on the mass rate of dry solids, the fraction of solids in the mixture, and the specific gravity of the sludge. The result, 0.000283678 m^3/s, represents the volume of wet sludge produced each second. This value is also expressed as 24.5098 cubic meters per day for operational planning. The calculation assumes that the density of water is exactly one thousand kilograms per cubic meter and that the specific gravity provided accurately reflects the bulk density of the wet sludge relative to water.

### Waste biomass production (observed yield), `enveng_sludge_production`

The observed yield and the daily mass of biological solids produced (to be wasted) from the kinetic yield, decay coefficient, SRT, flow and substrate removed.

$$P_x = \frac{Y}{1 + k_d \theta_c} Q(S_0 - S)$$

| symbol | | meaning | SI unit |
|---|---|---|---|
| `Y` | in | true yield, kg VSS per kg BOD | , |
| `kd` | in | endogenous decay coefficient | 1/s |
| `SRT` | in | solids retention time | s |
| `Q` | in | flow | m^3/s |
| `S0` | in | influent substrate | kg/m^3 |
| `S` | in | effluent substrate | kg/m^3 |
| `Y_obs` | out | observed yield | , |
| `Px` | out | biomass production | kg/s |
| `Px_kg_per_day` | out | in kg VSS/d | , |

**Applies when:** S < S0.

```
$ environmental-engineering solve enveng_sludge_production Y=0.5 "kd=0.06 1/day" "SRT=10 day"
"Q=10000 m^3/day" "S0=250 mg/L" "S=5 mg/L"
Y_obs = 0.3125
Px = 0.008861400463 kg/s
Px_kg_per_day = 765.625
  by: Waste biomass production (observed yield)  (Y_obs = Y/(1 + kd SRT); Px = Y_obs Q (S0 -
S))
  Y = 0.5
  kd = 6.944444444e-07 1/s
  SRT = 864000 s
  Q = 0.1157407407 m^3/s
  S0 = 0.25 kg/m^3
  S = 0.005 kg/m^3
  assumes: S < S0
```

**What the worked example means:** This calculation determines the daily mass of biological solids produced and the observed yield based on kinetic parameters. The observed yield, calculated as 0.3125, reflects the true yield adjusted for endogenous decay over the solids retention time. Using this yield with the flow rate and substrate removal, the model computes the total biomass production at 0.0088614 kg/s. This value represents the mass of volatile suspended solids generated per second, which converts to 765.625 when expressed on a daily basis. Understanding these outputs helps engineers size waste sludge handling systems accurately.

## Water quality (21)

### BOD exerted after a time (first-order), `enveng_bod_exerted`

The BOD exerted after time t and the oxygen demand remaining, from the ultimate BOD and the first-order rate constant.

$$BOD_t = L_0 \left(1 - e^{-kt}\right)$$

| symbol | | meaning | SI unit |
|---|---|---|---|
| `L0` | in | ultimate (carbonaceous) BOD | kg/m^3 |
| `k` | in | BOD rate constant (base e) | 1/s |
| `t` | in | incubation or travel time | s |
| `BOD_t` | out | BOD exerted after t | kg/m^3 |
| `L_t` | out | BOD remaining after t | kg/m^3 |
| `BOD_t_mgL` | out | BOD exerted in mg/L | , |

**Applies when:** first-order kinetics.

```
$ environmental-engineering solve enveng_bod_exerted "L0=250 mg/L" "k=0.23 1/day" "t=5 day"
BOD_t = 0.1708408077 kg/m^3
L_t = 0.07915919234 kg/m^3
BOD_t_mgL = 170.8408077
  by: BOD exerted after a time (first-order)  (BOD_t = L0 (1 - e^(-k t)), L_t = L0 e^(-k t))
  L0 = 0.25 kg/m^3
  k = 2.662037037e-06 1/s
  t = 432000 s
  assumes: first-order kinetics
```

**What the worked example means:** This calculation determines the biochemical oxygen demand exerted over a specific period and the amount remaining, based on first-order kinetics. Using an ultimate BOD of 250 mg/L and a rate constant of 0.23 1/day for a duration of 5 day, the model computes that 0.170841 kg/m^3 has been consumed. Consequently, the residual oxygen demand is 0.0791592 kg/m^3. This result indicates how much organic matter has been oxidized by microorganisms during the incubation or travel time.

**BOD from a dilution test (with optional seed correction),** `enveng_bod_dilution_test`

The BOD of a sample from the dissolved oxygen of the diluted sample before and after incubation and the sample fraction, with the seed correction when the dilution water is seeded.

$$BOD = \frac{(D_1 - D_2) - (B_1 - B_2)f}{P}$$

| symbol | | meaning | SI unit |
|---|---|---|---|
| `D1` | in | DO of the diluted sample immediately after preparation | kg/m^3 |
| `D2` | in | DO of the diluted sample after incubation | kg/m^3 |
| `P` | in | decimal fraction of sample in the bottle (sample volume / bottle volume) | , |
| `B1` | in | DO of the seed control before incubation (optional) | kg/m^3 |
| `B2` | in | DO of the seed control after incubation (optional) | kg/m^3 |
| `f` | in | ratio of seed in the sample bottle to seed in the control (optional) | , |
| `BOD` | out | BOD of the sample | kg/m^3 |
| `BOD_mgL` | out | BOD in mg/L | , |

**Applies when:** D1 > D2; B1, B2 and f given together or not at all.

```
$ environmental-engineering solve enveng_bod_dilution_test "D1=8.5 mg/L" "D2=4 mg/L" P=0.05
BOD = 0.09 kg/m^3
BOD_mgL = 90
  by: BOD from a dilution test (with optional seed correction)  (BOD = ((D1 - D2) - (B1 - B2)
f)/P (unseeded: BOD = (D1 - D2)/P))
  D1 = 0.0085 kg/m^3
  D2 = 0.004 kg/m^3
  P = 0.05
  assumes: D1 > D2; B1, B2 and f given together or not at all
```

**What the worked example means:** This calculation determines the Biochemical Oxygen Demand of a water sample using dilution data. It subtracts the oxygen depletion of the seed control from the depletion observed in the diluted sample, then divides by the fraction of the sample present. The result, expressed as 0.09 kg/m^3, represents the amount of oxygen consumed by microorganisms to decompose organic matter in the sample over the incubation period.

**BOD rate constant and ultimate BOD fitted to BOD data,** `enveng_bod_rate_fit`

The first-order rate constant k and ultimate BOD L0 that best fit a series of BOD measurements (nonlinear least squares of BOD = L0 (1 - e^(-k t))).

$$\min_{k, L_0} \sum_i \left[ y_i - L_0 \left(1 - e^{-k t_i}\right) \right]^2$$

| symbol | | meaning | SI unit |
|---|---|---|---|
| `t` | in | incubation times in days | , |
| `y` | in | measured BOD values in mg/L (same order) | , |
| `k` | out | fitted rate constant | 1/s |
| `k_per_day` | out | fitted rate constant in 1/d | , |
| `L0` | out | fitted ultimate BOD | kg/m^3 |
| `L0_mgL` | out | fitted ultimate BOD in mg/L | , |
| `rss` | out | residual sum of squares in (mg/L)^2 | , |

**Applies when:** at least three distinct positive times; a single rate constant between 0.001 and 20 per day fits.

```
$ environmental-engineering solve enveng_bod_rate_fit "t=[1, 2, 3, 5, 7, 10]" "y=[41, 74, 98,
132, 152, 168]"
k = 3.002440202e-06 1/s
k_per_day = 0.2594108335
L0 = 0.1815717527 kg/m^3
L0_mgL = 181.5717527
rss = 0.5327137697
  by: BOD rate constant and ultimate BOD fitted to BOD data  (for each k the best L0 is sum(y
f)/sum(f^2) with f = 1 - e^(-k t); k is the root of the stationarity condition sum((y - L0 f) t
e^(-k t)) = 0 (scan and bisection))
  assumes: at least three distinct positive times; a single rate constant between 0.001 and 20
per day fits
```

**What the worked example means:** This calculation determines the best-fit parameters for the first-order biochemical oxygen demand model using nonlinear least squares. By minimizing the squared differences between measured values and the theoretical curve, it finds a single rate constant of 0.259411 per day and an ultimate BOD concentration of 181.572 mg/L that most accurately describe the decomposition process observed over time.

**Concentration after mixing a discharge with a river, `enveng_mixing_two_streams`**

The fully mixed concentration (BOD, DO, a pollutant) and flow just below a wastewater outfall from the flows and concentrations of the discharge and the river.

$$C = \frac{Q_w C_w + Q_r C_r}{Q_w + Q_r}$$

| symbol | | meaning | SI unit |
|---|---|---|---|
| `Qw` | in | wastewater (discharge) flow | m^3/s |
| `Cw` | in | concentration in the wastewater | kg/m^3 |
| `Qr` | in | river flow upstream | m^3/s |
| `Cr` | in | concentration in the river upstream | kg/m^3 |
| `C` | out | mixed concentration | kg/m^3 |
| `C_mgL` | out | in mg/L | , |
| `Q` | out | combined flow | m^3/s |

**Applies when:** complete mixing; conservative over the mixing zone.

```
$ environmental-engineering solve enveng_mixing_two_streams "Qw=0.5 m^3/s" "Cw=200 mg/L" "Qr=5
m^3/s" "Cr=2 mg/L"
C = 0.02 kg/m^3
C_mgL = 20
Q = 5.5 m^3/s
  by: Concentration after mixing a discharge with a river  (C = (Qw Cw + Qr Cr)/(Qw + Qr))
  Qw = 0.5 m^3/s
  Cw = 0.2 kg/m^3
  Qr = 5 m^3/s
  Cr = 0.002 kg/m^3
  assumes: complete mixing; conservative over the mixing zone
```

**What the worked example means:** This calculation determines the fully mixed concentration downstream of a wastewater outfall by combining the discharge and river flows. Using the wastewater flow of 0.5 m^3/s at concentration 200 mg/L and the upstream river flow of 5 m^3/s at concentration 2 mg/L, the resulting mixture has a concentration of 0.02 kg/m^3 and a total flow of 5.5 m^3/s. This value represents the immediate impact on water quality assuming complete mixing occurs.

**Concentration in a well-mixed lake over time, `enveng_lake_transient`**

The concentration in a completely mixed lake a time t after a change in load, from the initial concentration, the load, outflow, volume and first-order decay.

$$C(t) = C_{ss} + (C_0 - C_{ss})e^{-(Q/V+k)t}$$

| symbol | | meaning | SI unit |
|---|---|---|---|
| `W` | in | pollutant load after the change | kg/s |
| `Q` | in | outflow rate | m^3/s |
| `k` | in | first-order decay constant | 1/s |
| `V` | in | lake volume | m^3 |
| `C0` | in | concentration at t = 0 | kg/m^3 |
| `t` | in | elapsed time | s |
| `C` | out | concentration at t | kg/m^3 |
| `C_mgL` | out | in mg/L | , |
| `C_ss` | out | new steady state | kg/m^3 |

**Applies when:** completely mixed; constant flow, volume and load.

```
$ environmental-engineering solve enveng_lake_transient "W=100 kg/day" "Q=10 m^3/s" "k=0.2
1/day" "V=10000000 m^3" "C0=0 mg/L" "t=5 day"
C = 2.657713748e-05 kg/m^3
C_mgL = 0.02657713748
C_ss = 3.491620112e-05 kg/m^3
  by: Concentration in a well-mixed lake over time  (V dC/dt = W - Q C - k C V, so C = Css +
(C0 - Css) e^(-(Q/V + k) t))
  W = 0.001157407407 kg/s
  Q = 10 m^3/s
  k = 2.314814815e-06 1/s
  V = 10000000 m^3
  C0 = 0 kg/m^3
  t = 432000 s
  assumes: completely mixed; constant flow, volume and load
```

**What the worked example means:** This calculation determines the pollutant concentration in a completely mixed lake after a change in load. It uses the initial concentration 0 mg/L, the new load 100 kg/day, outflow rate 10 m^3/s, decay constant 0.2 1/day, and volume 1e+07 m^3. The resulting concentration at elapsed time 5 day is 2.65771e-05 kg/m^3, which corresponds to 0.0265771. The long-term equilibrium value, or new steady state, is calculated as 3.49162e-05 kg/m^3.

**Critical (minimum) dissolved oxygen of the oxygen sag, `enveng_streeter_phelps_critical_point`**

The critical time at which the oxygen deficit of a river is largest, the critical deficit and the minimum dissolved oxygen (Streeter-Phelps).

$$t_c = \frac{1}{k_a - k_d} \ln\left[\frac{k_a}{k_d}\left(1 - \frac{D_0(k_a - k_d)}{k_d L_0}\right)\right], \quad D_c = \frac{k_d}{k_a} L_0 e^{-k_d t_c}$$

| symbol | | meaning | SI unit |
|---|---|---|---|
| `kd` | in | deoxygenation rate constant | 1/s |
| `ka` | in | reaeration rate constant | 1/s |
| `L0` | in | ultimate BOD of the mixed river water at the outfall | kg/m^3 |
| `D0` | in | oxygen deficit at the outfall (DOsat - DO) | kg/m^3 |
| `DOsat` | in | saturation dissolved oxygen | kg/m^3 |
| `t_c` | out | critical travel time | s |
| `t_c_days` | out | critical time in days | , |
| `D_c` | out | critical (largest) deficit | kg/m^3 |
| `DO_min` | out | minimum dissolved oxygen | kg/m^3 |
| `DO_min_mgL` | out | minimum dissolved oxygen in mg/L | , |

**Applies when:** the critical deficit stays below saturation; if the deficit falls from the outfall on, t_c = 0 and D_c = D0.

```
$ environmental-engineering solve enveng_streeter_phelps_critical_point "kd=0.23 1/day"
"ka=0.46 1/day" "L0=10 mg/L" "D0=1 mg/L" "DOsat=9.1 mg/L"
t_c = 220803.3385 s
t_c_days = 2.555594195
D_c = 0.002777777778 kg/m^3
DO_min = 0.006322222222 kg/m^3
DO_min_mgL = 6.322222222
  by: Critical (minimum) dissolved oxygen of the oxygen sag  (t_c = ln[(ka/kd)(1 - D0 (ka -
kd)/(kd L0))]/(ka - kd); D_c = (kd/ka) L0 e^(-kd t_c))
  kd = 2.662037037e-06 1/s
  ka = 5.324074074e-06 1/s
  L0 = 0.01 kg/m^3
  D0 = 0.001 kg/m^3
  DOsat = 0.0091 kg/m^3
  assumes: the critical deficit stays below saturation; if the deficit falls from the outfall
on, t_c = 0 and D_c = D0
```

**What the worked example means:** This calculation determines the worst-case water quality conditions downstream of an outfall using the Streeter-Phelps model. It computes the critical travel time 2.55559 when the oxygen deficit reaches its maximum value of 0.00277778 kg/m^3. Consequently, the dissolved oxygen level drops to its lowest point, recorded here as 6.32222. This result helps engineers assess whether the river can recover from the pollution load or if further treatment is needed.

**Dissolved oxygen at a distance downstream (Streeter-Phelps), `enveng_oxygen_sag_distance`**

The oxygen deficit and dissolved oxygen at a distance downstream of an outfall from the river velocity (travel time x/v) and the Streeter-Phelps sag.

$$D(x) = \frac{k_d L_0}{k_a - k_d}\left(e^{-k_d x/v} - e^{-k_a x/v}\right) + D_0 e^{-k_a x/v}$$

| symbol | | meaning | SI unit |
|---|---|---|---|
| `kd` | in | deoxygenation rate constant | 1/s |
| `ka` | in | reaeration rate constant | 1/s |
| `L0` | in | ultimate BOD of the mixed river water at the outfall | kg/m^3 |
| `D0` | in | oxygen deficit at the outfall (DOsat - DO) | kg/m^3 |
| `x` | in | distance downstream | m |
| `v` | in | mean river velocity | m/s |
| `DOsat` | in | saturation dissolved oxygen | kg/m^3 |
| `D` | out | oxygen deficit at x | kg/m^3 |
| `DO` | out | dissolved oxygen at x | kg/m^3 |
| `D_mgL` | out | deficit in mg/L | , |
| `DO_mgL` | out | dissolved oxygen in mg/L | , |
| `t` | out | travel time | s |

**Applies when:** plug flow at constant velocity; deficit stays below saturation.

```
$ environmental-engineering solve enveng_oxygen_sag_distance "kd=0.23 1/day" "ka=0.46 1/day"
"L0=10 mg/L" "D0=1 mg/L" "x=30 km" "v=0.3 m/s" "DOsat=9.1 mg/L"
D = 0.002378123211 kg/m^3
DO = 0.006721876789 kg/m^3
D_mgL = 2.378123211
DO_mgL = 6.721876789
t = 100000 s
  by: Dissolved oxygen at a distance downstream (Streeter-Phelps)  (t = x/v, then the Streeter-
Phelps deficit)
  kd = 2.662037037e-06 1/s
  ka = 5.324074074e-06 1/s
  L0 = 0.01 kg/m^3
  D0 = 0.001 kg/m^3
  x = 30000 m
  v = 0.3 m/s
  DOsat = 0.0091 kg/m^3
  assumes: plug flow at constant velocity; deficit stays below saturation
```

**What the worked example means:** The model calculates the oxygen deficit at a specific distance downstream using the Streeter-Phelps equation. It considers the travel time derived from river velocity and distance, along with deoxygenation and reaeration rates. The resulting deficit is 2.37812, which corresponds to a dissolved oxygen concentration of 6.72188. This value indicates how much oxygen has been consumed by biological activity relative to the saturation level at that point in the river.

**Dissolved oxygen saturation concentration (Benson-Krause), `enveng_do_saturation`**

The saturation concentration of dissolved oxygen in water in equilibrium with moist air at 1 atm, from the temperature and salinity (Benson and Krause equation as adopted in Standard Methods).

$$\ln C_s = -139.34411 + \frac{1.575701 \times 10^5}{T} - \frac{6.642308 \times 10^7}{T^2} + \frac{1.2438 \times 10^{10}}{T^3} - \frac{8.621949 \times 10^{11}}{T^4} - S$$

| symbol | | meaning | SI unit |
|---|---|---|---|
| `T` | in | water temperature | K |
| `S` | in | salinity in g/kg (0 for fresh water) | , |
| `DOsat` | out | saturation dissolved oxygen | kg/m^3 |
| `DOsat_mgL` | out | in mg/L | , |

**Applies when:** 0-40 degC; 0-40 g/kg salinity; 1 atm moist air.

```
$ environmental-engineering solve enveng_do_saturation "T=20 degC" S=0
DOsat = 0.009092426043 kg/m^3
DOsat_mgL = 9.092426043
  by: Dissolved oxygen saturation concentration (Benson-Krause)  (ln C = -139.34411 +
1.575701e5/T - 6.642308e7/T^2 + 1.2438e10/T^3 - 8.621949e11/T^4 - S(1.7674e-2 - 10.754/T +
2140.7/T^2))
  T = 293.15 K
  S = 0
  assumes: 0-40 degC; 0-40 g/kg salinity; 1 atm moist air
```

**What the worked example means:** This calculation determines the maximum amount of oxygen that water can hold when in equilibrium with moist air at one atmosphere. It uses the Benson-Krause equation, which depends on

the absolute temperature and salinity of the water. For the given conditions, the resulting saturation level is 0.00909243 kg/m^3, which corresponds to 9.09243. This value represents the upper limit for dissolved oxygen availability in this specific aquatic environment.

**First-order decay of a pollutant downstream (plug flow), `enveng_river_first_order_decay`**

The concentration of a decaying pollutant (BOD, coliforms, a chemical) a distance downstream in a river in plug flow, and the travel time.

$$C = C_0\, e^{-kx/u}$$

| symbol | | meaning | SI unit |
|---|---|---|---|
| `C0` | in | concentration at the start (after mixing) | kg/m^3 |
| `k` | in | first-order decay constant | 1/s |
| `x` | in | distance downstream | m |
| `u` | in | mean velocity | m/s |
| `C` | out | concentration at x | kg/m^3 |
| `C_mgL` | out | in mg/L | , |
| `t` | out | travel time x/u | s |
| `fraction_remaining` | out | C/C0 | , |

**Applies when:** plug flow; first-order decay.

```
$ environmental-engineering solve enveng_river_first_order_decay "C0=20 mg/L" "k=0.3 1/day"
"x=40 km" "u=0.25 m/s"
C = 0.01147506841 kg/m^3
C_mgL = 11.47506841
t = 160000 s
fraction_remaining = 0.5737534207
  by: First-order decay of a pollutant downstream (plug flow)  (C = C0 e^(-k x/u))
  C0 = 0.02 kg/m^3
  k = 3.472222222e-06 1/s
  x = 40000 m
  u = 0.25 m/s
  assumes: plug flow; first-order decay
```

**What the worked example means:** This calculation determines the pollutant concentration at a specific distance downstream using first-order decay kinetics under plug flow conditions. Starting from an initial mixed concentration of 20 mg/L, the substance decays as it travels 40 km meters at a mean velocity of 0.25 m/s. The resulting concentration is 0.0114751 kg/m^3, equivalent to 11.4751 in milligrams per liter. The fraction of the original pollutant remaining after this journey is 0.573753, and the time required to cover the distance is 160000 s. This model helps estimate how much of a biodegradable or chemically reactive pollutant persists in a river segment.

**Flow-weighted mixing of several streams, `enveng_mixing_several_streams`**

The flow-weighted concentration (or temperature) of several streams combined, and the total flow; flows in any one unit, concentrations in any one unit.

$$C = \frac{\sum_i Q_i C_i}{\sum_i Q_i}$$

| symbol | | meaning | SI unit |
|---|---|---|---|
| `Q` | in | flows of the streams (any one unit, e.g. m^3/s) | , |
| `C` | in | concentrations of the streams (any one unit, e.g. mg/L) | , |
| `C_mix` | out | mixed concentration, in the unit of C | , |
| `Q_total` | out | total flow, in the unit of Q | , |

**Applies when:** complete mixing; conservative substance.

```
$ environmental-engineering solve enveng_mixing_several_streams "Q=[2, 0.5, 1]" "C=[3, 120,
40]"
C_mix = 30.28571429
Q_total = 3.5
  by: Flow-weighted mixing of several streams  (C = sum(Q_i C_i)/sum(Q_i))
  assumes: complete mixing; conservative substance
```

**What the worked example means:** This calculation determines the final concentration of a mixture by weighting each input stream's concentration by its respective flow rate. The numerator sums the product of every individual flow and concentration, while the denominator sums all flows to find the total volume moving through the system. The result, 30.2857, represents the uniform concentration achieved when these streams combine completely, assuming no chemical reactions or losses occur during the mixing process.

**Geometric mean of bacterial counts (with mean and standard deviation), `enveng_geometric_mean_counts`**

The geometric mean of a set of positive values such as coliform counts (MPN or CFU per 100 mL), the n-th root of their product, with the arithmetic mean, the sample standard deviation and the mean of the base-10 logarithms.

$$GM = \left(\prod_{i=1}^{n} x_i\right)^{1/n}, \qquad s = \sqrt{\frac{\sum (x_i - \bar{x})^2}{n-1}}$$

| symbol | | meaning | SI unit |
|---|---|---|---|
| `x` | in | positive values, e.g. coliform counts per 100 mL | , |
| `geometric_mean` | out | geometric mean (prod x)^(1/n) | , |
| `mean` | out | arithmetic mean | , |
| `sd` | out | sample standard deviation (n - 1) | , |
| `mean_log10` | out | mean of log10 x | , |
| `count` | out | number of values | , |

**Applies when:** every value > 0; at least two values.

```
$ environmental-engineering solve enveng_geometric_mean_counts "x=[10, 100, 1000]"
geometric_mean = 100
mean = 370
sd = 547.4486277
mean_log10 = 2
count = 3
  by: Geometric mean of bacterial counts (with mean and standard deviation)  (GM = exp(mean(ln
x)); mean = sum x / n; s = sqrt(sum (x - mean)^2/(n - 1)))
  assumes: every value > 0; at least two values
```

In the worked example, x = [10, 100, 1000] go through the equation above and give geometric_mean = 100, mean = 370, sd = 547.4486277, mean_log10 = 2. It applies only when every value > 0; at least two values: outside that it says what is wrong instead of returning a number.

**pH and pOH from the hydrogen or hydroxide ion concentration, `enveng_ph_poh`**

The pH, the pOH and both ion concentrations of a water sample at 25 degC from either its hydrogen ion concentration [H+] or its hydroxide ion concentration [OH-] (pH = -log10[H+], pOH = -log10[OH-], pH + pOH = 14).

$$\mathrm{pH} = -\log_{10}[\mathrm{H}^+], \quad \mathrm{pOH} = -\log_{10}[\mathrm{OH}^-], \quad \mathrm{pH} + \mathrm{pOH} = 14$$

| symbol | | meaning | SI unit |
|---|---|---|---|
| `H` | in | hydrogen ion concentration [H+] (optional) | mol/m^3 |
| `OH` | in | hydroxide ion concentration [OH-] (optional) | mol/m^3 |
| `pH` | out | pH = -log10[H+] | , |
| `pOH` | out | pOH = -log10[OH-] = 14 - pH | , |
| `H_molL` | out | hydrogen ion concentration, mol/L | , |
| `OH_molL` | out | hydroxide ion concentration, mol/L | , |

**Applies when:** 25 degC: Kw = 1e-14, pH + pOH = 14; exactly one of H, OH; 1e-14 <= concentration <= 1 mol/L (dilute: activity = concentration).

```
$ environmental-engineering solve enveng_ph_poh "OH=1e-05 mol/L"
pH = 9
pOH = 5
H_molL = 1e-09
OH_molL = 1e-05
  by: pH and pOH from the hydrogen or hydroxide ion concentration  (pH = -log10[H+]; [OH-] =
Kw/[H+]; pOH = 14 - pH (or from [OH-] the other way round))
  OH = 0.01 mol/m^3
  assumes: 25 degC: Kw = 1e-14, pH + pOH = 14; exactly one of H, OH; 1e-14 <= concentration <=
1 mol/L (dilute: activity = concentration)
```

In the worked example, OH = 1e-05 mol/L go through the equation above and give pH = 9, pOH = 5, H_molL = 1e-09, OH_molL = 1e-05. It applies only when 25 degC: Kw = 1e-14, pH + pOH = 14; exactly one of H, OH; 1e-14 <= concentration <= 1 mol/L (dilute: activity = concentration): outside that it says what is wrong instead of returning a number.

**Pollutant mass load of a flow, `enveng_pollutant_load`**

The mass load (rate) carried by a flow at a concentration, in kg/s, kg per day and tonnes per year.

$$W = Q\,C$$

| symbol | | meaning | SI unit |
|---|---|---|---|
| `Q` | in | flow rate | m^3/s |
| `C` | in | concentration | kg/m^3 |
| `W` | out | mass load | kg/s |
| `W_kg_per_day` | out | load in kg per day | , |

| symbol | | meaning | SI unit |
|---|---|---|---|
| `W_t_per_year` | out | load in tonnes per (365.25-day) year | , |

**Applies when:** Q > 0.

```
$ environmental-engineering solve enveng_pollutant_load "Q=0.5 m^3/s" "C=30 mg/L"
W = 0.015 kg/s
W_kg_per_day = 1296
W_t_per_year = 473.364
  by: Pollutant mass load of a flow  (W = Q C)
  Q = 0.5 m^3/s
  C = 0.03 kg/m^3
  assumes: Q > 0
```

**What the worked example means:** This calculation determines the pollutant mass load carried by a flow. By multiplying the flow rate 0.5 m^3/s by the concentration 30 mg/L, this gives the instantaneous load of 0.015 kg/s. This result is then converted into daily units of 1296 and annual units of 473.364 to provide practical measures for environmental management and reporting.

### Steady-state concentration in a well-mixed lake (first-order decay), `enveng_lake_steady_state`

The steady concentration of a pollutant in a lake treated as a completely mixed reactor with inflow load, outflow and first-order decay, and the hydraulic residence time.

$$C = \frac{W}{Q + kV}$$

| symbol | | meaning | SI unit |
|---|---|---|---|
| `W` | in | total pollutant input (load) | kg/s |
| `Q` | in | outflow rate | m^3/s |
| `k` | in | first-order decay constant (0 if conservative) | 1/s |
| `V` | in | lake volume | m^3 |
| `C` | out | steady-state concentration | kg/m^3 |
| `C_mgL` | out | in mg/L | , |
| `tau` | out | hydraulic residence time V/Q | s |

**Applies when:** completely mixed; steady state.

```
$ environmental-engineering solve enveng_lake_steady_state "W=100 kg/day" "Q=10 m^3/s" "k=0.2
1/day" "V=10000000 m^3"
C = 3.491620112e-05 kg/m^3
C_mgL = 0.03491620112
tau = 1000000 s
  by: Steady-state concentration in a well-mixed lake (first-order decay)  (0 = W - Q C - k C
V, C = W/(Q + k V))
  W = 0.001157407407 kg/s
  Q = 10 m^3/s
  k = 2.314814815e-06 1/s
  V = 10000000 m^3
  assumes: completely mixed; steady state
```

**What the worked example means:** This calculation determines the equilibrium pollutant level in a lake modeled as a completely mixed reactor. By balancing the input load 100 kg/day against losses via outflow 10 m^3/s and first-order decay 0.2 1/day within volume 1e+07 m^3, the model computes the steady-state concentration 3.49162e-05 kg/m^3. The result, also expressed as 0.0349162, represents the long-term average quality of the water body under these conditions. Additionally, the hydraulic residence time 1e+06 s is calculated to indicate how long water remains in the system on average.

### Stream reaeration coefficient (O'Connor-Dobbins, Churchill, Owens-Gibbs), `enveng_reaeration_coefficient`

The reaeration rate constant of a stream at 20 degC from its mean velocity and depth by a chosen empirical formula.

$$k_a = \frac{3.93\,u^{0.5}}{H^{1.5}}\ \text{(O'Connor-Dobbins)}$$

| symbol | | meaning | SI unit |
|---|---|---|---|
| `method` | in | o_connor_dobbins, churchill or owens_gibbs | , |
| `u` | in | mean stream velocity | m/s |
| `H` | in | mean depth | m |
| `ka` | out | reaeration rate constant at 20 degC | 1/s |
| `ka_per_day` | out | in 1/d | , |

**Applies when:** u in m/s and H in m inside the formula constants.

```
$ environmental-engineering solve enveng_reaeration_coefficient method=o_connor_dobbins "u=0.3
m/s" "H=2 m"
ka = 8.808347541e-06 1/s
ka_per_day = 0.7610412275
  by: Stream reaeration coefficient (O'Connor-Dobbins, Churchill, Owens-Gibbs)  (ka = 3.93
u^0.5/H^1.5 (O'Connor-Dobbins), 5.026 u/H^1.67 (Churchill), 5.32 u^0.67/H^1.85 (Owens-Gibbs),
per day)
  u = 0.3 m/s
  H = 2 m
  method = o_connor_dobbins
  assumes: u in m/s and H in m inside the formula constants
```

**What the worked example means:** This calculation determines the stream reaeration coefficient at twenty degrees Celsius using the selected empirical method. It uses the mean stream velocity of 0.3 m/s and the mean depth of 2 m. The resulting rate constant is 8.80835e-06 1/s, which represents how quickly oxygen transfers from the atmosphere into the water column. This value is also expressed as 0.761041 in daily units for standard modeling applications.

### Streeter-Phelps oxygen deficit and dissolved oxygen at a travel time, `enveng_streeter_phelps_deficit`

The dissolved-oxygen deficit and the dissolved oxygen of a river at a travel time downstream of a BOD discharge (Streeter-Phelps oxygen sag).

$$D = \frac{k_d L_0}{k_a - k_d}\left(e^{-k_d t} - e^{-k_a t}\right) + D_0 e^{-k_a t}$$

| symbol | | meaning | SI unit |
|---|---|---|---|
| `kd` | in | deoxygenation rate constant | 1/s |
| `ka` | in | reaeration rate constant | 1/s |
| `L0` | in | ultimate BOD of the mixed river water at the outfall | kg/m^3 |
| `D0` | in | oxygen deficit at the outfall (DOsat - DO) | kg/m^3 |
| `t` | in | travel time from the outfall | s |
| `DOsat` | in | saturation dissolved oxygen | kg/m^3 |
| `D` | out | oxygen deficit at t | kg/m^3 |
| `DO` | out | dissolved oxygen at t | kg/m^3 |
| `D_mgL` | out | deficit in mg/L | , |
| `DO_mgL` | out | dissolved oxygen in mg/L | , |

**Applies when:** first-order deoxygenation and reaeration; plug flow; deficit stays below saturation.

```
$ environmental-engineering solve enveng_streeter_phelps_deficit "kd=0.23 1/day" "ka=0.46
1/day" "L0=10 mg/L" "D0=1 mg/L" "t=2 day" "DOsat=9.1 mg/L"
D = 0.002726165085 kg/m^3
DO = 0.006373834915 kg/m^3
D_mgL = 2.726165085
DO_mgL = 6.373834915
  by: Streeter-Phelps oxygen deficit and dissolved oxygen at a travel time  (D = kd L0/(ka -
kd) (e^(-kd t) - e^(-ka t)) + D0 e^(-ka t); for ka = kd, D = (kd L0 t + D0) e^(-kd t))
  kd = 2.662037037e-06 1/s
  ka = 5.324074074e-06 1/s
  L0 = 0.01 kg/m^3
  D0 = 0.001 kg/m^3
  t = 172800 s
  DOsat = 0.0091 kg/m^3
  assumes: first-order deoxygenation and reaeration; plug flow; deficit stays below saturation
```

**What the worked example means:** This calculation determines the oxygen deficit and dissolved oxygen levels in a river at a specific travel time downstream of a discharge. Using the Streeter-Phelps model, which balances deoxygenation from biochemical demand against reaeration from the atmosphere, the system computes that the deficit reaches 0.00272617 kg/m^3 while the actual dissolved oxygen stands at 0.00637383 kg/m^3. These values indicate how much oxygen has been consumed relative to saturation and what remains available for aquatic life.

**Temperature correction of a rate constant (theta model),** `enveng_rate_constant_temperature`

A first-order rate constant (BOD, reaeration, decay) at temperature T from its 20 degC value and the temperature coefficient theta.

$$k_T = k_{20}\,\theta^{\,T-20}$$

| symbol | | meaning | SI unit |
|---|---|---|---|
| `k20` | in | rate constant at 20 degC | 1/s |
| `theta` | in | temperature coefficient (1.047 BOD, 1.024 reaeration) | , |
| `T` | in | water temperature | K |
| `kT` | out | rate constant at T | 1/s |
| `kT_per_day` | out | rate constant in 1/d | , |

**Applies when:** 0.9 < theta < 1.2; -5 degC <= T <= 60 degC.

```
$ environmental-engineering solve enveng_rate_constant_temperature "k20=0.23 1/day" theta=1.047
"T=25 degC"
kT = 3.349249506e-06 1/s
kT_per_day = 0.2893751573
  by: Temperature correction of a rate constant (theta model)  (kT = k20 theta^(T - 20))
  k20 = 2.662037037e-06 1/s
  theta = 1.047
  T = 298.15 K
  assumes: 0.9 < theta < 1.2; -5 degC <= T <= 60 degC
```

**What the worked example means:** This calculation adjusts a first-order rate constant from its standard reference value at twenty degrees Celsius to the actual water temperature using the theta model. By applying the temperature coefficient theta, the method accounts for how biological and chemical reaction speeds change with thermal conditions. The result shows that at the current temperature, the effective rate constant is 3.34925e-06 1/s per second, equivalent to 0.289375 per day, allowing for accurate modeling of processes like biochemical oxygen demand decay or reaeration.

**Theoretical oxygen demand of a solution (mg/L), `enveng_thod_solution`**

The theoretical oxygen demand in mg/L of a solution of an organic compound CaHbOcNd at a given concentration, its complete oxidation to CO2 and water (nitrogen to ammonia, or to nitrate with nitrification): ThOD = C x (g O2 per g compound).

$$\mathrm{ThOD} = C\,\frac{\left(a + \frac{b}{4} - \frac{c}{2} - \frac{3d}{4}\right) M_{O_2}}{M}$$

| symbol | | meaning | SI unit |
|---|---|---|---|
| `formula` | in | molecular formula with C, H, O, N only (e.g. C6H12O6) | , |
| `C` | in | concentration of the compound | kg/m^3 |
| `nitrification` | in | include the oxygen to nitrify ammonia (default no) (optional) | , |
| `ThOD` | out | theoretical oxygen demand of the solution | kg/m^3 |
| `ThOD_mgL` | out | ThOD, mg/L | , |
| `thod_ratio` | out | g O2 per g compound | , |
| `o2_per_mol` | out | mol O2 per mol compound | , |
| `molar_mass` | out | molar mass, g/mol | , |
| `C_molL` | out | molar concentration of the compound, mol/L | , |

**Applies when:** elements C, H, O, N only, b >= 3d; complete oxidation; nitrogen to NH3 unless nitrification = yes.

```
$ environmental-engineering solve enveng_thod_solution formula=C6H12O6 "C=100 mg/L"
ThOD = 0.1065703767 kg/m^3
ThOD_mgL = 106.5703767
thod_ratio = 1.065703767
o2_per_mol = 6
molar_mass = 180.15588
C_molL = 0.0005550748607
  by: Theoretical oxygen demand of a solution (mg/L)  (ThOD = C (a + b/4 - c/2 - 3d/4 [+ 2d])
M_O2 / M)
  C = 0.1 kg/m^3
  formula = C6H12O6
  assumes: elements C, H, O, N only, b >= 3d; complete oxidation; nitrogen to NH3 unless
nitrification = yes
```

In the worked example, formula = C6H12O6, C = 100 mg/L go through the equation above and give ThOD = 0.1065703767 kg/m^3, ThOD_mgL = 106.5703767, thod_ratio = 1.065703767, o2_per_mol = 6. It applies only when elements C, H, O, N only, b >= 3d; complete oxidation; nitrogen to NH3 unless nitrification = yes: outside that it says what is wrong instead of returning a number.

**Theoretical oxygen demand of an organic compound, `enveng_theoretical_oxygen_demand`**

The theoretical oxygen demand (g O2 per g compound) of a compound CaHbOcNd oxidised to CO2 and water, with its nitrogen released as ammonia or, with nitrification, oxidised to nitrate.

$$\mathrm{ThOD} = \frac{\left(a + \frac{b}{4} - \frac{c}{2} - \frac{3d}{4}\right) M_{O_2}}{M}$$

| symbol | | meaning | SI unit |
|---|---|---|---|
| `formula` | in | molecular formula with C, H, O, N only (e.g. C6H12O6) | , |
| `nitrification` | in | include the oxygen to nitrify ammonia (yes) or not (no) | , |
| `thod` | out | theoretical oxygen demand, g O2 per g compound | , |
| `o2_per_mol` | out | mol O2 per mol compound | , |
| `molar_mass` | out | molar mass, g/mol | , |

**Applies when:** elements C, H, O, N only; b >= 3d.

```
$ environmental-engineering solve enveng_theoretical_oxygen_demand formula=C6H12O6
nitrification=no
thod = 1.065703767
o2_per_mol = 6
molar_mass = 180.15588
  by: Theoretical oxygen demand of an organic compound  (CaHbOcNd + (a + b/4 - c/2 - 3d/4) O2 -
> a CO2 + (b - 3d)/2 H2O + d NH3; nitrification adds 2d O2)
  formula = C6H12O6
  nitrification = no
  assumes: elements C, H, O, N only; b >= 3d
```

**What the worked example means:** This calculation determines the theoretical oxygen demand, expressed as grams of oxygen per gram of compound, for the molecular formula C6H12O6. The process assumes complete oxidation of carbon to carbon dioxide and hydrogen to water, with nitrogen released as ammonia unless nitrification is enabled. The result, 1.0657, indicates the maximum oxygen required if all organic matter were

biologically or chemically oxidized under these specific conditions, based on a molar mass of 180.156 and an oxygen consumption of 6 moles per mole of compound.

**Total alkalinity as CaCO3 from the carbonate species, `enveng_total_alkalinity`**

The total alkalinity of a water, [HCO3-] + 2[CO3 2-] + [OH-] - [H+] in equivalents per litre, expressed as mg/L of CaCO3 (50,000 mg/eq), from the ion concentrations in mg/L or mol/L or the pH; with one of [H+], [OH-] given, the other follows from Kw = 1e-14 (25 degC).

$$\mathrm{Alk} = \left([\mathrm{HCO_3^-}] + 2[\mathrm{CO_3^{2-}}] + [\mathrm{OH^-}] - [\mathrm{H^+}]\right) \times 50\,000\ \mathrm{mg\,CaCO_3/eq}$$

| symbol | | meaning | SI unit |
|---|---|---|---|
| `HCO3` | in | bicarbonate [HCO3-] as mass concentration (optional) | kg/m^3 |
| `CO3` | in | carbonate [CO3 2-] as mass concentration (optional) | kg/m^3 |
| `OH` | in | hydroxide [OH-] as mass concentration (optional) | kg/m^3 |
| `H` | in | hydrogen ion [H+] as mass concentration (optional) | kg/m^3 |
| `HCO3_M` | in | bicarbonate [HCO3-] as molar concentration (optional) | mol/m^3 |
| `CO3_M` | in | carbonate [CO3 2-] as molar concentration (optional) | mol/m^3 |
| `OH_M` | in | hydroxide [OH-] as molar concentration (optional) | mol/m^3 |
| `H_M` | in | hydrogen ion [H+] as molar concentration (optional) | mol/m^3 |
| `pH` | in | pH of the water (gives [H+]) (optional) | , |
| `alkalinity` | out | total alkalinity, mg/L as CaCO3 | , |
| `alk_meq_L` | out | total alkalinity, meq/L | , |
| `alk_eq_L` | out | total alkalinity, eq/L | , |
| `HCO3_caco3` | out | bicarbonate alkalinity, mg/L as CaCO3 | , |
| `CO3_caco3` | out | carbonate alkalinity, mg/L as CaCO3 | , |
| `OH_caco3` | out | hydroxide alkalinity, mg/L as CaCO3 | , |
| `H_caco3` | out | hydrogen ion (subtracted), mg/L as CaCO3 | , |
| `H_molL` | out | [H+] used, mol/L | , |
| `OH_molL` | out | [OH-] used, mol/L | , |

**Applies when:** at least one of HCO3, CO3, OH; each ion once (mass or molar form; H also as pH); neither H nor OH given: both neglected; one given: the other from Kw = 1e-14 (25 degC); as CaCO3 with 50,000 mg/eq; molar masses HCO3 61.017, CO3 60.009, OH 17.007, H 1.008 g/mol.

```
$ environmental-engineering solve enveng_total_alkalinity "HCO3=122 mg/L" "CO3=30 mg/L" pH=10
alkalinity = 154.9649805
alk_meq_L = 3.09929961
alk_eq_L = 0.00309929961
HCO3_caco3 = 99.97240106
CO3_caco3 = 49.99258443
OH_caco3 = 5
H_caco3 = 5e-06
H_molL = 1e-10
OH_molL = 0.0001
  by: Total alkalinity as CaCO3 from the carbonate species  (Alk (eq/L) = [HCO3-] + 2[CO3 2-] +
[OH-] - [H+] (mol/L); Alk (mg/L as CaCO3) = 50,000 x Alk (eq/L))
  HCO3 = 0.122 kg/m^3
  CO3 = 0.03 kg/m^3
  pH = 10
  assumes: at least one of HCO3, CO3, OH; each ion once (mass or molar form; H also as pH);
neither H nor OH given: both neglected; one given: the other from Kw = 1e-14 (25 degC); as
CaCO3 with 50,000 mg/eq; molar masses HCO3 61.017, CO3 60.009, OH 17.007, H 1.008 g/mol
```

In the worked example, HCO3 = 122 mg/L, CO3 = 30 mg/L, pH = 10 go through the equation above and give alkalinity = 154.9649805, alk_meq_L = 3.09929961, alk_eq_L = 0.00309929961, HCO3_caco3 = 99.97240106. It applies only when at least one of HCO3, CO3, OH; each ion once (mass or molar form; H also as pH); neither H nor OH given: both neglected; one given: the other from Kw = 1e-14 (25 degC); as CaCO3 with 50,000 mg/eq; molar masses HCO3 61.017, CO3 60.009, OH 17.007, H 1.008 g/mol: outside that it says what is wrong instead of returning a number.

**Ultimate BOD from a BOD test, `enveng_ultimate_bod`**

The ultimate carbonaceous BOD from a BOD measured after t days (e.g. BOD5) and the rate constant.

$$L_0 = \frac{BOD_t}{1 - e^{-kt}}$$

| symbol | | meaning | SI unit |
|---|---|---|---|
| `BOD_t` | in | BOD measured after time t (e.g. BOD5) | kg/m^3 |
| `k` | in | BOD rate constant (base e) | 1/s |
| `t` | in | incubation time | s |
| `L0` | out | ultimate BOD | kg/m^3 |
| `L0_mgL` | out | ultimate BOD in mg/L | , |

**Applies when:** first-order kinetics.

```
$ environmental-engineering solve enveng_ultimate_bod "BOD_t=200 mg/L" "k=0.23 1/day" "t=5 day"
L0 = 0.2926701219 kg/m^3
L0_mgL = 292.6701219
  by: Ultimate BOD from a BOD test  (L0 = BOD_t/(1 - e^(-k t)))
  BOD_t = 0.2 kg/m^3
  k = 2.662037037e-06 1/s
  t = 432000 s
  assumes: first-order kinetics
```

In the worked example, BOD_t = 200 mg/L, k = 0.23 1/day, t = 5 day go through the equation above and give L0 = 0.2926701219 kg/m^3, L0_mgL = 292.6701219. It applies only when first-order kinetics: outside that it says what is wrong instead of returning a number.

## Water treatment (32)

### Activated carbon dose from the Freundlich isotherm, `enveng_freundlich_carbon_dose`

The equilibrium adsorbent loading from the Freundlich isotherm q = K Ce^(1/n) (q in mg/g, Ce in mg/L) and the batch carbon dose that lowers C0 to Ce.

$$q_e = K C_e^{1/n}, \quad \frac{m}{V} = \frac{C_0 - C_e}{q_e}$$

| symbol | | meaning | SI unit |
|---|---|---|---|
| `C0` | in | initial concentration | kg/m^3 |
| `Ce` | in | target equilibrium concentration | kg/m^3 |
| `K` | in | Freundlich capacity factor, (mg/g)(L/mg)^(1/n) | , |
| `n` | in | Freundlich exponent n (isotherm exponent 1/n) | , |
| `qe` | out | equilibrium loading, mg/g | , |
| `dose` | out | adsorbent dose | kg/m^3 |
| `dose_mgL` | out | dose in mg/L | , |

**Applies when:** Ce < C0; batch equilibrium.

```
$ environmental-engineering solve enveng_freundlich_carbon_dose "C0=10 mg/L" "Ce=1 mg/L" K=50
n=2
qe = 50
dose = 0.18 kg/m^3
dose_mgL = 180
  by: Activated carbon dose from the Freundlich isotherm  (qe = K Ce^(1/n); dose = (C0 -
Ce)/qe)
  C0 = 0.01 kg/m^3
  Ce = 0.001 kg/m^3
  K = 50
  n = 2
  assumes: Ce < C0; batch equilibrium
```

**What the worked example means:** This calculation determines the amount of activated carbon needed to reduce a contaminant concentration from an initial value of 10 mg/L to a target equilibrium value of 1 mg/L. First, the Freundlich isotherm equation uses the capacity factor 50 and exponent 2 to find the equilibrium loading 50, which represents how much contaminant each gram of carbon can hold. Then, the mass balance equation divides the difference between the initial and final concentrations by this loading to yield the necessary dose 0.18 kg/m^3. This result indicates the practical quantity of adsorbent required for batch treatment under these specific conditions.

### Adsorbent dose from the Langmuir isotherm, `enveng_langmuir_carbon_dose`

The equilibrium loading from the Langmuir isotherm q = qmax b Ce/(1 + b Ce) (q in mg/g, Ce in mg/L) and the batch adsorbent dose that lowers C0 to Ce.

$$q_e = \frac{q_{max} b C_e}{1 + b C_e}, \qquad \frac{m}{V} = \frac{C_0 - C_e}{q_e}$$

| symbol | | meaning | SI unit |
|---|---|---|---|
| `C0` | in | initial concentration | kg/m^3 |
| `Ce` | in | target equilibrium concentration | kg/m^3 |
| `qmax` | in | Langmuir monolayer capacity, mg/g | , |
| `b` | in | Langmuir constant, L/mg | , |
| `qe` | out | equilibrium loading, mg/g | , |
| `dose` | out | adsorbent dose | kg/m^3 |
| `dose_mgL` | out | dose in mg/L | , |

**Applies when:** Ce < C0.

```
$ environmental-engineering solve enveng_langmuir_carbon_dose "C0=10 mg/L" "Ce=1 mg/L" qmax=150
b=0.8
qe = 66.66666667
dose = 0.135 kg/m^3
dose_mgL = 135
  by: Adsorbent dose from the Langmuir isotherm  (qe = qmax b Ce/(1 + b Ce); dose = (C0 -
Ce)/qe)
  C0 = 0.01 kg/m^3
  Ce = 0.001 kg/m^3
  qmax = 150
  b = 0.8
  assumes: Ce < C0
```

**What the worked example means:** This calculation determines the mass of adsorbent per unit volume of solution required to reduce the initial concentration from 10 mg/L to the target equilibrium concentration of 1 mg/L. It first computes the amount of contaminant held per gram of adsorbent at equilibrium, which is 66.6667, using the Langmuir isotherm parameters for monolayer capacity 150 and affinity constant 0.8. The resulting dose is 0.135 kg/m^3, or equivalently 135, indicating how much solid material must be added to achieve the desired water quality.

**Chemical feed rate: mass per day = Q x dose / purity (kg/d and lb/d), `enveng_chemical_feed_rate`**

The feed rate of a treatment chemical (coagulant, chlorine product, lime, polymer, ...) for a flow and a dose, mass per day = Q x dose / purity, in kg/d and lb/d, with the active chemical alone; the flow in m^3/d or MGD, the dose in mg/L, the purity (active fraction of the product) as a fraction or %, 1 when not given.

$$\dot{m} = \frac{Q \cdot \text{dose}}{p}, \qquad \dot{m}\,[\text{lb/d}] = \frac{8.3454\, Q[\text{MGD}]\, \text{dose}[\text{mg/L}]}{p}$$

| symbol | | meaning | SI unit |
|---|---|---|---|
| `Q` | in | flow treated (m^3/d, or MGD) | m^3/s |
| `dose` | in | dose of the active chemical | kg/m^3 |
| `purity` | in | active fraction of the product (fraction or %; default 1) (optional) | , |
| `feed` | out | feed rate of the product Q dose/purity | kg/s |

| symbol | | meaning | SI unit |
|---|---|---|---|
| `feed_kg_per_day` | out | product feed, kg/d | , |
| `feed_lb_per_day` | out | product feed, lb/d | , |
| `active_kg_per_day` | out | active chemical fed, kg/d | , |
| `active_lb_per_day` | out | active chemical fed, lb/d | , |
| `Q_MGD` | out | flow, million US gallons a day | , |

**Applies when:** 0 < purity <= 1 (1 unless given); dose as the active chemical; 1 lb = 0.45359237 kg, 1 US gal = 3.785411784 L.

```
$ environmental-engineering solve enveng_chemical_feed_rate "Q=10000 m^3/day" "dose=25 mg/L"
feed = 0.002893518519 kg/s
feed_kg_per_day = 250
feed_lb_per_day = 551.1556555
active_kg_per_day = 250
active_lb_per_day = 551.1556555
Q_MGD = 2.641720524
  by: Chemical feed rate: mass per day = Q x dose / purity (kg/d and lb/d)  (feed = Q x dose /
purity; lb/d = kg/d / 0.45359237 (= 8.3454 x MGD x mg/L / purity))
  Q = 0.1157407407 m^3/s
  dose = 0.025 kg/m^3
  assumes: 0 < purity <= 1 (1 unless given); dose as the active chemical; 1 lb = 0.45359237 kg,
1 US gal = 3.785411784 L
```

In the worked example, Q = 10000 m^3/day, dose = 25 mg/L go through the equation above and give feed = 0.002893518519 kg/s, feed_kg_per_day = 250, feed_lb_per_day = 551.1556555, active_kg_per_day = 250. It applies only when 0 < purity <= 1 (1 unless given); dose as the active chemical; 1 lb = 0.45359237 kg, 1 US gal = 3.785411784 L: outside that it says what is wrong instead of returning a number.

### Chlorine dose and feed rate, `enveng_chlorine_dose_feed`

The chlorine dose (demand plus residual), the chlorine feed rate for a flow and the feed of a product with a stated available-chlorine fraction (e.g. 65 % calcium hypochlorite).

$$\dot{m} = Q\,(\text{demand} + \text{residual}), \quad \dot{m}_{product} = \frac{\dot{m}}{p}$$

| symbol | | meaning | SI unit |
|---|---|---|---|
| `Q` | in | flow treated | m^3/s |
| `demand` | in | chlorine demand | kg/m^3 |
| `residual` | in | required free chlorine residual | kg/m^3 |
| `purity` | in | available chlorine fraction of the product (1 for chlorine gas; fraction or %) | , |
| `dose` | out | chlorine dose | kg/m^3 |
| `dose_mgL` | out | dose in mg/L | , |
| `feed` | out | chlorine feed rate | kg/s |
| `feed_kg_per_day` | out | chlorine feed in kg/d | , |
| `product_kg_per_day` | out | product feed in kg/d | , |

**Applies when:** 0 < purity <= 1.

```
$ environmental-engineering solve enveng_chlorine_dose_feed "Q=20000 m^3/day" "demand=2.5 mg/L"
"residual=0.5 mg/L" purity=1
dose = 0.003 kg/m^3
dose_mgL = 3
feed = 0.0006944444444 kg/s
feed_kg_per_day = 60
product_kg_per_day = 60
  by: Chlorine dose and feed rate  (dose = demand + residual; feed = Q dose; product =
feed/purity)
  Q = 0.2314814815 m^3/s
  demand = 0.0025 kg/m^3
  residual = 0.0005 kg/m^3
  purity = 1
  assumes: 0 < purity <= 1
```

**What the worked example means:** This calculation determines the total chlorine dose by adding the chlorine demand to the required free chlorine residual. The computed dose of 0.003 kg/m^3 is then multiplied by the flow treated to find the pure chlorine feed rate, which equals 0.000694444 kg/s. Finally, the feed rate for the specific chemical product is derived by dividing the pure chlorine feed by the available chlorine fraction of the product, yielding 60. This ensures the correct amount of commercial-grade chlorine is dosed to achieve the desired residual.

### Clean-bed head loss through a granular filter (Ergun / Carman-Kozeny), `enveng_filter_clean_bed_headloss`

The head loss through a clean granular filter bed from the grain size, sphericity, porosity, depth and filtration rate (Ergun equation, the Carman-Kozeny form with friction factor 150/Re + 1.75).

$$\frac{h}{L} = \frac{150\mu(1-\varepsilon)^2 v}{\rho g \varepsilon^3 (\psi d)^2} + \frac{1.75(1-\varepsilon)v^2}{g\varepsilon^3 \psi d}$$

| symbol | | meaning | SI unit |
|---|---|---|---|
| `d` | in | grain diameter | m |
| `eps` | in | bed porosity (fraction) | , |
| `psi` | in | grain sphericity (fraction, 1 for spheres) | , |
| `L` | in | bed depth | m |
| `v` | in | filtration (approach) velocity | m/s |
| `rho` | in | water density | kg/m^3 |
| `mu` | in | water viscosity | Pa*s |
| `dP` | out | pressure drop | Pa |
| `h` | out | head loss | m |
| `Re` | out | bed Reynolds number rho v psi d/(mu (1 - eps)) | , |

**Applies when:** 0 < eps < 1; 0 < psi <= 1; clean bed.

```
$ environmental-engineering solve enveng_filter_clean_bed_headloss "d=0.5 mm" eps=0.4 psi=0.85
"L=0.7 m" "v=8 m/h" "rho=998.2 kg/m^3" "mu=0.001002 Pa*s"
dP = 7414.170928 Pa
h = 0.7573983471 m
Re = 1.568104532
  by: Clean-bed head loss through a granular filter (Ergun / Carman-Kozeny)  (dP/L = 150 mu (1-
eps)^2 v/(eps^3 (psi d)^2) + 1.75 rho (1-eps) v^2/(eps^3 psi d); h = dP/(rho g))
  d = 0.0005 m
  eps = 0.4
  psi = 0.85
  L = 0.7 m
  v = 0.002222222222 m/s
  rho = 998.2 kg/m^3
  mu = 0.001002 Pa*s
  assumes: 0 < eps < 1; 0 < psi <= 1; clean bed
```

**What the worked example means:** This calculation determines the head loss through a clean granular filter bed using the Ergun equation, which combines viscous and inertial resistance terms. The result depends on the grain diameter 0.5 mm, sphericity 0.85, porosity 0.4, filtration velocity 8 m/h, and water properties density 998.2 kg/m^3 and viscosity 0.001002 Pa*s. For the given inputs, the total head loss is 0.757398 m meters, corresponding to a pressure drop of 7414.17 Pa Pascals over the bed depth of 0.7 m meters. This value represents the energy required to push water through the empty pore spaces of the filter media before any solids are captured.

### Disinfection by the Chick-Watson law, `enveng_chick_watson`

The survival fraction and log inactivation of organisms exposed to a disinfectant concentration C for time t under ln(N/N0) = -k C^n t (k in (L/mg)^n per minute, C in mg/L, t in minutes inside the law).

$$\ln \frac{N}{N_0} = -k C^n t$$

| symbol | | meaning | SI unit |
|---|---|---|---|
| `k` | in | Chick-Watson coefficient in (L/mg)^n per minute | , |
| `n` | in | coefficient of dilution (1 for most disinfectants) | , |
| `C` | in | disinfectant concentration | kg/m^3 |
| `t` | in | contact time | s |
| `survival` | out | N/N0 | , |
| `log_inactivation` | out | log10 inactivation | , |

**Applies when:** constant concentration; 0.3 <= n <= 3.

```
$ environmental-engineering solve enveng_chick_watson k=0.1 n=1 "C=1 mg/L" "t=30 min"
survival = 0.04978706837
log_inactivation = 1.302883446
  by: Disinfection by the Chick-Watson law  (ln(N/N0) = -k C^n t with C in mg/L and t in
minutes)
  k = 0.1
  n = 1
  C = 0.001 kg/m^3
  t = 1800 s
  assumes: constant concentration; 0.3 <= n <= 3
```

**What the worked example means:** This calculation determines the effectiveness of a disinfectant using the Chick-Watson law. Given a coefficient of dilution 1, a rate constant 0.1, a concentration 1 mg/L, and a contact time 30 min, the model computes the resulting survival fraction 0.0497871 and the corresponding log inactivation 1.30288. These values indicate how many organisms remain alive relative to the initial population after exposure.

### Disinfection CT value of a contact basin, `enveng_disinfection_ct`

The effective contact time t10 of a basin (baffling factor times the theoretical detention time) and the CT value of a disinfectant residual.

$$CT = C\,t_{10} = C\,BF\,\frac{V}{Q}$$

| symbol | | meaning | SI unit |
|---|---|---|---|
| `C` | in | disinfectant residual at the outlet | kg/m^3 |
| `V` | in | basin volume | m^3 |
| `Q` | in | peak flow | m^3/s |
| `BF` | in | baffling factor t10/T (0.1 unbaffled to 1 plug flow) | , |
| `t10` | out | effective contact time | s |
| `CT` | out | CT value | kg*s/m^3 |
| `CT_mg_min_L` | out | CT in mg min/L | , |

**Applies when:** 0 < BF <= 1.

```
$ environmental-engineering solve enveng_disinfection_ct "C=1 mg/L" "V=1000 m^3" "Q=20000
m^3/day" BF=0.5
t10 = 2160 s
CT = 2.16 kg*s/m^3
CT_mg_min_L = 36
  by: Disinfection CT value of a contact basin  (t10 = BF V/Q; CT = C t10)
  C = 0.001 kg/m^3
  V = 1000 m^3
  Q = 0.2314814815 m^3/s
  BF = 0.5
  assumes: 0 < BF <= 1
```

**What the worked example means:** This calculation determines the disinfection efficacy potential of a contact basin by first finding the effective contact time, t10. This is done by multiplying the baffling factor, 0.5, which accounts for flow short-circuiting, by the theoretical detention time derived from the basin volume, 1000 m^3, divided by the peak flow rate, 20000 m^3/day. The result, 2160 s, represents the actual time water spends in contact with the disinfectant. The CT value, 2.16 kg*s/m^3, is then computed by multiplying this effective contact time by the residual disinfectant concentration at the outlet, 1 mg/L. A higher CT value indicates greater disinfection power.

### Filter bed expansion during backwash, `enveng_backwash_bed_expansion`

The expanded porosity, expanded depth and percentage expansion of a filter bed fluidised by backwash, from the backwash rate and the grains' settling velocity (expanded porosity (vb/vs)^0.22).

$$\varepsilon_e = \left(\frac{v_b}{v_s}\right)^{0.22}, \quad L_e = \frac{L(1-\varepsilon)}{1-\varepsilon_e}$$

| symbol | | meaning | SI unit |
|---|---|---|---|
| `L` | in | unexpanded bed depth | m |
| `eps` | in | unexpanded porosity (fraction) | , |
| `vb` | in | backwash rate | m/s |
| `vs` | in | settling velocity of the grains | m/s |
| `eps_e` | out | expanded porosity | , |
| `Le` | out | expanded bed depth | m |
| `expansion` | out | fractional expansion Le/L - 1 | , |

**Applies when:** the bed is fluidised: eps < (vb/vs)^0.22 < 1.

```
$ environmental-engineering solve enveng_backwash_bed_expansion "L=0.7 m" eps=0.42 "vb=50 m/h"
"vs=300 m/h"
eps_e = 0.6742279077
Le = 1.246270045 m
expansion = 0.780385778
  by: Filter bed expansion during backwash  (eps_e = (vb/vs)^0.22; Le = L (1 - eps)/(1 -
eps_e))
  L = 0.7 m
  eps = 0.42
  vb = 0.01388888889 m/s
  vs = 0.08333333333 m/s
  assumes: the bed is fluidised: eps < (vb/vs)^0.22 < 1
```

**What the worked example means:** This calculation determines the state of a filter bed during backwashing. Using the unexpanded bed depth of 0.7 m and porosity of 0.42, along with a backwash rate of 50 m/h and grain settling velocity of 300 m/h, it computes the expanded porosity as 0.674228 and the new bed depth as 1.24627 m. The resulting fractional expansion is 0.780386, indicating how much the bed has lifted and loosened to facilitate cleaning.

### Filtration (hydraulic loading) rate with one filter out of service, `enveng_filtration_rate`

The filtration rate of a bank of identical filters, with all in service and with one out of service (for backwashing).

$$v = \frac{Q}{NA}, \quad v' = \frac{Q}{(N-1)A}$$

| symbol | | meaning | SI unit |
|---|---|---|---|
| `Q` | in | plant flow | m^3/s |
| `A` | in | area of one filter | m^2 |
| `N` | in | number of filters | , |
| `v` | out | filtration rate with all filters | m/s |
| `v_m_per_h` | out | in m/h | , |
| `v_one_out` | out | rate with one filter out of service | m/s |

| symbol | | meaning | SI unit |
|---|---|---|---|
| `v_one_out_m_per_h` | out | in m/h | , |

**Applies when:** N >= 2 identical filters.

```
$ environmental-engineering solve enveng_filtration_rate "Q=40000 m^3/day" "A=50 m^2" N=6
v = 0.001543209877 m/s
v_m_per_h = 5.555555556
v_one_out = 0.001851851852 m/s
v_one_out_m_per_h = 6.666666667
  by: Filtration (hydraulic loading) rate with one filter out of service  (v = Q/(N A); v' =
Q/((N - 1) A))
  Q = 0.462962963 m^3/s
  A = 50 m^2
  assumes: N >= 2 identical filters
```

**What the worked example means:** This calculation determines the hydraulic loading rate for a bank of identical filters. Using the total plant flow 40000 m^3/day and the area of each individual filter 50 m^2, this gives the rate when all 6 units are operating, resulting in 0.00154321 m/s. The rate is then recalculated with one unit offline for backwashing, which increases the load on the remaining active filters to 0.00185185 m/s. These values indicate how much water passes through each square meter of filter media per second.

**Flow through a filter from its loading rate and bed area, `enveng_filter_flow_from_loading`**

The flow a filter treats from its hydraulic (surface) loading rate and the bed area, Q = v A, in m^3/h and m^3/d.

$$Q = v\,A$$

| symbol | | meaning | SI unit |
|---|---|---|---|
| `v` | in | filtration (hydraulic loading) rate | m/s |
| `A` | in | filter bed area | m^2 |
| `Q` | out | flow v A | m^3/s |
| `Q_m3_per_h` | out | in m^3/h | , |
| `Q_m3_per_day` | out | in m^3/d | , |

**Applies when:** v, A > 0.

```
$ environmental-engineering solve enveng_filter_flow_from_loading "v=120 m/day" "A=50 m^2"
Q = 0.06944444444 m^3/s
Q_m3_per_h = 250
Q_m3_per_day = 6000
  by: Flow through a filter from its loading rate and bed area  (Q = v A)
  v = 0.001388888889 m/s
  A = 50 m^2
  assumes: v, A > 0
```

In the worked example, v = 120 m/day, A = 50 m^2 go through the equation above and give Q = 0.06944444444 m^3/s, Q_m3_per_h = 250, Q_m3_per_day = 6000. It applies only when v, A > 0: outside that it says what is wrong instead of returning a number.

**Flow velocity in a rectangular channel (continuity, grit chamber), `enveng_channel_flow_velocity`**

The mean (horizontal) velocity of flow in a rectangular channel or grit chamber from the flow, the width and the water depth (v = Q/(W d)), or the flow from the velocity (Q = A v).

$$v = \frac{Q}{A} = \frac{Q}{W\,d}, \qquad Q = A\,v$$

| symbol | | meaning | SI unit |
|---|---|---|---|
| `W` | in | channel width | m |
| `d` | in | water depth | m |
| `Q` | in | flow rate (optional) | m^3/s |
| `v` | in | mean flow velocity (optional) | m/s |
| `A` | out | flow cross-section W d | m^2 |
| `v` | out | mean (horizontal) velocity Q/A | m/s |
| `Q` | out | flow rate A v | m^3/s |

**Applies when:** W, d > 0; exactly one of Q, v, positive; rectangular section, uniform mean velocity.

```
$ environmental-engineering solve enveng_channel_flow_velocity "Q=0.3 m^3/s" "W=1 m" "d=1 m"
A = 1 m^2
v = 0.3 m/s
Q = 0.3 m^3/s
  by: Flow velocity in a rectangular channel (continuity, grit chamber)  (A = W d; v = Q/A (or
Q = A v))
  Q = 0.3 m^3/s
  W = 1 m
  d = 1 m
  assumes: W, d > 0; exactly one of Q, v, positive; rectangular section, uniform mean velocity
```

In the worked example, Q = 0.3 m^3/s, W = 1 m, d = 1 m go through the equation above and give A = 1 m^2, v = 0.3 m/s, Q = 0.3 m^3/s. It applies only when W, d > 0; exactly one of Q, v, positive; rectangular section, uniform mean velocity: outside that it says what is wrong instead of returning a number.

**Fraction of discrete particles removed in an ideal settling basin, `enveng_ideal_basin_removal`**

The overflow rate of an ideal (Camp) settling basin and the fraction of particles of one settling velocity it removes (all if vs >= v0, otherwise vs/v0).

$$R = \min\left(1, \frac{v_s}{v_0}\right)$$

| symbol | | meaning | SI unit |
|---|---|---|---|
| `vs` | in | settling velocity of the particles | m/s |
| `Q` | in | flow | m^3/s |
| `A` | in | basin surface area | m^2 |
| `v0` | out | overflow rate Q/A | m/s |
| `R` | out | fraction removed | , |

**Applies when:** ideal basin, discrete (type I) settling.

```
$ environmental-engineering solve enveng_ideal_basin_removal "vs=0.5 m/h" "Q=10000 m^3/day"
"A=400 m^2"
v0 = 0.0002893518519 m/s
R = 0.48
  by: Fraction of discrete particles removed in an ideal settling basin  (v0 = Q/A; R = min(1,
vs/v0))
  vs = 0.0001388888889 m/s
  Q = 0.1157407407 m^3/s
  A = 400 m^2
  assumes: ideal basin, discrete (type I) settling
```

**What the worked example means:** This calculation determines the removal efficiency for discrete particles in an ideal sedimentation tank. First, the overflow rate is computed by dividing the flow rate 10000 m^3/day by the surface area 400 m^2, resulting in 0.000289352 m/s. Then, the fraction removed 0.48 is found by comparing this overflow rate to the particle settling velocity 0.5 m/h. If the settling velocity exceeds the overflow rate, all particles are captured; otherwise, only a portion equal to their relative speeds is removed.

**Fraction of free chlorine as hypochlorous acid (pH), `enveng_hypochlorous_acid_fraction`**

The fractions of free chlorine present as HOCl and as OCl- at a pH, from the acid dissociation constant (given, or from the temperature by Morris' correlation).

$$\alpha_{HOCl} = \frac{1}{1 + 10^{\,pH - pK_a}}$$

| symbol | | meaning | SI unit |
|---|---|---|---|
| `pH` | in | pH of the water | , |
| `pKa` | in | pKa of HOCl (7.54 at 25 degC) (optional) | , |
| `T` | in | water temperature (to compute pKa instead) (optional) | K |
| `alpha_HOCl` | out | fraction as HOCl | , |
| `alpha_OCl` | out | fraction as OCl- | , |
| `pKa` | out | pKa used | , |

**Applies when:** either pKa or T; 0-35 degC for the correlation.

```
$ environmental-engineering solve enveng_hypochlorous_acid_fraction pH=7 pKa=7.54
alpha_HOCl = 0.7761545753
alpha_OCl = 0.2238454247
pKa = 7.54
  by: Fraction of free chlorine as hypochlorous acid (pH)  (alpha_HOCl = 1/(1 + 10^(pH - pKa));
pKa = 3000/T - 10.0686 + 0.0253 T (Morris))
  pH = 7
  pKa = 7.54
  assumes: either pKa or T; 0-35 degC for the correlation
```

**What the worked example means:** This calculation determines how free chlorine distributes between its two forms in water. Using the provided pH and temperature or pKa values, it computes that 0.776155 of the total free chlorine exists as hypochlorous acid, while 0.223845 exists as hypochlorite ion. The effective acidity constant used for this split was 7.54. Hypochlorous acid is generally the more potent disinfectant, so knowing this fraction helps assess treatment efficiency under current conditions.

### Freundlich isotherm constants fitted to adsorption data, `enveng_freundlich_fit`

The Freundlich constants K and 1/n fitted by linear regression of log q against log Ce, with the coefficient of determination.

$$\log q_e = \log K + \frac{1}{n}\log C_e$$

| symbol | | meaning | SI unit |
|---|---|---|---|
| `Ce` | in | equilibrium concentrations in mg/L | , |
| `qe` | in | equilibrium loadings in mg/g (same order) | , |
| `K` | out | Freundlich K, (mg/g)(L/mg)^(1/n) | , |
| `inv_n` | out | isotherm exponent 1/n | , |
| `n` | out | Freundlich n | , |
| `r2` | out | coefficient of determination of the log-log fit | , |

**Applies when:** at least three positive points.

```
$ environmental-engineering solve enveng_freundlich_fit "Ce=[0.5, 1, 2, 5, 10]" "qe=[28, 40,
55, 86, 120]"
K = 39.47623429
inv_n = 0.483576413
n = 2.067925509
r2 = 0.9998093398
  by: Freundlich isotherm constants fitted to adsorption data  (least squares of log10 q =
log10 K + (1/n) log10 Ce)
  assumes: at least three positive points
```

**What the worked example means:** This calculation determines the parameters for the Freundlich adsorption isotherm by performing a linear least-squares regression on the logarithmic form of the equation. The input data consists of equilibrium concentrations Ce and corresponding loadings qe. The resulting intercept yields the capacity factor K, which is 39.4762, while the slope provides the intensity factor 1/n, calculated as 0.483576. The goodness of fit for this log-log relationship is indicated by the coefficient of determination, r2, which equals 0.999809. These constants allow prediction of how much contaminant will adhere to the solid phase at various liquid concentrations.

### Hydraulic detention time, `enveng_detention_time`

The hydraulic detention (residence) time of a tank or basin from its volume and the flow.

$$\theta = \frac{V}{Q}$$

| symbol | | meaning | SI unit |
|---|---|---|---|
| `V` | in | tank volume | m^3 |
| `Q` | in | flow rate | m^3/s |
| `theta` | out | detention time V/Q | s |
| `theta_h` | out | detention time in hours | , |

**Applies when:** V, Q > 0.

```
$ environmental-engineering solve enveng_detention_time "V=1000 m^3" "Q=12000 m^3/day"
theta = 7200 s
theta_h = 2
  by: Hydraulic detention time  (theta = V/Q)
  V = 1000 m^3
  Q = 0.1388888889 m^3/s
  assumes: V, Q > 0
```

**What the worked example means:** This calculation determines the average time water remains in a tank or basin. It uses the total volume of the container, given as 1000 m^3, and the flow rate entering the system, given as 12000 m^3/day. The result, 7200 s, represents how long a parcel of water stays inside before exiting. This metric helps engineers understand contact time for treatment processes like sedimentation or disinfection.

**Langmuir isotherm constants fitted to adsorption data, `enveng_langmuir_fit`**

The Langmuir capacity qmax and constant b fitted by linear regression of Ce/q against Ce (linearised Langmuir form), with the coefficient of determination.

$$\frac{C_e}{q_e} = \frac{1}{q_{max} b} + \frac{C_e}{q_{max}}$$

| symbol | | meaning | SI unit |
|---|---|---|---|
| `Ce` | in | equilibrium concentrations in mg/L | , |
| `qe` | in | equilibrium loadings in mg/g (same order) | , |
| `qmax` | out | monolayer capacity, mg/g | , |
| `b` | out | Langmuir constant, L/mg | , |
| `r2` | out | coefficient of determination of the linearised fit | , |

**Applies when:** positive slope and intercept of Ce/q against Ce.

```
$ environmental-engineering solve enveng_langmuir_fit "Ce=[0.5, 1, 2, 5, 10]" "qe=[40, 66, 97,
133, 152]"
qmax = 177.9004838
b = 0.5896757373
r2 = 0.9999728233
  by: Langmuir isotherm constants fitted to adsorption data  (least squares of Ce/q = 1/(qmax
b) + Ce/qmax)
  assumes: positive slope and intercept of Ce/q against Ce
```

**What the worked example means:** This calculation determines the Langmuir adsorption parameters by performing linear regression on the transformed data. The equilibrium concentration divided by the equilibrium loading is plotted against the equilibrium concentration. From this linear relationship, the slope yields the reciprocal of the maximum capacity, while the intercept provides the ratio of one to the product of the maximum capacity and the affinity constant. The resulting monolayer capacity of 177.9 represents the theoretical limit of adsorbate uptake per mass of adsorbent when all surface sites are occupied. The Langmuir constant of 0.589676 indicates the affinity between the adsorbate and the adsorbent surface. The coefficient of determination of 0.999973 quantifies how well the linearized model fits the experimental observations.

**Log inactivation from the CT ratio, `enveng_ct_log_inactivation`**

The CT achieved, the ratio to the CT required for 3-log (99.9 %) inactivation from the regulatory tables, and the log inactivation credited (3 times the ratio).

$$\log\,\text{inactivation} = 3\,\frac{Ct_{10}}{CT_{99.9}}$$

| symbol | | meaning | SI unit |
|---|---|---|---|
| `C` | in | disinfectant residual | kg/m^3 |
| `t10` | in | effective contact time | s |
| `CT999` | in | CT required for 3-log inactivation (from the tables) | kg*s/m^3 |
| `CT` | out | CT achieved | kg*s/m^3 |
| `ratio` | out | CT achieved / CT for 3-log | , |
| `log_inactivation` | out | log inactivation credited | , |

**Applies when:** first-order (Chick-Watson) inactivation with n = 1.

```
$ environmental-engineering solve enveng_ct_log_inactivation "C=1.2 mg/L" "t10=40 min"
"CT999=104 mg*min/L"
CT = 2.88 kg*s/m^3
ratio = 0.4615384615
log_inactivation = 1.384615385
  by: Log inactivation from the CT ratio  (log inactivation = 3 CT/CT_99.9)
  C = 0.0012 kg/m^3
  t10 = 2400 s
  CT999 = 6.24 kg*s/m^3
  assumes: first-order (Chick-Watson) inactivation with n = 1
```

**What the worked example means:** This calculation determines the disinfection effectiveness by comparing the achieved contact time concentration product to the regulatory requirement for three-log removal. The achieved CT value, calculated as the product of the disinfectant residual and effective contact time, is 2.88 kg*s/m^3 kg*s/m^3. This is divided by the required CT for ninety-nine point nine percent inactivation, which is 104 mg*min/L mg*min/L, yielding a ratio of 0.461538. Multiplying this ratio by three gives the final credited log inactivation of 1.38462. A higher ratio indicates greater microbial reduction relative to the standard benchmark.

### Log removal and percentage removal, `enveng_log_removal`

The log10 removal (log reduction value) and the percentage removal from the counts or concentrations before and after treatment.

$$LR = \log_{10}\frac{N_0}{N}$$

| symbol | | meaning | SI unit |
|---|---|---|---|
| `N0` | in | count or concentration before treatment | , |
| `N` | in | count or concentration after treatment (same unit) | , |
| `log_removal` | out | log10 removal | , |
| `percent_removal` | out | percentage removed | , |

**Applies when:** 0 < N < N0.

```
$ environmental-engineering solve enveng_log_removal N0=1000000 N=100
log_removal = 4
percent_removal = 99.99
  by: Log removal and percentage removal  (LR = log10(N0/N); percent = (1 - N/N0) 100)
  N0 = 1000000
  N = 100
  assumes: 0 < N < N0
```

**What the worked example means:** This calculation determines the effectiveness of the treatment by comparing the initial concentration or count before treatment, represented as N0, with the final value after treatment, denoted as N. The log removal is computed using the base ten logarithm of the ratio of these two values, while the percentage removal reflects the proportion of contaminants eliminated relative to the starting amount. For the input values provided, the resulting log removal is 4 and the percentage removal is 99.99, indicating the scale of reduction achieved.

**Mean velocity gradient G (rapid mixing and flocculation), `enveng_velocity_gradient`**

The root-mean-square velocity gradient G of a mixing or flocculation tank from the power dissipated, the viscosity and the volume, and the Camp number G t when the detention time is given.

$$G = \sqrt{\frac{P}{\mu V}}$$

| symbol | | meaning | SI unit |
|---|---|---|---|
| `P` | in | power dissipated in the water | W |
| `mu` | in | water viscosity | Pa*s |
| `V` | in | tank volume | m^3 |
| `t` | in | detention time (for G t) (optional) | s |
| `G` | out | mean velocity gradient | 1/s |
| `Gt` | out | Camp number G t (dimensionless) | , |

**Applies when:** P, mu, V > 0.

```
$ environmental-engineering solve enveng_velocity_gradient "P=2000 W" "mu=0.001002 Pa*s" "V=50
m^3"
G = 199.8002995 1/s
  by: Mean velocity gradient G (rapid mixing and flocculation)  (G = sqrt(P/(mu V)); Camp
number G t)
  P = 2000 W
  mu = 0.001002 Pa*s
  V = 50 m^3
  assumes: P, mu, V > 0
```

In the worked example, P = 2000 W, mu = 0.001002 Pa*s, V = 50 m^3 go through the equation above and give G = 199.8002995 1/s. It applies only when P, mu, V > 0: outside that it says what is wrong instead of returning a number.

**Membrane permeate flux J = Qp/A (LMH and m/d), or the area for a flux, `enveng_membrane_flux`**

The permeate flux of a membrane, J = Q_permeate / A, in L/(m^2 h) (LMH) and m/d, from the permeate flow and the membrane area; or the membrane area needed for a design flux, or the permeate flow a flux gives (any two of the three).

$$J = \frac{Q_p}{A}$$

| symbol | | meaning | SI unit |
|---|---|---|---|
| `Qp` | in | permeate flow (optional) | m^3/s |
| `A` | in | membrane area (optional) | m^2 |
| `J` | in | permeate flux (L/(m^2 h) or LMH, m/d) (optional) | m/s |
| `J` | out | permeate flux Qp/A | m/s |
| `J_LMH` | out | flux, L/(m^2 h) (LMH) | , |
| `J_m_per_day` | out | flux, m/d | , |
| `A` | out | membrane area Qp/J | m^2 |
| `Qp` | out | permeate flow J A | m^3/s |
| `Qp_m3_per_day` | out | permeate flow, m^3/d | , |

**Applies when:** exactly two of Qp, A, J; the flux of the permeate (product water) over the active membrane area.

```
$ environmental-engineering solve enveng_membrane_flux "Qp=750 m^3/day" "A=1500 m^2"
J = 5.787037037e-06 m/s
J_LMH = 20.83333333
J_m_per_day = 0.5
A = 1500 m^2
Qp = 0.008680555556 m^3/s
Qp_m3_per_day = 750
  by: Membrane permeate flux J = Qp/A (LMH and m/d), or the area for a flux  (J = Qp/A; A =
Qp/J; Qp = J A; 1 m/s = 3.6e6 L/(m^2 h))
  Qp = 0.008680555556 m^3/s
  A = 1500 m^2
  assumes: exactly two of Qp, A, J; the flux of the permeate (product water) over the active
membrane area
```

In the worked example, Qp = 750 m^3/day, A = 1500 m^2 go through the equation above and give J = 5.787037037e-06 m/s, J_LMH = 20.83333333, J_m_per_day = 0.5, A = 1500 m^2. It applies only when exactly two of Qp, A, J; the flux of the permeate (product water) over the active membrane area: outside that it says what is wrong instead of returning a number.

### Membrane recovery, rejection, flux and concentrate, `enveng_membrane_mass_balance`

The water recovery, solute rejection, permeate flux (also in L/(m^2 h)) and the concentrate flow and concentration of a membrane unit from the feed and permeate flows and concentrations and the membrane area.

$$r = \frac{Q_p}{Q_f},\ R = 1 - \frac{C_p}{C_f},\ C_c = \frac{Q_f C_f - Q_p C_p}{Q_f - Q_p}$$

| symbol | | meaning | SI unit |
|---|---|---|---|
| `Qf` | in | feed flow | m^3/s |

| symbol | | meaning | SI unit |
|---|---|---|---|
| `Qp` | in | permeate flow | m^3/s |
| `Cf` | in | feed concentration | kg/m^3 |
| `Cp` | in | permeate concentration | kg/m^3 |
| `A` | in | membrane area | m^2 |
| `recovery` | out | Qp/Qf | , |
| `rejection` | out | 1 - Cp/Cf | , |
| `J` | out | permeate flux | m/s |
| `J_LMH` | out | flux in L/(m^2 h) | , |
| `Qc` | out | concentrate flow | m^3/s |
| `Cc` | out | concentrate concentration | kg/m^3 |
| `Cc_mgL` | out | concentrate concentration in mg/L | , |

**Applies when:** Qp < Qf; Cp < Cf; steady state.

```
$ environmental-engineering solve enveng_membrane_mass_balance "Qf=1000 m^3/day" "Qp=750
m^3/day" "Cf=2000 mg/L" "Cp=50 mg/L" "A=1500 m^2"
recovery = 0.75
rejection = 0.975
J = 5.787037037e-06 m/s
J_LMH = 20.83333333
Qc = 0.002893518519 m^3/s
Cc = 7.85 kg/m^3
Cc_mgL = 7850
  by: Membrane recovery, rejection, flux and concentrate  (r = Qp/Qf; R = 1 - Cp/Cf; J = Qp/A;
Qc = Qf - Qp; Cc = (Qf Cf - Qp Cp)/Qc)
  Qf = 0.01157407407 m^3/s
  Qp = 0.008680555556 m^3/s
  Cf = 2 kg/m^3
  Cp = 0.05 kg/m^3
  A = 1500 m^2
  assumes: Qp < Qf; Cp < Cf; steady state
```

**What the worked example means:** This calculation determines the performance metrics of a membrane separation process based on feed and permeate flow rates, concentrations, and membrane area. The water recovery is found to be 0.75, indicating the fraction of feed converted to permeate. Solute rejection is 0.975, reflecting the membrane's ability to retain contaminants. The permeate flux is 20.8333 liters per square meter per hour, while the concentrate stream has a flow rate of 0.00289352 m^3/s and a concentration of 7850. These values characterize the efficiency and operating point of the system under steady-state conditions.

### Microbial inactivation by Chick's law (first order), `enveng_chick_law`

The surviving organisms, survival fraction, log removal and percentage removal after a contact time under first-order (Chick's law) inactivation or die-off.

$$\frac{N}{N_0} = e^{-kt}$$

| symbol | | meaning | SI unit |
|---|---|---|---|
| `N0` | in | initial organism concentration (count per volume, e.g. per 100 mL) | , |

| symbol | | meaning | SI unit |
|---|---|---|---|
| `k` | in | inactivation (die-off) rate constant | 1/s |
| `t` | in | contact time | s |
| `N` | out | surviving organisms, same unit as N0 | , |
| `survival` | out | N/N0 | , |
| `log_removal` | out | log10 removal | , |
| `percent_removal` | out | percentage removed | , |

**Applies when:** first-order kinetics.

```
$ environmental-engineering solve enveng_chick_law N0=1000000 "k=0.2 1/min" "t=30 min"
N = 2478.752177
survival = 0.002478752177
log_removal = 2.605766891
percent_removal = 99.75212478
  by: Microbial inactivation by Chick's law (first order)  (N = N0 e^(-k t); log removal = k
t/ln 10)
  N0 = 1000000
  k = 0.003333333333 1/s
  t = 1800 s
  assumes: first-order kinetics
```

**What the worked example means:** This calculation applies Chick's law to determine microbial inactivation under first-order kinetics. Given an initial organism count of 1e+06, a die-off rate constant of 0.2 1/min, and a contact time of 30 min, the model computes the final population as 2478.75. The survival fraction is 0.00247875, which corresponds to a log removal value of 2.60577 and a percentage removal of 99.7521. These results indicate how effectively the disinfection process reduces the microbial load during the specified exposure period.

**Minimum disinfection contact time from a required CT, `enveng_contact_time_from_ct`**

The minimum contact time a disinfectant residual needs to reach a required CT value (t = CT / C).

$$t = \frac{CT_{req}}{C}$$

| symbol | | meaning | SI unit |
|---|---|---|---|
| `CT` | in | required CT value | kg*s/m^3 |
| `C` | in | disinfectant residual | kg/m^3 |
| `t` | out | minimum contact time CT/C | s |
| `t_min` | out | in minutes | , |
| `t_h` | out | in hours | , |

**Applies when:** CT, C > 0; constant residual over the contact time.

```
$ environmental-engineering solve enveng_contact_time_from_ct "CT=60 mg*min/L" "C=1.5 mg/L"
t = 2400 s
t_min = 40
t_h = 0.6666666667
  by: Minimum disinfection contact time from a required CT  (t = CT_required / C)
  CT = 3.6 kg*s/m^3
  C = 0.0015 kg/m^3
  assumes: CT, C > 0; constant residual over the contact time
```

In the worked example, CT = 60 mg*min/L, C = 1.5 mg/L go through the equation above and give t = 2400 s, t_min = 40, t_h = 0.6666666667. It applies only when CT, C > 0; constant residual over the contact time: outside that it says what is wrong instead of returning a number.

**Osmotic pressure of a dilute solution (van 't Hoff), `enveng_osmotic_pressure`**

The osmotic pressure of a dilute salt solution from its mass concentration, molar mass, van 't Hoff factor and temperature.

$$\pi = i \frac{C}{M} RT$$

| symbol | | meaning | SI unit |
|---|---|---|---|
| `C` | in | solute mass concentration | kg/m^3 |
| `M` | in | solute molar mass | kg/mol |
| `i` | in | van 't Hoff factor (ions per formula unit, 2 for NaCl) | , |
| `T` | in | temperature | K |
| `pi` | out | osmotic pressure | Pa |
| `pi_bar` | out | osmotic pressure in bar | , |

**Applies when:** dilute (ideal) solution.

```
$ environmental-engineering solve enveng_osmotic_pressure "C=35 g/L" "M=58.44 g/mol" i=2
"T=298.15 K"
pi = 2969318.824 Pa
pi_bar = 29.69318824
  by: Osmotic pressure of a dilute solution (van 't Hoff)  (pi = i (C/M) R T)
  C = 35 kg/m^3
  M = 0.05844 kg/mol
  i = 2
  T = 298.15 K
  assumes: dilute (ideal) solution
```

**What the worked example means:** This calculation determines the osmotic pressure of a dilute salt solution using the van 't Hoff equation. It combines the solute mass concentration, molar mass, temperature, and the van 't Hoff factor to estimate the pressure required to stop solvent flow across a semipermeable membrane. The resulting osmotic pressure is 2.96932e+06 Pa, equivalent to 29.6932 bar.

**Power needed for a target velocity gradient, `enveng_mixing_power`**

The power that must be dissipated in a tank to give a target mean velocity gradient G.

$$P = G^2 \mu V$$

| symbol | | meaning | SI unit |
|---|---|---|---|
| `G` | in | target velocity gradient | 1/s |
| `mu` | in | water viscosity | Pa*s |
| `V` | in | tank volume | m^3 |
| `P` | out | power required | W |

**Applies when:** G, mu, V > 0.

```
$ environmental-engineering solve enveng_mixing_power "G=700 1/s" "mu=0.001002 Pa*s" "V=10 m^3"
P = 4909.8 W
  by: Power needed for a target velocity gradient  (P = G^2 mu V)
  G = 700 1/s
  mu = 0.001002 Pa*s
  V = 10 m^3
  assumes: G, mu, V > 0
```

**What the worked example means:** This calculation determines the power that must be dissipated within a tank to achieve a specific mean velocity gradient. By using the formula involving the square of the gradient, the fluid viscosity, and the total volume, for a gradient of 700 1/s and a volume of 10 m^3, with water viscosity at 0.001002 Pa*s, the necessary power output is 4909.8 W. This value represents the energy input needed to maintain the desired mixing intensity throughout the specified liquid volume.

**Sedimentation basin sizing from overflow rate and detention time, `enveng_sedimentation_basin_sizing`**

The surface area, volume, side-water depth and (circular) diameter of a settling basin from the design flow, the design overflow rate and the detention time.

$$A = \frac{Q}{v_0},\ V = Q\theta,\ h = v_0\theta$$

| symbol | | meaning | SI unit |
|---|---|---|---|
| `Q` | in | design flow | m^3/s |
| `v0` | in | design overflow rate (surface loading) | m/s |
| `theta` | in | design detention time | s |
| `A` | out | surface area Q/v0 | m^2 |
| `V` | out | volume Q theta | m^3 |
| `depth` | out | depth V/A = v0 theta | m |
| `D` | out | diameter of one circular basin | m |

**Applies when:** one basin.

```
$ environmental-engineering solve enveng_sedimentation_basin_sizing "Q=15000 m^3/day" "v0=40
m/day" "theta=2 h"
A = 375 m^2
V = 1250 m^3
depth = 3.333333333 m
D = 21.85096861 m
  by: Sedimentation basin sizing from overflow rate and detention time  (A = Q/v0; V = Q theta;
depth = V/A; D = sqrt(4A/pi))
  Q = 0.1736111111 m^3/s
  v0 = 0.000462962963 m/s
  theta = 7200 s
  assumes: one basin
```

**What the worked example means:** This calculation determines the physical dimensions of a single sedimentation basin based on the design flow, overflow rate, and detention time. The required surface area is found by dividing the flow rate by the overflow rate, yielding 375 m^2. The total volume needed is the product of the flow rate and the detention time, giving 1250 m^3. Consequently, the side water depth is derived from the ratio of volume to area, which equals 3.33333 m. For a circular tank, this area corresponds to a diameter of 21.851 m. These values ensure that particles have sufficient time and space to settle out of the water before it exits the basin.

### Stokes settling velocity of a discrete particle, `enveng_stokes_settling_velocity`

The terminal settling velocity of a small spherical particle in water by Stokes' law, with the particle Reynolds number that confirms laminar settling.

$$v_s = \frac{g(\rho_p - \rho_w)d^2}{18\mu}$$

| symbol | | meaning | SI unit |
|---|---|---|---|
| `d` | in | particle diameter | m |
| `rho_p` | in | particle density | kg/m^3 |
| `rho_w` | in | water density | kg/m^3 |
| `mu` | in | water viscosity | Pa*s |
| `vs` | out | settling velocity | m/s |
| `Re` | out | particle Reynolds number | , |
| `vs_m_per_h` | out | in m/h | , |

**Applies when:** Re <= 1; rho_p > rho_w.

```
$ environmental-engineering solve enveng_stokes_settling_velocity "d=0.05 mm" "rho_p=2650
kg/m^3" "rho_w=998.2 kg/m^3" "mu=0.001002 Pa*s"
vs = 0.002245318318 m/s
Re = 0.1118401569
vs_m_per_h = 8.083145943
  by: Stokes settling velocity of a discrete particle  (vs = g (rho_p - rho_w) d^2/(18 mu))
  d = 5e-05 m
  rho_p = 2650 kg/m^3
  rho_w = 998.2 kg/m^3
  mu = 0.001002 Pa*s
  assumes: Re <= 1; rho_p > rho_w
```

**What the worked example means:** This calculation determines the terminal settling velocity of a discrete spherical particle in water using Stokes' law. Given a particle diameter of 0.05 mm, a particle density of 2650 kg/m^3, and water properties including a density of 998.2 kg/m^3 and viscosity of 0.001002 Pa*s, the resulting settling velocity is 0.00224532 m/s m/s. The calculated particle Reynolds number is 0.11184, which verifies that the flow around the particle remains laminar, satisfying the condition for the validity of this method.

**Surface overflow rate of a circular clarifier, `enveng_overflow_rate_circular`**

The surface area and the surface overflow rate (surface loading) of a circular settling tank from the flow and the tank diameter.

$$v_0 = \frac{Q}{\pi D^2/4}$$

| symbol | | meaning | SI unit |
|---|---|---|---|
| `Q` | in | flow rate | m^3/s |
| `D` | in | tank diameter | m |
| `A` | out | surface area | m^2 |
| `v0` | out | overflow rate Q/A | m/s |
| `v0_m_per_day` | out | overflow rate in m^3/(m^2 d) | , |

**Applies when:** Q, D > 0.

```
$ environmental-engineering solve enveng_overflow_rate_circular "Q=10000 m^3/day" "D=20 m"
A = 314.1592654 m^2
v0 = 0.0003684142201 m/s
v0_m_per_day = 31.83098862
  by: Surface overflow rate of a circular clarifier  (A = pi D^2/4; v0 = Q/A)
  Q = 0.1157407407 m^3/s
  D = 20 m
  assumes: Q, D > 0
```

**What the worked example means:** This calculation determines the physical dimensions and loading characteristics of a circular clarifier. Using the provided flow rate 10000 m^3/day and tank diameter 20 m, the surface area 314.159 m^2 is computed first. This area is then used to find the surface overflow rate 0.000368414 m/s, which represents the velocity at which water rises through the tank. A lower overflow rate generally indicates better settling conditions for suspended solids.

**Terminal settling velocity in any flow regime (drag coefficient iteration), `enveng_terminal_settling_velocity`**

The terminal settling velocity of a sphere in a fluid for laminar, transitional or turbulent settling, solving the force balance with the drag law Cd = 24/Re + 3/sqrt(Re) + 0.34.

$$v_t = \sqrt{\frac{4gd(\rho_p - \rho_f)}{3C_D\rho_f}}, \quad C_D = \frac{24}{Re} + \frac{3}{\sqrt{Re}} + 0.34$$

| symbol | | meaning | SI unit |
|---|---|---|---|
| `d` | in | particle diameter | m |

| symbol | | meaning | SI unit |
|---|---|---|---|
| `rho_p` | in | particle density | kg/m^3 |
| `rho_f` | in | fluid density | kg/m^3 |
| `mu` | in | fluid viscosity | Pa*s |
| `vt` | out | terminal settling velocity | m/s |
| `Re` | out | particle Reynolds number | , |
| `Cd` | out | drag coefficient | , |

**Applies when:** Re <= 2e5; rho_p > rho_f.

```
$ environmental-engineering solve enveng_terminal_settling_velocity "d=0.5 mm" "rho_p=2650
kg/m^3" "rho_f=998.2 kg/m^3" "mu=0.001002 Pa*s"
vt = 0.09058056284 m/s
Re = 45.11852187
Cd = 1.318558136
  by: Terminal settling velocity in any flow regime (drag coefficient iteration)  (vt^2 Cd(Re)
= (4/3) g d (rho_p - rho_f)/rho_f with Cd = 24/Re + 3/sqrt(Re) + 0.34, solved by bisection
below the Stokes velocity)
  d = 0.0005 m
  rho_p = 2650 kg/m^3
  rho_f = 998.2 kg/m^3
  mu = 0.001002 Pa*s
  assumes: Re <= 2e5; rho_p > rho_f
```

**What the worked example means:** This calculation determines the steady speed at which a particle falls through a fluid, balancing gravitational pull against drag and buoyancy. Using the particle diameter 0.5 mm, densities 2650 kg/m^3 and 998.2 kg/m^3, and viscosity 0.001002 Pa*s, the algorithm iterates to find the velocity where forces equalize. The resulting terminal velocity is 0.0905806 m/s, corresponding to a Reynolds number of 45.1185 and a drag coefficient of 1.31856. This approach works for laminar, transitional, or turbulent regimes up to a Reynolds number of 200000.

### Transmembrane pressure of a crossflow membrane, `enveng_transmembrane_pressure`

The transmembrane pressure of a crossflow membrane element, TMP = (P_feed + P_concentrate)/2 - P_permeate, and the feed-channel pressure drop.

$$TMP = \frac{P_f + P_c}{2} - P_p$$

| symbol | | meaning | SI unit |
|---|---|---|---|
| `P_feed` | in | feed pressure | Pa |
| `P_conc` | in | concentrate (retentate) pressure | Pa |
| `P_perm` | in | permeate pressure | Pa |
| `TMP` | out | transmembrane pressure | Pa |
| `TMP_kPa` | out | in kPa | , |
| `TMP_bar` | out | in bar | , |
| `dP` | out | feed-channel pressure drop P_feed - P_conc | Pa |
| `dP_kPa` | out | pressure drop, kPa | , |

**Applies when:** the same pressure basis for all three (gauge or absolute); P_conc <= P_feed; TMP > 0.

```
$ environmental-engineering solve enveng_transmembrane_pressure "P_feed=800 kPa" "P_conc=700
kPa" "P_perm=50 kPa"
TMP = 700000 Pa
TMP_kPa = 700
TMP_bar = 7
dP = 100000 Pa
dP_kPa = 100
  by: Transmembrane pressure of a crossflow membrane  (TMP = (P_feed + P_conc)/2 - P_perm)
  P_feed = 800000 Pa
  P_conc = 700000 Pa
  P_perm = 50000 Pa
  assumes: the same pressure basis for all three (gauge or absolute); P_conc <= P_feed; TMP > 0
```

In the worked example, P_feed = 800 kPa, P_conc = 700 kPa, P_perm = 50 kPa go through the equation above and give TMP = 700000 Pa, TMP_kPa = 700, TMP_bar = 7, dP = 100000 Pa. It applies only when the same pressure basis for all three (gauge or absolute); P_conc <= P_feed; TMP > 0: outside that it says what is wrong instead of returning a number.

### Water hardness as CaCO3, `enveng_water_hardness`

Calcium, magnesium and total hardness expressed as mg/L of CaCO3 from the calcium and magnesium concentrations, with the usual classification (soft to very hard).

$$TH = Ca\,\frac{50.04}{20.04} + Mg\,\frac{50.04}{12.15}$$

| symbol | | meaning | SI unit |
|---|---|---|---|
| `Ca` | in | calcium concentration (as Ca) | kg/m^3 |
| `Mg` | in | magnesium concentration (as Mg) | kg/m^3 |
| `total_hardness` | out | total hardness, mg/L as CaCO3 | , |
| `calcium_hardness` | out | calcium hardness, mg/L as CaCO3 | , |
| `magnesium_hardness` | out | magnesium hardness, mg/L as CaCO3 | , |
| `classification` | out | soft (< 60), moderately hard (60-120), hard (120-180), very hard (>= 180) | , |

**Applies when:** only Ca and Mg contribute.

```
$ environmental-engineering solve enveng_water_hardness "Ca=60 mg/L" "Mg=15 mg/L"
total_hardness = 211.6074929
calcium_hardness = 149.8381656
magnesium_hardness = 61.7693273
classification = very hard
  by: Water hardness as CaCO3  (hardness = C (EW_CaCO3/EW_ion), EW_CaCO3 = 50.04, EW_Ca =
20.04, EW_Mg = 12.15)
  Ca = 0.06 kg/m^3
  Mg = 0.015 kg/m^3
  assumes: only Ca and Mg contribute
```

**What the worked example means:** This calculation determines water hardness by converting the measured concentrations of calcium and magnesium into an equivalent mass of calcium carbonate. The calcium hardness is found to be 149.838, while the magnesium hardness is 61.7693, resulting in a total hardness value of 211.607.

Based on these values, the water sample is classified as very hard, indicating its suitability for various uses and potential scaling issues.

**Weir loading rate of a circular clarifier, `enveng_weir_loading_rate`**

The weir length and weir loading rate of a circular clarifier with a peripheral weir.

$$WL = \frac{Q}{\pi D}$$

| symbol | | meaning | SI unit |
|---|---|---|---|
| `Q` | in | flow rate | m^3/s |
| `D` | in | weir (tank) diameter | m |
| `weir_length` | out | weir length pi D | m |
| `WL` | out | weir loading rate Q/(pi D) | m^2/s |
| `WL_m3_per_m_day` | out | weir loading in m^3/(m d) | , |

**Applies when:** peripheral weir.

```
$ environmental-engineering solve enveng_weir_loading_rate "Q=10000 m^3/day" "D=20 m"
weir_length = 62.83185307 m
WL = 0.001842071101 m^2/s
WL_m3_per_m_day = 159.1549431
  by: Weir loading rate of a circular clarifier  (WL = Q/(pi D))
  Q = 0.1157407407 m^3/s
  D = 20 m
  assumes: peripheral weir
```

**What the worked example means:** This calculation determines the weir loading rate for a circular clarifier with a peripheral weir. It uses the flow rate 10000 m^3/day and the tank diameter 20 m to find the total weir length, which is 62.8319 m. The resulting loading rate is 0.00184207 m^2/s, representing the volume of water passing over each meter of weir per second.

Questions and reports: support@ai999x.com. AI999X, The Future Is Self-Improving.